



Facility Management Professional™



Finance and Business

Leadership and Strategy

Operations and Maintenance

Project Management

Operations and Maintenance

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Acknowledgments

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Introduction

In 2009, the International Facility Management Association (IFMA) conducted a global job task analysis (GJTA) to identify the tasks and knowledge that are important for competent performance by facility managers across the world. As part of that effort, IFMA identified operations and maintenance (O&M) as a competency area, described the role of the facility manager as related to operations and maintenance, and identified key tasks, knowledge statements and competencies that support O&M.

IFMA FMP Credential Program

IFMA's Facility Management Professional (FMP) credential is an assessment-based certificate program demonstrating a proven comprehension of the foundations of facility management (FM). In September 2010, the FMP credential was updated and enhanced to align with the most recent IFMA global job task analysis, ensuring that it teaches and tests the knowledge demanded by today's global employers in a convenient print and online self-study program.

The FMP Credential Program teaches and assesses your comprehension of topics in these four knowledge domains:

- Operations and maintenance
- Project management
- Finance and business
- Leadership and strategy

This course focuses on the area of operations and maintenance. Combine this print manual with the online study tools (including quizzes, case studies, Flashcards, progress reports and more) to help you apply and retain the FMP concepts. Successfully complete all four FMP courses and final assessments, submit your FMP application to IFMA for approval and receive your credential.

Role of facility managers as related to operations and maintenance

According to the IFMA GJTA:

The primary role of facility managers is to manage/oversee an operating facility. To do this, facility managers must have a working knowledge of building systems, structure, interiors and exteriors and grounds so the facility and all of its required systems function efficiently, reliably, safely, securely and in a manner consistent with existing regulations and standards.



In addition, the various aspects of O&M are constantly changing due to wear or deterioration, new regulations or requirements, operational modifications, occupant expectations, and many other individual but interrelated circumstances. Managing a facility means continually managing change.

**Note on term
“entire
organization”**

The term “entire organization” is used throughout this course to apply to the facility management organization’s parent or client organization (corporation, government body, nonprofit, etc.).



Chapter 1: Introduction to Operations and Maintenance

After completing this chapter, students will be able to:

- Explain the role of operations and maintenance in facility management.
 - Discuss the scope of the facility manager's duties and responsibilities in support of the organization's missions and goals.
 - Describe how to develop, recommend and manage/oversee a strategic facility plan and associated O&M tactical plans.
 - Determine appropriate O&M schedules for a comprehensive list of facility assets.
-

✦ Topic 1: Operations and Maintenance Overview

This course addresses operations and maintenance as a whole. It provides insights into O&M tasks and knowledge so that the facility manager can:

- Assess the condition of the facility systems, structure, interiors, exteriors and grounds.
- Manage/oversee facility operations and maintenance (hard) services.
- Manage/oversee occupant (soft) services such as custodial, parking and security.
- Develop specifications and performance requirements for services and select the best resources.
- Measure, manage and plan for performance, efficiency, sustainability and mission-related expectations.

Definitions

The following are some key definitions related to facility O&M.

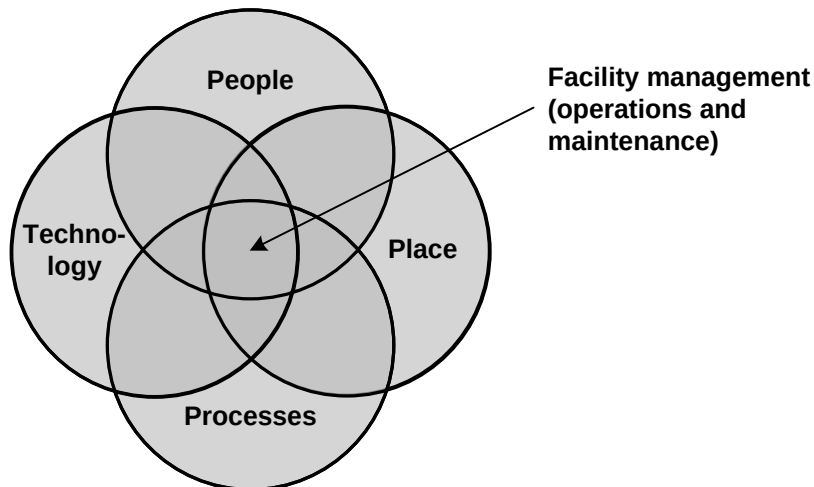
- **Operations** involves assuring that the facility's infrastructure and how it is used and managed provides a satisfactory work environment, is in compliance with laws and regulations, meets financial performance goals, reflects efficient utility services and costs, and protects the surrounding community and environment.
- **Maintenance** deals with assuring that all of the elements of the infrastructure are serviced so they operate efficiently and are reliable and safe. It includes scheduling and conducting regular, periodic predictive, preventive and corrective maintenance activities.



Thus, operations keeps the facility functioning for its intended purpose and maintenance is the necessary work that sustains the asset's anticipated useful life.

- **Building infrastructure** consists of building systems, building structures, interior elements (including furniture, fixtures and equipment), exterior envelope, separate structures and grounds.
- **Facility management** (from an operations and maintenance perspective) is at the center of occupant needs, business and communication processes, management and workflow technologies and the physical infrastructure itself, as shown in Exhibit 1-1.

Exhibit 1-1: Interacting Responsibilities



Facility manager duties and responsibilities

Operations and maintenance represents a cost center, as it adds directly to the cost of an organization but only indirectly to its profit. O&M produces a total expense second only to the salary or pay of occupants in a typical facility. This makes operating and managing facilities critical in the overall success of the organization.

Facility managers must know how to adapt as well as respond. Directly responsible for the physical asset and its function, facility managers bring together methods and procedures to recognize, maintain, adjust and improve that functionality. They measure the success of those efforts with productivity and performance results rather than just through activity requirements.



Within established policies, the facility manager should be qualified to direct operations and maintenance activities across the entire organization. The facility manager should be accountable for and able to demonstrate the lowest overall total cost of ownership of O&M to provide the optimum level of service that meets the needs of the organization and those who occupy the facility. To realize this, facility managers draw upon not only their experience but also their business capability to plan and organize.

An individual who is new to managing a facility or is going to be its manager needs to become familiar and comfortable in his or her understanding and should:

- Understand the overall nature of the entire organization and its culture. If there is a long-term plan or an O&M operating plan, the new facility manager must become familiar with it, what it expects (goals) and what has been done (realization of goals).
- Know what the mission-critical issues are—what the owner and executive decision makers consider most important. Is it productivity? Customer satisfaction? Quality? Security?
- Understand operational and management issues relative to financial planning, budgeting, capital projects and so forth.
- Be familiar with the state of service contracts and agreements.
- Understand the automated management system or other computerized maintenance management system, if any, that is used for facility O&M.
- Be prepared and able to organize, track, discuss and interpret service activities for executive-level presentation to decision makers.

A facility manager is expected to operate and maintain the facility in a manner that supports the goals and mission of the entire organization. He or she must interpret and understand the needs of the organization and occupiers while balancing competing requirements. In this way, the facility manager is able to:

- Provide safe, healthy and productive facility environments.
- Develop the lowest possible systems/component failure and highest possible reliability at the lowest overall total cost.
- Develop, implement and manage recycling and sustainability programs.
- Develop, recommend and manage facility operational planning requirements, such as temperature control, lighting and equipment replacement.
- Anticipate change and plan in advance for necessary adaptations.
- Comply with all relevant international, national and local codes and regulations.



- Provide more efficient use for higher occupant productivity, lower overtime expense and lower energy consumption.
- Facilitate the use of automated facility management technology (such as computer-based workflow, operations and analysis functions).
- Provide efficient, cost-effective support services (contracted or internal staff) throughout the entire organization.
- Hire, train, manage and assess the technical performance and occupant service performance of O&M staff.
- Produce suitable and publicly satisfactory overall appearance and aesthetics.
- Supply appropriate fire, life safety and emergency response readiness.
- Act as a consultant to other departments on facility-related matters.
- Consider human factors as an integral component of facility operations and maintenance.

✦ Topic 2: Begin with a Plan

To accomplish their duties and responsibilities, facility managers should have a detailed and complete management plan. Planning is important in anticipating both long-term needs and short-term requirements. It becomes the basis for how facility O&M is successfully applied to all assets of the infrastructure so that occupants/customers are satisfied and the organization's mission is accomplished. It helps the facility manager to determine:

- The level of service expected and wanted.
- Operations and maintenance policies and procedures.
- Operational requirements.
- Maintenance program requirements in detail and specified through service level agreements.
- Operational and capital improvement projects based on total cost of ownership and life-cycle cost analysis.
- Availability and capability of facility management organization resources.
- Services and tasks to contract.
- Programs for quality control and assurance.
- Key performance indicators and balanced scorecard measurements.

All of these aspects are defined and discussed in greater detail throughout this course. They are merely introduced here at the highest level.

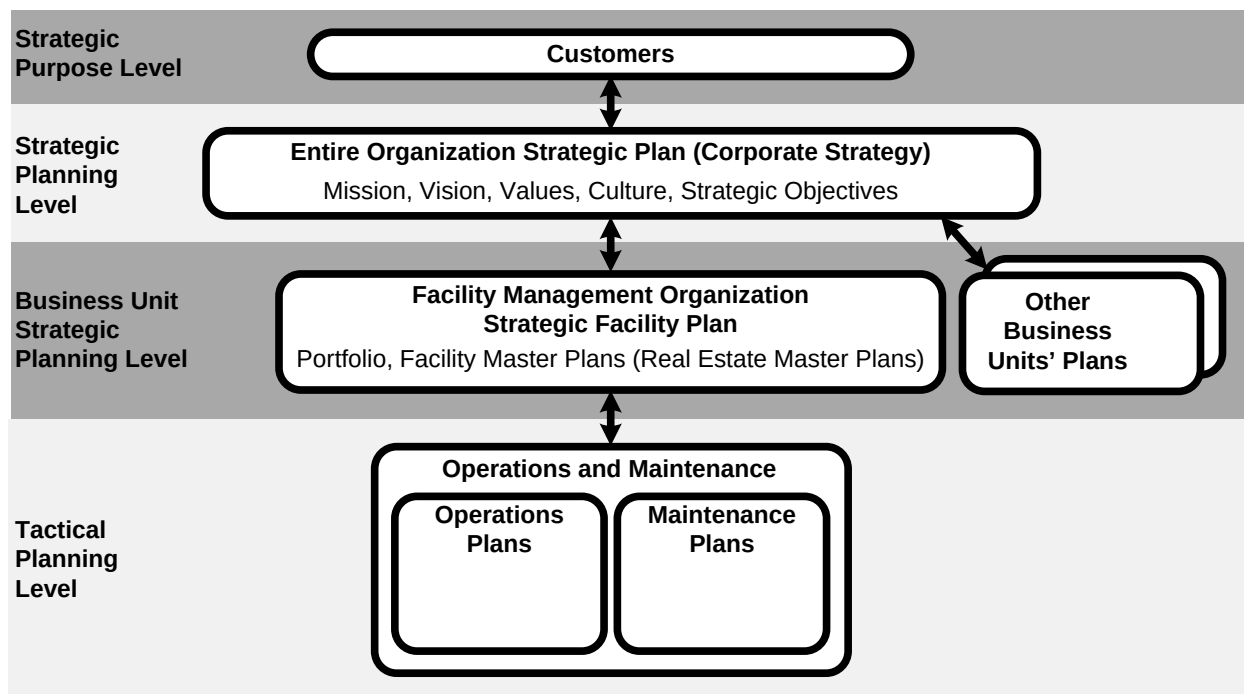
Plans generally begin at a high strategic level and proceed to the ever-greater detail of tactics to accomplish the strategy. A tactical plan is a detailed set of



steps needed to accomplish a goal in the strategic plan. For O&M, basic levels of plans may include the following:

- The strategic purpose and strategic planning levels (the top two levels shown in Exhibit 1-2) consider the requirements and expectations of occupants or customers and align them with the mission, vision, values and culture of the entire organization.
- By understanding the entire organization's identified strategy, the facility manager works within the strategic goals to support or develop a three- to five-year strategic facility plan (shown as the third level in the exhibit).
- O&M tactical plans of the steps needed to accomplish the goals in the strategic plan are created. Tactical plans are usually short-range activities (generally covering a year or less). For facility management these tactics are represented through maintenance schedules, operational activities, service delivery outcomes, equipment replacement timelines and other O&M tasks.

Exhibit 1-2: Strategic Planning Overview Model



The strategic planning overview model shown in Exhibit 1-2 demonstrates how an organization describes and aligns its mission and core values to its operational perceptions specific to operations and maintenance. Other parallel aspects, such as preparing budgets, developing design and construction plans and preparing for capital renewal, are discussed in other courses in the IFMA FMP Credential Program, such as *Project Management* and *Finance and Business*.

The facility manager develops an O&M approach through several aspects:

- Determine physical assets as an inventory.
- Determine maintenance and repair cycle.
- Determine maintenance schedule.
- Determine applicable regulations, codes and standards.
- Determine agreed-upon occupant services and requirements.

Determine physical assets as an inventory

To apply a strategic facility plan and the resulting O&M plans, the facility manager first needs to know what materials, components, equipment and systems exist in the facility—that is, all infrastructure assets. It is critical to know what a facility actually contains in order to manage and maintain it. In addition, the facility manager needs to understand and incorporate plans for capital renewal and repairs in a complete program for managing the physical infrastructure.

A facility's physical assets are not static. They are continually changing due to wear, age, obsolescence or other time-related factors. Because of this, facility management can also be thought of as change management. The facility manager needs to anticipate change based on tendencies and insight as well as actual equipment, component or service conditions. To anticipate these possibilities, facility managers need a thorough, detailed understanding of the current condition of all the infrastructure's physical elements.

For both of the reasons described above—to know what exists and to anticipate its deterioration—the facility manager needs a complete and accurate inventory/register of facility assets.

If a complete, good-quality inventory/register is not available, the facility manager should initiate an inspection to create one. During the inspection, observers look for abnormalities in building systems such as:

- Poor working order.
- Missing items.
- Deficiencies.
- Illumination problems.
- Spills, evidence of leakage, gas releases, insufficient air movement.
- Corrosion.
- Excessive or unusual noise, vibration.
- Safety or security hazards such as missing spark arrestors, open vents, trip hazards, intrusion potential (access grates or grills).



A facility inspection has other purposes beyond verifying that equipment and systems operation is occurring as intended. The facility manager may also use the inspection to:

- Confirm the effectiveness of current preventive, predictive and corrective programs.
- Interact with occupants to collect their observations and comments relative to the mission and vision of the entire organization.

Once a facility inspection is completed and the inventory/register is documented, it can also become part of a contractual arrangement through a lease.

The inventory/register is the basis for operations and maintenance plans that recognize all the physical elements contained in the entire organization's systems, structures, interiors, exteriors and grounds. These physical elements are related across processes or aspects that include:

- Security and safety (emergency preparedness and business continuity).
- Technology (security and information systems).
- Environmental stewardship and sustainability.
- Project management.
- Quality (performance to specifications).
- Finance and business.
- Leadership and strategy.

Each of these is an IFMA core competency area comparable to O&M. However, for the purposes of this course, it is important to recognize that having an inventory/register of physical assets is the necessary foundation upon which all other tasks and these separate competencies are built.

Determine maintenance and repair cycle

Facilities need to be properly maintained, repaired and operated to meet occupant/visitor needs and regulatory requirements. The decisions on when to apply predictive, preventive or no scheduled maintenance are determined by, among other things, safety, overall cost and available resources, occupant satisfaction and productivity. In this way, facility maintenance may be considered a return on performance rather than simply a property investment.

The proper operation and maintenance of a facility should be a precise and orderly management of preventive and corrective activities. The facility



manager orchestrates these activities in a controlled, predetermined cycle that incorporates the range of operational aspects, such as:

- Equipment and systems operations, maintenance and repair.
- Renewal, alterations, and relocations or moves.
- Energy management.
- Waste management and recycling.
- Fire/life safety, disaster recovery and security.

Each of these major operational areas is more thoroughly discussed in other topics in this course. However, they may all be part of a planning cycle that generally consists of determining expectations, identifying current conditions, assigning resources, preparing a work plan and gathering feedback.

Determine maintenance schedule

The maintenance schedule represents an O&M tactical approach that is essentially time-related, where the activities that occur the soonest are known and the more distant activities are less known. Activities expected in the short term have challenges and requirements that are understood. Activities that may occur later (long term) need planning and preparation in advance. The short-term strategy determines what to do to maintain and improve current O&M management and services. The long-term strategy considers all the variables and aspects of sustainability, quality, occupant services, provisions for equipment and capital renewal estimates. It identifies priorities, compares costs, determines services and reviews historical data.

Whether the planning is short-term or long-term, it is important to include the qualitative aspects of maintenance activities such as ownership goals, staff or contractor expertise, occupant expectations, priorities and so on. These soft elements relate to an O&M approach that recognizes and incorporates both planned and unplanned maintenance aspects.

- **Planned maintenance** is any maintenance activity for which a predetermined job procedure has been documented and for which all labor, materials, tools and equipment required to carry out the task have been estimated and their availability assured before commencement of the task.
- **Unplanned or unscheduled maintenance** is any maintenance work that has not been included on an approved maintenance schedule prior to its commencement. It generally addresses failed equipment or components or an unsatisfactory level of service. (Level of service is discussed in Chapter 5.)



Maintenance may be planned or unplanned, but ideally the facility manager employs scheduled, preventive maintenance and repair activities to lower cost and prevent costly breakdowns, with a small proportion of unplanned or corrective maintenance. It is not possible to eliminate unplanned maintenance. Best practice is to find the optimum balance between preventive and corrective maintenance to realize the greatest reliability at the lowest cost.

With a plan in mind, the facility manager can execute an orderly approach to the business of operating and maintaining building systems, structures, interiors, exteriors and grounds. Properly maintained facilities meet the expectations of occupants, the obligations of owners, and the requirements of regulators. It should not matter whether facilities are owned or leased. Meeting the minimum expectations of all parties should be important to the facility manager.

Determine applicable regulations, codes and standards

Regulations, codes and standards occur across the world, and many share common requirements internationally. They can prescribe conditions for nearly any aspect of a facility and include nearly everything for which a facility manager is responsible.

- **Regulations** are laws or rules prescribed by an authority to regulate conduct; an official governmental regulatory act that transmits a stated purpose.
- **Codes** are systems of regulations that define scoping requirements. Internationally and in the United States, model codes (such as those provided by International Code Council, or ICC) are adopted, amended and enforced at the local level. These codes refer to many other codes and standards that cover building, environmental, safety, barrier-free access and other requirements.
- **Standards** state the minimum level of acceptability and incorporate the technical installation requirements of the code when/where referenced.

Building codes are performance and prescriptive requirements for building construction, operations and maintenance. They establish predictable and consistent minimum standards that are applied to the quality and durability of construction materials and building systems. Construction must meet minimum standards that determine what is practical and adequate for protecting life, safety and the welfare of the public.

Building codes can embrace all aspects of building construction, including fire and life safety issues; structural design; security; mechanical, electrical and plumbing systems; energy conservation; sustainability; and accessibility. They



provide safeguards to ensure uniformity in the construction and facilities industries.

While codes provide a means to reduce risks to an acceptable level, no code can totally eliminate all potential hazards. Government agencies or authorities that have jurisdiction may interpret and enforce codes differently from country to country and at different times. Local regulators or even facility owners or executives may require exceptions that exceed nominal code requirements to make them more appropriate to the purpose or mission of the facility.

In addition, regulations, codes and standards themselves can change regularly. New references may be published each year and updated even more frequently.

Facility managers should be aware of all applicable requirements, should be current in their understanding, and should include those codes, regulations and standards in all considerations of building systems, structures, interiors, exteriors and grounds.

Codes, regulations and standards change so frequently and continuously that it is not practical to list them in a document such as this. A useful, international, relatively current appendix is available through the IFMA FMP Credential Program Resource Center.

Determine agreed-upon occupant services and requirements

To create a safe, productive and occupant-pleasing environment, the facility manager needs to understand what services and requirements occupants and facility owners/decision makers need and expect. These needs and expectations form the basis for operations plans and routine facility management activities, and the facility manager must work within the approved funds and staff resources available to supply them.

It is important that services and their requirements are not only understood but also agreed upon among the people who use the services and those who provide them. This mutual understanding provides the basis for satisfaction on all sides.

Considerations about customers or occupants are generally included at the highest strategic level. The entire organization includes these considerations as part of the strategic facility plan mentioned earlier.

Planning and measuring approaches such as service level agreements and performance measurements support this mutual understanding. These aspects are discussed more completely in Chapters 5 and 6.



Progress Check Questions

Directions: Read each question and respond in the space provided. Answers and page references follow the questions.

Match the following facility management terms with their description.

- | | |
|-----------------------------------|---|
| 1. _____ Operations | a. Materials, equipment and systems found in infrastructure |
| 2. _____ Maintenance | b. Assuring how facility infrastructure is used |
| 3. _____ Asset inventory/register | c. Assuring how infrastructure elements are serviced |
-
4. If a complete, good-quality inventory/register is not available, what should the facility manager do?
- () a. Begin to create one as new assets are acquired and installed.
- () b. Initiate an inspection to create one.
- () c. Observe how individual assets are functioning based on work logs and condition reports.
- () d. There is nothing a facility manager can do.
-
5. A maintenance and repair cycle is typically
- () a. an examination of facility assets and capital improvement expectations.
- () b. an energy management review.
- () c. precise and orderly management of preventive and corrective activities.
- () d. a methods and resources schedule.
-
6. Maintenance activities may be routinely divided into
- () a. operable and inoperable.
- () b. predictive and unnecessary.
- () c. strategic and tactical.
- () d. planned and unplanned.



Progress check answers

1. b (p. 3)
2. c (p. 3)
3. a (p. 8)
4. b (p. 8)
5. c (p. 9)
6. d (p. 10)



Chapter 2: Assess Facility Needs

After completing this chapter, students will be able to:

- Identify and describe building structures, systems, interiors, exteriors and grounds.
 - Discuss common deficiencies in building structures, systems, interiors, exteriors and grounds.
 - Identify security, health and safety considerations relative to building structures, systems, interiors, exteriors and grounds.
-

★ Topic 1: Assess Condition of Building Structure

Typical building structures and common deficiencies

Building structures are the construction elements that support the architecture. They enable buildings to resist forces of users (occupants, visitors), nature (wind, rain, earthquakes) and the equipment that maintains the structure's living comfort. Elements of building structures include those listed in Exhibit 2-1.

Exhibit 2-1: Elements of Building Structures

Building Structures	Elements
Foundations	• Drilled piers and shafts, also known as caissons • Driven and bored piles (steel, wood, precast concrete) • Shallow spread and strip footings (simple concrete footings bearing on ground support columns, concrete foundation walls or the grade beams above them) • Special foundation systems (such as structural mats, which are thick, deeply positioned slabs) In some countries the foundation is considered to be the area surrounding the footings such as an earth or rock foundation or concrete footing. For the purposes of this course, foundations describe load-bearing elements at or below grade.
Substructure (below-grade) frames	• Structural concrete framing and walls • Steel framing
Superstructure	• Cast-in-place concrete (typically reinforced and post-tension) • Prefabricated/precast concrete columns, beams and joists • Steel frame with trusses, beams and joists • Wood for low-rise construction
Floor slabs (as part of superstructure)	• Slab-on-grade • Elevated or suspended structural slabs



Generally building structures are not visible to observation.

An inspection to observe the condition of a building structure can detect deficiencies early so that repair can restore the structure's optimum usefulness. Structural audits may require hiring experts, who may use specialized instruments, investigative tools and measurements to diagnose possible problems. The facility manager should know when to contract professional resources such as structural engineers to use their experience for assessing structural conditions.

An assessment of the facility structure can include other aspects such as environmental sustainability, quality, technology and safety. The list of possible causes and structural deficiencies is virtually limitless.

Examples of deficiencies to look for in a building's structure may include:

- Cracks in exterior or interior walls or in exposed concrete at the base of the building; arching in floors or concrete.
- Cracks running diagonally along mortar joints in veneers.
- Differential displacement of floor slabs or joints.
- Separation of corners at windows, doors or joints where walls meet; caulked joints pulling apart along the bottoms of windows and doors.
- Standing water or pools collected near the facility during or after a rain.
- Evidence of moisture penetration such as leaking, dampness, staining or condensation.
- Nail or screw heads popping out of interior partitions.
- Unusual cracks in floors; ridges in tile or below carpet that produce wear points and safety issues.
- Deterioration evident as corrosion, chemical reactions, decay or insect infestation.

Note that not all cracks are undesirable; some may be useful or even designed to permit expansion or spacing. However, any crack that produces unevenness across the separation could indicate structural problems.

Foundations

Let's take a closer look at one particular building structure—the foundation. Foundations are the structural components of the building frame that accept and transfer the massive load (or weight) of the building unit into the surrounding soils.

Foundations themselves consist of different elements that interact to provide stability, bear loads, protect against moisture penetration, and maintain integrity against potential erosion, seismic events and freeze/thaw cycles.



Foundations include items such as the following:

- Foundation walls or supports are usually of poured concrete, concrete blocks, pilings or piers.
- The footing is the enlarged concrete pad at the base of the foundation wall that transfers the weight of the structure to the soil/rock.
- The footing drain tile is typically a pipe perforated with openings or “weep holes” to allow water to drain.
- The floor slab is a layer of concrete that forms a floor at any level of a building.
- The grade beam supports the exterior wall of a superstructure (for buildings without a basement) by directly bearing on the column footings or may be self-supporting as a strap footing.
- The rebar is reinforcing carbon steel bar contained by the concrete pour to add strength and stability to the wall or floor.
- Gravel fill may be placed beneath a basement floor slab to permit drainage and help maintain a dry floor. Vapor barriers generally separate the gravel from the concrete slab-on-grade.
- Backfill is earth that is replaced and compacted around the foundation once the concrete walls have been built.
- The drainage board is a below-grade sheet, board or membrane that may insulate as well as enhance drainage.
- Filter fabric is often placed over the drain pipe or drain tile to keep fine soil or silt from accumulating and clogging drain openings.
- The waterproofing membrane is a liquid coating that cures to a flexible rubber and is applied to waterproof below-grade concrete walls.

Exhibit 2-2 on the next page illustrates some typical foundation elements.

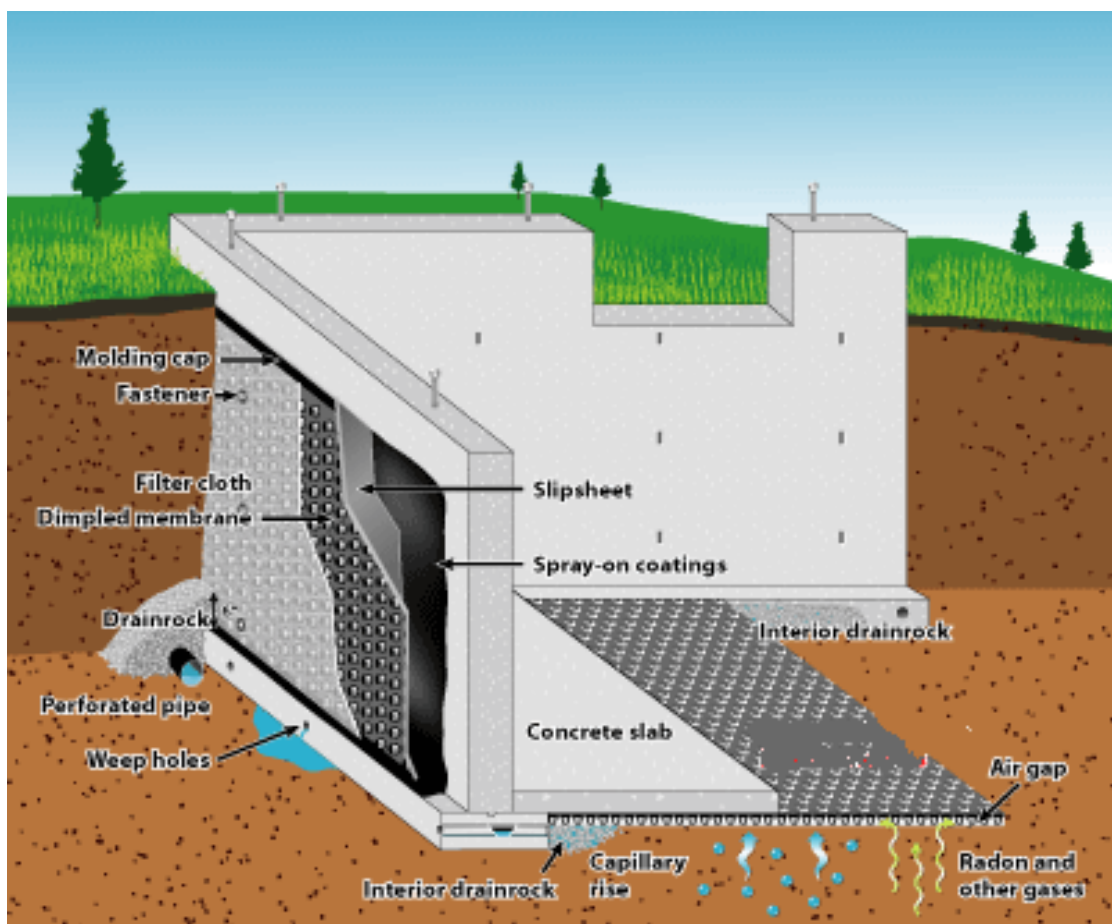
It may not require a specialist to recognize foundation problems such as cracks (including cracks around doors and windows) or door/window openings that bind because of settling. However, if any inspection reveals evidence of foundation problems, the facility manager should hire a consultant or specialist contractor.

To remediate or prevent foundation deficiencies:

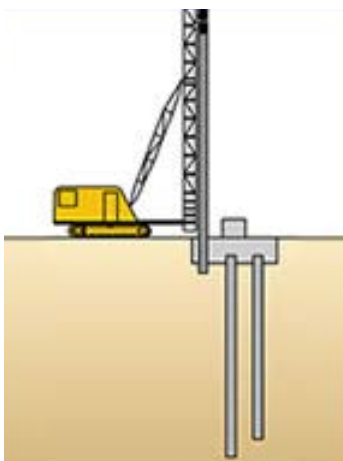
- Grade soil to ensure that it slopes away from the building to drain water away from foundations and below-grade walls.
- Drain roof downspouts so that rain water is directed away from building walls.
- Plant trees away from, not adjacent, to a building.



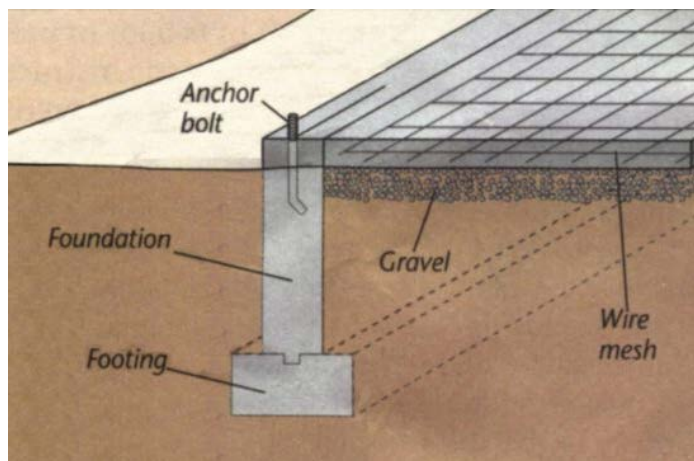
Exhibit 2-2: General Foundation Elements



Deep foundation



Shallow foundation



All aspects of a facility's infrastructure—structures, systems, interiors, exteriors and grounds—should undergo a thorough assessment so that the facility manager:

- Is aware of the condition of everything related to the facility and can develop a comprehensive work plan that indicates when to maintain, repair or perhaps eventually replace components.
- Can identify when (frequency) to schedule preventive maintenance and how to recognize and correct out-of-line conditions.
- Can maintain systems and components within operational specifications to sustain and improve efficiency.

These assessments should be part of the facility manager's comprehensive audit and should include sustainability, quality, technology, and safety aspects. Throughout this course, sample assessments are included that provide guidance for a facility manager in assessing the state of various facility components. Exhibit 2-3 on the next page is the first of these; it illustrates a partial sample of a foundation system assessment.

Health and safety considerations

It is important to recognize and correct deficiencies in structures because they can seriously affect the health, comfort and safety of occupants. The following are some examples:

- Moisture in walls or beneath roofs can promote mold, algae or other conditions that affect air quality.
- Decay or deterioration of wood or other natural building products caused by moisture, insect infestation or exposure can compromise the load-bearing capacity and make the building structure unsafe.
- Cracks in floors or walls may indicate unsafe shifts in the structure itself and perhaps unsafe design loads.
- Unwanted openings or deterioration around doors or windows can significantly reduce energy efficiency and occupant comfort.
- Doors that do not fit squarely, that bind or are misaligned may not fully close. This can produce energy leaks as well as security issues if an opening cannot close enough to be locked.
- Uneven tread width, height, and depth and loose, weak or missing railings can cause trip and fall hazards or other safety issues.
- Uneven floors, thresholds or railing heights can also cause hazardous walking conditions.
- Leaking or standing water can damage equipment, electrical systems, furnishings and finishes and not only produce unpleasant aesthetics but unsafe conditions.



Exhibit 2-3: Foundation System Assessment

Facility, system and location identification _____ Inspector name and date of assessment _____		Components: footings, grade beams, walls, waterproofing, drains, rebar, insulation
General causes or concerns		
Deterioration ☐ Surface spalling ☐ Corrosion, chemical reaction ☐ Decay ☐ Shrinkage ☐ Settlement ☐ Weather wear ☐ Frost heave ☐ Poor crawl space ventilation	Moisture penetration ☐ Inadequate drainage ☐ Surface pooling ☐ Water table changes ☐ Condensation ☐ Leakage ☐ Unexpected hydraulic pressure	Compliance ☐ Local regulations/codes ☐ Area requirements ☐ National/international standards ☐ Required permits
Design loads ☐ Building equipment ☐ Vibration potential ☐ Seismic resistance ☐ Structural or occupancy changes	Drainage ☐ Reduced soil cover ☐ Foundation settling ☐ Improper/inadequate surface slope ☐ Insufficient drainage	Maintenance history _____ _____ _____ _____ _____ _____
Component evaluation		
Description (type, location, serial number/ID)	Condition (deficiency described)	
_____	_____	
_____	_____	
_____	_____	
_____	_____	
_____	_____	
_____	_____	
_____	_____	
_____	_____	
_____	_____	
_____	_____	



Natural disasters

Risks from natural disasters affect most of the world's buildings. The combined potential for unsafe conditions because of seismic disruption (earthquakes, volcanoes), storms (hurricanes, typhoons, tornadoes), floods, landslides, lightning-induced fires, and other causes make any building vulnerable. While there is little from an O&M perspective that can be done to reduce the potential for these hazards, the facility manager can retain experts to analyze and improve resistance to the effects of nature's power, especially when planning capital improvements or renovations.

✦ **Topic 2: Assess Exterior Structures and Elements**

Typical facility exteriors and common deficiencies

Facility exteriors generally include the building envelope as well as structures separate from the main building.

“Building envelope” is a common term for the system that keeps the weather and noise out and the heat and cooling in. The building envelope generally includes elements such as facades, curtain walls, exterior walls, windows, skylights and roofs.

A curtain wall, or exterior insulation finish system (EIFS), is a facade that encloses the building. It is attached to the building's skeletal structure as a means to provide architectural design as well as a barrier to exterior elements. As a veneer or outer covering, curtain walls may be made of aluminum, glass, stone, metal, concrete, brick, stucco or other materials. They span multiple floors and are designed to accommodate thermal expansion and building sway. They may also provide thermal efficiency to more cost-effectively heat, cool or illuminate the building.

Separate or independent facility exterior structures can include:

- Outbuildings, sheds, guard houses.
- Fitness, exercise and recreation courts, pools, rinks, fields.
- Animal shelters.
- Vaults, greenhouses, conservatories.
- Band shells, pavilions, outdoor dining facilities that are not part of the main facility.

These exterior structures may also require attention to sustainability, technology, security and safety, as would the main facility infrastructure.

The appearance of exterior structures imparts an image to the public and occupants. The overall aesthetics of materials and finishes, the condition of



surfaces, cleanliness, and features provide an obvious impression of the entire infrastructure. Lighting, signage, layout and general maintenance all imply to visitors what importance and value the owners and thus the facility manager place on the building, its occupants and their purpose.

A facility that is not occupied by the owner may not have as stringent appearance requirements. Tenants may be held to accept specific minimum standards, or they may negotiate for different standards as part of a lease agreement. In any case, a facility's appearance influences perceptions on its quality and even its capabilities.

Some common deficiencies to look for among exterior elements and structures in general include:

- Deteriorating foundation, wall, roof, and other surface conditions (cracks, peeling, rotting, moisture/condensation).
- Spills, stains, improperly stored materials or items.
- Ill-fitting, misshaped or otherwise out-of-alignment windows, doors, building corners or rooflines.
- Poor joints, anchoring, ventilation, lighting.
- Insufficient size, capacity, access or egress.
- Deteriorating exterior sealants and mortar joints.

Exterior structures require much the same O&M and facility management as primary building structures to the extent that these supplemental structures are heated, cooled, supplied with water, enclosed and occupied.

Two types of exterior structures that deserve some further attention are roof systems and signage.

Roof systems

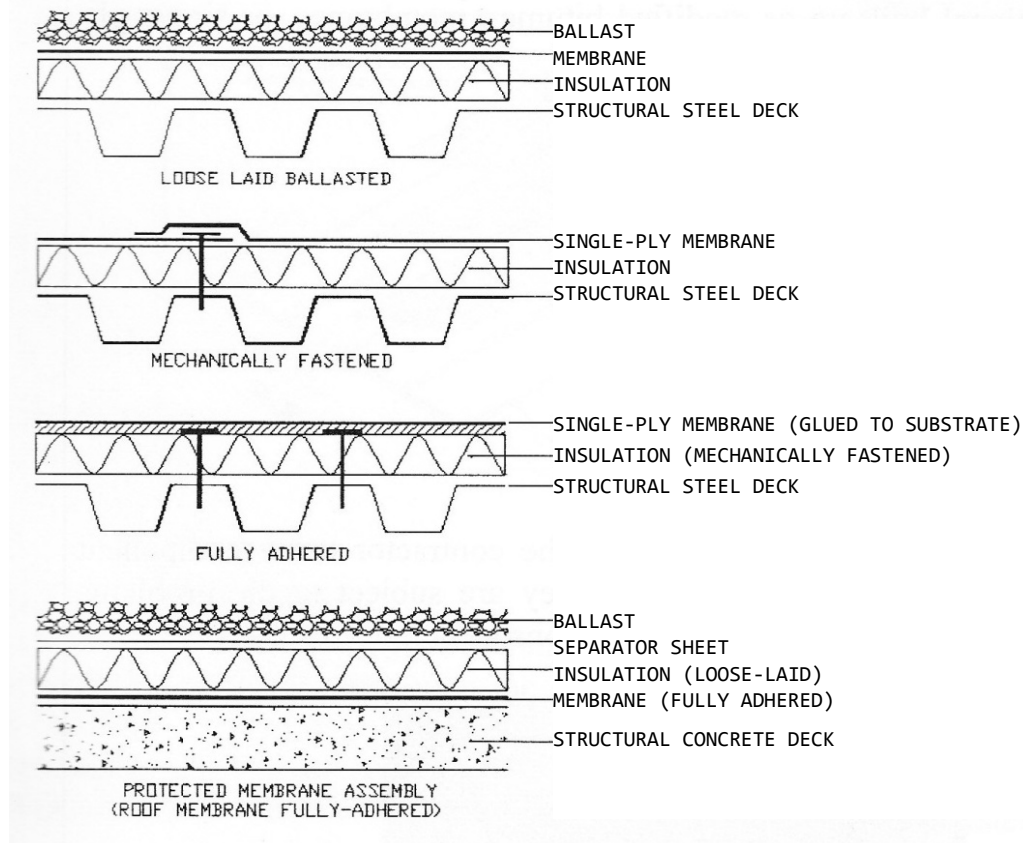
Roof systems are of particular concern to facility managers because they must maintain a weatherproof membrane and withstand the range of potential weather conditions. A roof that is not properly designed, constructed and maintained will likely affect the life cycle of the structure or require expensive, early replacement.

Roofs are generally either pitched (steep slope) or low-slope (appear to be flat). Pitched roofs often consist of shingles, slate tiles, thatch or standing-seam metal panels. Most commercial structures have primarily low-slope roofs with waterproofing membranes.



There are many different types of low-slope roofing constructions and materials. Some of the more common types of roofs are shown in Exhibit-2-4.

Exhibit 2-4: Common Roof Constructions



Loose-laid ballasted roofs consist of a membrane weighted down by a layer of small round stones, gravel, paving slabs or even soil and plantings (green roof systems).

Mechanically fastened roofs use a single-ply membrane affixed to the roof deck using screws, nails or adhesive and mechanically welded or otherwise fabricated to produce an impervious layer.

Fully adhered roofs use a membrane that is bonded to the roof deck to form a single sheet.

Protected membrane assembly roofs adhere the membrane fully to the roof deck beneath an insulation layer, separate sheet and ballast on the surface, thus putting the waterproof layer beneath the insulation rather than on the surface, so that some manufacturers refer to them as upside-down roofs.

Bituminous roofs use materials such as felts, small minerals and fillers such as limestone or sand as built-up layers to produce a waterproof surface. Modified bitumen membranes may also be used to build bituminous roofs using factory-fabricated layers.

Single-ply roof systems contain thermoset membranes commonly made of rubber polymers. Common thermoset membranes include ethylene propylene diene monomer (EPDM), chlorosulfonated polyethylene (CSPE) and Hypolon. These membranes require adhesives to form a watertight seal where the membrane overlaps.

Thermoplastic membranes may also be used on single-ply roofs. A common thermoplastic is polyvinyl chloride (PVC) that is made to be flexible and in which seams are welded using heat or chemicals. Other thermoplastics include chlorinated polyethylene (CPE) and thermoplastic olefin (TPO).

The average life span of most roof systems constructed in the past several decades is between 10 and 20 years; this can be extended with proper design, construction and maintenance. Although there are many different types of roofing, all roofs need maintenance to simply withstand the abuse of normal aging. This is because undetected small problems can become large and expensive dilemmas that can compromise the building structure and interior and even the facility's mission.

Some common deficiencies to look for in single-ply roofs are:

- Blisters, ridges, curling, wrinkles, holes, tears, punctures, cuts, abrasions or cracking in roof membranes.
- Loss of protective surfacing or adhesion to deck surfaces.
- Surface degradation because of weather exposure or defects.
- Membrane shrinkage
- Evidence of oil or chemical damage.
- Inadequate slope or drainage causing ponded water.

Flashing sheets serve the same purpose as built-up roofing and are molded around required roof openings to maintain a waterproof seal at these vulnerable points. Some common deficiencies to look for in flashings are:

- Inadequate allowance for movement.
- Movement of roof drains or vent pipes.
- Poor attachment.
- Material shrinkage.



- Poor adhesion or coverage protections (shrinkage).
- Punctures, abrasions or flashing height that is not sufficient to maintain integrity.

Signage

Signage is a significant exterior aspect for facility management. The variety and regulatory requirements alone make the placement, design, maintenance and use of signage complex as well as important. Signage may include:

- Plaques, dimensional lettered signs, panels.
- Illuminated, nonilluminated, photoluminescent (light-absorbing and light-emitting), light-reflective.
- Traffic signage, guidance signage.

Signs are important elements relating to exterior structures as well as grounds, as illustrated in Exhibit 2-5.

Exhibit 2-5: Exterior Signage



Assessing the quality, positioning and professional presentation of signs is particularly relevant to facilitating individuals finding their way around a facility.

Signage and identification of exterior structures enhance a facility's image when:

- Buildings and entries are clearly visible to passing traffic and pedestrians.
- Buildings are clearly identified from all directions with clear, visible, professional and well-constructed signs.
- Signs clearly direct traffic between and among buildings.
- Signs are properly sized, clearly lettered and at proper heights for easy reading.
- Signs are well maintained.
- Signs for exits, entrances, emergency and directional purposes are lighted or otherwise distinguishable in darkness.

Exhibit 2-6 on the next page is an example of what could be included in an assessment of signage on exterior structures.

Health and safety considerations

Relative to exteriors, safety concerns include the risk of falls, the risk of falling objects and storm preparedness.

Risk of falls

There is an ever-present risk of falls when working near the edge of an unprotected roof area or when maintaining facades or stairs. The facility manager must recognize and be informed about aspects related to fall protection.

This includes generally knowing about:

- Activities and situations that may require added alertness and fall protection.
- Proper use of ladders, platforms, scaffolds and other elevation-raising devices.
- Proper use and maintenance of safety belts, harnesses, lanyards and other fall-arrest systems and personal protective equipment.
- Applicable safety codes and regulations.

Risk of falling objects

Facade ordinances exist in some major cities to protect the public and surrounding property from the risk of falling objects that might occur because of poorly maintained or improperly repaired exteriors. These codes or regulations generally establish baseline cycles of inspection, maintenance and repair.

The facility manager must be aware of applicable facade ordinances and support their application.



Exhibit 2-6: Signage Assessment

Facility, system and location identification _____ Inspector name and date of assessment _____		Components: identification, directional and traffic signs; nameplates, posts/poles, marks and notices, pavement graphics
General causes or concerns		
Visibility ☐ Not seen from all directions ☐ Not visible to all passing traffic ☐ Low reflectance levels ☐ Not apparent at night ☐ Properly oriented	Quality ☐ Poor general condition ☐ Poor overall appearance ☐ Not professionally made ☐ Temporary appearance ☐ Wear or fading ☐ Elements missing ☐ Damage	Compliance ☐ Local regulations/codes ☐ Area requirements ☐ National/international standards ☐ Required permits
Clarity/legibility ☐ Proper letter size ☐ Poor placement ☐ Poor positioning	Safety ☐ Unclear direction ☐ May confuse traffic	Maintenance history _____ _____ _____ _____ _____ _____ _____
Maintenance ☐ Deteriorated paint ☐ Defacement ☐ Corrosion/rust ☐ Discoloration	Does not meet image ☐ Aesthetic minimum ☐ Occupant expectations ☐ Owner expectations ☐ Neighborhood expectations	
Component evaluation		
Description (type, location, serial number/ID)	Condition (deficiency described)	
_____	_____	
_____	_____	
_____	_____	
_____	_____	
_____	_____	
_____	_____	
_____	_____	
_____	_____	
_____	_____	
_____	_____	



Storm preparedness One safety issue that deserves specific attention relative to exterior structures is storm preparedness. Exterior structures are often not enclosed or constructed with the same load-bearing or weather-shielding elements as an occupant facility.

The facility manager plans for natural threats by being providing routine inspection of exterior structures for storm readiness (such as shutters; tie-downs; landscape runoff; containment features; removing, storing and capping roof vents; inspecting roof drains and reinforcing roofing systems; securing and protecting windows and doors; removing or securing potential projectiles such as trash cans, loose signs, construction materials, ladders, chairs, sand, gravel and so forth).

✦ **Topic 3: Assess Condition of Building Systems**

Typical building systems and common deficiencies

The term “building systems” generally refers to the complete, physical facility including all equipment, materials and components. However, it can also refer to the organized combination of related parts that actively provide important functions for the facility.

Under that definition, building systems typically have the following characteristics:

- They are in place; that is, once installed they tend to be permanent features of the facility.
- They usually appear on blueprints or schematics or can be flowcharted.

There are several specific, permanent building systems that provide critical operating features:

- Transport systems (including elevators and lifts)
- Fire suppression and life safety systems
- Plumbing systems
- Heating, ventilation and air conditioning (HVAC) systems
- Electrical systems
- Electronic communications and information technology systems
- Safety and security systems

Transport systems Transport systems include conveyance for people, mail, trash and waste, food, freight and hazardous materials. Transport or conveyance systems for people include elevators, escalators, lifts and moving walks. Permanent stairs are generally not included, as they are part of the building interior or exterior structure.



Transport systems are first installed when a building is constructed. They can be complicated, involving sophisticated mechanical, electrical and electronic technology.

Elevators are generally of two types:

- Traction (electric) elevators use geared or gearless mechanisms for vertical movement. They permit virtually limitless rise for tall buildings and high rise speeds but can be of high cost.
- Hydraulic (oil) elevators can be of several types and offer a lower initial cost but relatively high renovation costs and lower rise speeds than traction elevators. They are generally limited to heights of six stories or fewer.

Due to security, evacuation and confinement issues, elevators and enclosed escalators generally require two-way communications, redundancy and protective features. Two of the more frequently occurring liability issues produced by elevators are:

- Trip hazards caused when a cab does not stop level with the building floor.
- Knocking people down when improper kinetic energy settings cause doors to run into people as they enter or exit a cab.

Legislative changes may require modifications to elevators/lifts not only to support users but also to protect the health and safety of service personnel.

The facility manager can support a transport system that is operating to occupants' satisfaction by being mindful of:

- An adequate and specific maintenance contract.
- Recognized and understood inspection reports.
- Condition of guide rails, accessories and surfaces.
- Satisfactory (occupant-acceptable) operational noise, floor-to-floor travel times, and door opening/closing duration (traffic accommodation).
- Visible, legible and adequate signage.
- Security and communication weaknesses (alarm bells, phone operation).
- Standby and emergency power.
- Fire control requirements.
- Quality, condition and maintenance of equipment.
- Codes and regulatory requirements.

Renovations are expensive and disruptive yet may be needed every 20 to 25 years or as elevators end their efficient life span. Older buildings modernize elevators to compete with newer buildings and to meet changing occupant expectations.



Fire suppression and life safety systems

Elevators, escalators and moving walks are complex systems. Their maintenance is generally best in the hands of professional contractors who are expert in operations, regulatory and code requirements, safety and security features. Complete, regular, periodic checks and maintenance by a licensed contractor are most likely the facility manager's most reasonable O&M options.

Fire life safety or fire alarm and suppression functions ensure that the facility is a safe, secure place to work and is in compliance with applicable laws and regulations. Life safety refers to government regulations and building code requirements for a building relative to seismic, fire and accessibility standards.

Alarms monitor fire and smoke conditions and provide the means to alert or notify occupants. Safety systems also include safe, planned exits (egress) from buildings, including pathways, emergency lighting and other support. Codes generally mandate minimum protections.

Basic fire protection principles generally divide fire protection into passive and active systems.

- **Passive fire protection measures** help control the ignition, growth and spread of fire by using fire-resistive materials or providing physical barriers to the movement of flames or smoke. Passive systems are generally fixed-in-place walls, floors, ceilings, beams, columns and shaft enclosures that are built to a prescribed fire resistance rating. Control of egress (exit paths) may also be a part of the passive system.
- **Active fire protection measures** take direct physical action to suppress or eliminate a fire and control migration of smoke. These include fire sprinkler and smoke control systems that receive signals, alert occupants, and start fire control actions. Fire extinguishers are also generally considered part of an active fire protection system.

There are many different types of suppression systems:

- Wet pipe systems maintain water constantly within the system to discharge immediately onto the fire when a sprinkler is activated.
- Dry pipe systems use pipes filled with pressurized air or nitrogen instead of water to hold a remote dry pipe valve closed and prevent water from entering the pipe until fire causes one or more sprinklers to operate.
- Pre-action systems are similar to dry pipe systems in that water is not contained within the pipes. However, in this type of system, water is withheld by an electrically operated valve and controlled by independent flame, heat or smoke detection.



- Deluge systems, as the name implies, provide high-velocity suppression in high-hazard areas. They are similar to pre-action systems except that sprinkler heads are open and the pipe is not pressurized so that, upon smoke or heat detection, water discharges through the sprinkler heads.
- Chemical/gaseous systems use inert gases and liquid or dry chemical agents rather than water to extinguish a fire. A typical fire extinguisher is a form of chemical fire suppression. There are also gaseous or chemical fire suppression products such as heptafluoropropane (FM200®) or Energen systems that are introduced into the room or space to displace oxygen in a fire zone and thus remove or reduce the fire condition.
- Low oxygen (hypoxic) environments are created where fire or smoke can threaten irreplaceable assets yet fire suppression using water or chemicals could also create substantial damage. Oxygen content is reduced enough to suppress or prevent fire yet remains high enough for normal human activity.

Fire suppression systems should be regularly and rigorously inspected, tested, maintained and documented. International and national codes determine the requirements.

Maintenance, inspection and testing of alarm and suppression systems is complex, as demonstrated by Exhibit 2-7. As such, this oversight is generally contracted to competent life safety service providers.

Exhibit 2-7: Examples of Fire Suppression Elements in a System



Exhibit 2-8 provides an example of an assessment for fire protection deficiencies.

Exhibit 2-8: Active Fire System Assessment

Facility, system and location identification _____ Inspector name and date of assessment _____		Components: HVAC exhaust, smoke/fire detectors, alerts (sounders), controllers, sprinklers, pipes, standpipe hoses, extinguishers
General causes or concerns		
Condition	Maintenance	Compliance
☐ Hose/pipe integrity	☐ Dirty smoke detectors	☐ Local regulations/codes
☐ False alarms/trouble alarms	☐ Obstructed or damaged sprinkler heads	☐ Work with local authorities
☐ Alarms not heard	☐ Incorrect installation	☐ National/international standards
☐ Potential freezing	☐ Inadequate number of extinguishers	☐ Required permits
☐ Water in dry pipe systems	☐ Extinguishers	☐ Up to current code
☐ Inadequate flow (water pressure)	☐ Control valves/dampers closed or inoperable	Maintenance history
Inspection	☐ Air compressor or jockey pump failure	_____
☐ Improper, inadequate or incomplete tests	☐ Accidental discharge	_____
☐ Inexperienced inspectors	☐ Motors, fans	_____
☐ Poor or inconsistent documentation	☐ Lights/signs operational	_____
☐ No dry system inspection		_____
Component evaluation		
Description (type, location, serial number/ID)	Condition (deficiency described)	
_____	_____	
_____	_____	
_____	_____	
_____	_____	
_____	_____	
_____	_____	
_____	_____	



Plumbing systems Plumbing systems typically distribute water or other fluids, liquids and gases and also collect waste.

Plumbing systems are used in buildings to:

- Provide potable (drinkable) water.
- Provide chilled and hot water for cleaning.
- Provide water to operate HVAC equipment
- Deliver water, chemicals or gases for fire suppression systems.
- Remove (dirty) wastewater.
- Remove (gray) rainwater and groundwater.

Components in a typical plumbing system include:

- Distribution system piping to transport liquids/gases.
- Pumps to provide the means to move liquids/gases.
- Control components such as electronic or mechanical switches and gauges.
- Wastewater piping, drains, control valves and traps to control the flow of liquids.
- Plumbing fixtures as a means to deliver liquids (mostly water) or gases.
- Water heaters to increase water temperature.
- Filters and traps to clean or purify liquids/gases.
- Vents to provide gas relief and permit flow.
- Backflow preventers (or check valves) to prevent water/waste cross-contamination.

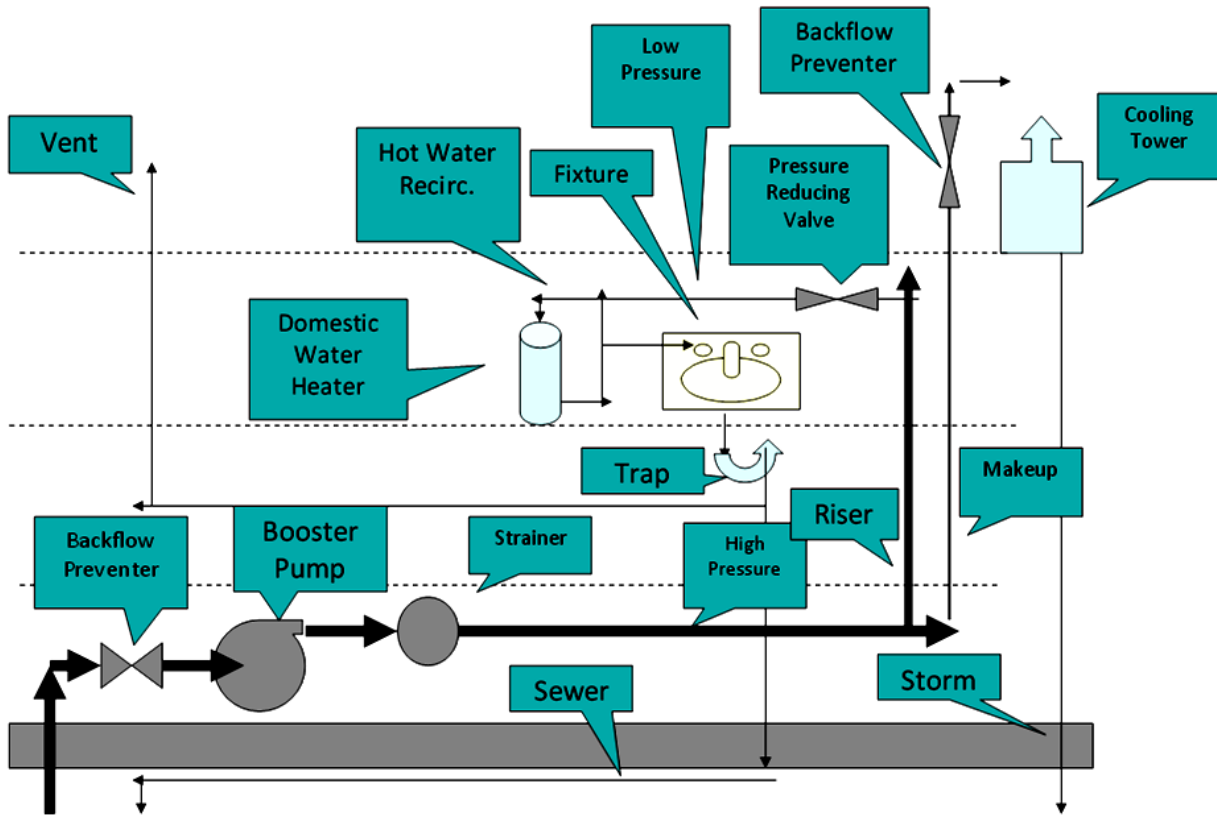
Exhibit 2-9 on the next page illustrates a typical building plumbing system. The heavy lines indicate supply lines and flow direction.

Plumbing systems are a long-term investment, most of which is out of sight in walls, ceilings and floors. They need to be durable and reliable for the anticipated life of the building. This means that they are usually designed to not become outdated or require replacement while major components are still serviceable.

In addition, disruptions or leaks can cause major damage to other building systems, interiors and exteriors or endanger the health of occupants. This makes durable design especially important.



Exhibit 2-9: Plumbing (Distribution) System



Examples of deficiencies to look for relative to plumbing systems include:

- Leaks, spots or stains on surfaces.
- Corroded piping
- Backflow and overflow potential.
- Discharges or evidence of system contamination.
- Clogged pipes or drains.
- Broken or inoperable valves or fixtures.
- Incorrect, inadequate or variable pressure.
- Slow or inconsistent flow.

Exhibit 2-10 is an example of some of the elements contained in a water plumbing systems audit. A similar assessment would be required to assess a plumbing system that manages other liquids or gases for heating, cooling or other purposes.

Exhibit 2-10: Water Plumbing System Assessment

[illegible]

Heating, ventilation and air conditioning systems

Heating, ventilation and air conditioning (HVAC) mechanical systems within a building control and maintain temperature, humidity and air quality. Depending upon the type of system and the geographical location, they may include equipment for steam and hot water generation, chilled fluids, indoor/outdoor air exchange, and heating/cooling distribution. They also contain control components such as electronic or mechanical switches and gauges that are generally managed by a computerized building operations and reporting system.

The elements of HVAC systems are summarized in Exhibit 2-11.

Exhibit 2-11: HVAC System Elements (continued on next page)

Element	Description
Heating	• Oil, gas or electric boilers that heat water or generate steam for distribution • Furnaces that burn fuel to heat water or air for distribution • Radiant heaters that deliver heat that is absorbed by objects rather than using forced air to transfer heat • Unit heaters that may be a room-sized device or central heating system that circulates warm air or steam through finned tubes over which a fan forces air to provide warmth • Heat exchangers that passively transfer heat from one material to another in two chambers separated by a single wall or partition • Solar heating, which uses sunlight to collect, store and generate warmth
Cooling	• Chillers that include compressor, evaporator, condenser, reservoir, thermal expansion valve and other components as a system to create cooling • Refrigerant compressors that turn low-pressure gases into high-pressure and high-temperature gases as a way to create a cooling effect • Cooling towers that spray or drop cooling water to release heat to the air through evaporation • Condensing units that condense refrigerant vapor into a liquid and discharge the resultant heat to the environment • Split systems, which consist of a single system that has components in two locations such as an outside compressor and an indoor furnace/air-handling unit • Package units or all-in-one systems in which all components are located in one cabinet



Exhibit 2-11: HVAC System Elements (concluded)

Element	Description
HVAC distribution	- Hydronic piping and pumps that deliver steam, hot water and chilled water to radiators, fan coil units and convectors and deliver the returning water to the primary generation unit - Ducts, plenums and chases that route or otherwise enclose spaces to distribute warm or cool air - Fans that drive or force warm and cool air through a distribution system - VAV (variable air volume) boxes that control the capacity and distribution of supply air into a specific zone - Terminal units, such as air-handling units to condition and circulate air, fan coil units to heat or cool and direct air, and other such devices - Exhaust systems or devices such as vents or fans to control odors or otherwise expel air, fumes or particles
Other HVAC-related aspects	- Thermal storage that uses technologies, reservoirs or other repositories to maintain a temperature higher (warmer) or lower (cooler) than the ambient environment - Heat recovery units that use heat produced in a process to enhance efficiency or in another way - Humidity control equipment that regulates the degree of saturation or quantity of water vapor in the air - Variable frequency drives, also known as variable speed drives, that control the rotational speed of an electric motor - Automated building management systems or building management and control systems that manage the complex interactions of systems and the building environment for human comfort and operating efficiency

Central plant designs combine any or all HVAC capabilities into one place as a more efficient way to generate interior thermal conditions for one or more buildings. Rather than producing steam, hot water, heat and cooling within individual buildings or in separate places within a building, the central plant provides for these needs from its own dedicated place.

Examples of deficiencies to look for relative to HVAC systems include:

- Inadequate heating or cooling within a building or area of a building.
- Uneven air supply causing uneven temperatures and/or drafts.
- Odors.



- Humidity and air quality problems.
- Lack of fresh outside air.
- Leaking return ducts in crawl spaces (poor thermal management).
- Unsafe electrical wiring or damaged/deteriorated equipment (burners, heat exchanger, components).
- Carbon monoxide and other toxic gases drawn through air intake.
- Refrigerant leaks.

Lack of proper maintenance is the chief cause of HVAC deficiencies. Without proper routine maintenance, HVAC systems can develop problems that reduce equipment life and efficiency and occupant comfort such as:

- Excessive component vibrations and poor fan alignment.
- Insufficient air movement to reach all spaces.
- Drafts in areas being cooled.
- Excessive noise, evidence of leaking, deterioration and corrosion.

Facility managers should be active and knowledgeable about environmentally friendly replacement substances, equipment alternatives and sustainable resources.

Electrical systems

Electrical systems generally include (but are not limited to) aspects of power generation, use and distribution such as:

- Service entrances and switchgear.
- Wiring (conductors, conduit, wireways, floor raceways).
- Transformers, distribution equipment and feeder circuits.
- Switches (disconnect, switch, fuse, on/off) and motor control centers.
- Motors and motor circuits.
- Protection devices (circuit breakers, fuses, proper grounding applications, ground fault interrupters, earth leakage circuit breakers, and ground fault circuit interrupters).
- Panelboards, switchboards or distribution boards (which organize and control feeder and branch circuits).
- Electrical controls and instrumentation.
- Power-generating and storage equipment (e.g., generators, power filters and conditioners, battery storage and transfer switches that divert, change or manage electrical loads).
- Power factor correction capabilities.
- Communications and security systems.
- Lighting.
- Emergency and standby power sources.



Electrical systems are critical to facility operation because they provide energy for heating, cooling, lighting and equipment operation. Preventive maintenance is important because the total cost of an unscheduled shutdown due to failure is usually much higher than the cost of scheduled equipment repair or replacement.

In addition, electrical failure that interrupts communications can dramatically reduce productivity. Failure that compromises any security system puts occupants at unnecessary risk. For these reasons, at a minimum the facility manager should be aware of the inventory of different types of electrical equipment and their characteristics and use as well as safety aspects, regulations and codes.

Diagnosing problems or even assessing the condition of electrical systems can be complex and may pose a danger to inspectors as well as system users. It is important to assign a trained and qualified person to perform any survey of a facility's electrical systems.

Examples of deficiencies to look for relative to electrical systems include:

- Damaged equipment.
- High or low voltage, unbalanced loads, poor power quality.
- Overloaded circuits.
- Metal corrosion, insulation deterioration, loose or incomplete connection.
- Dirty connections and equipment.
- Water or moisture problems.
- Improper component replacement.
- Improper grounding.
- Inadequate clearance and ventilation.

Improvements in electrical systems and advances in electrical devices have produced gains in sustainability, energy conservation and energy use. Some representative enhancements include:

- Power factor correction units that filter electrical current to increase working (or real) power and reduce nonworking (or reactive) power. The improved power factor can reduce energy bills and enhance the capacity of an electrical system.
- Converting older fluorescent and incandescent lighting to LED (light-emitting diode) lighting, compact fluorescent light (CFL) products, EE/ED series fluorescent lights and other high efficiency technologies.



- Variable speed drives and other electrical control devices to conserve energy use.
- Energy management systems in computer-aided tools that automatically control and monitor the heating, ventilating and lighting needs of a building or group of buildings. This energy tool may also be included as part of a complete building management system; this is discussed in Topic 4 of Chapter 3.

Electronic communications and information technology systems

As an overall definition, communication is the process of creating and sending messages with the goal to influence the opinions, actions and decisions of the intended audience. The process includes selecting the appropriate media to best reach the intended audience.

Systems used for communications and information technology (IT) can range from simple arrangements of devices to complex installations that are critical to the mission and require sophisticated operations and maintenance.

From an operational perspective, communication plays a role in several ways.

- Fire life safety requires prepared, clear, practiced and deliverable communication even when there is no electrical power, systems are down, and evacuations must occur from pre-arranged plans.
- New systems, materials or equipment should integrate with existing operations so that the functionality of the entire facility is optimized. This is supported when communications with contractors, staff and independent parties are consistent, informed and redundant.
- Operations are enhanced when the facility manager prepares messages, incorporates feedback and resolves questions in ways that satisfy everyone involved in operations, including staff, contractors, regulators, occupants, visitors, neighborhoods and communities.

Communications systems may operate to support data, voice, audio-video and electronic information systems with services that could include line, wireless, cable, high-speed and satellite delivery. A facility may devote entire rooms to communications and data-gathering equipment, requiring special consultant services for operations and maintenance.

Information technology systems include networks, cables, sensors, fiber optics, hardware, peripheral data equipment and software that may be centralized or have elements distributed. These technologies are used to support facility operations and communications as well as automate the facility manager's O&M activities. They may be used to create, manipulate



and store data. They may also generate reports related to facility maintenance as well as the electronics and fiber optics to control energy consumption, temperature, humidity, lighting and the operation of buildings and grounds.

These operational aspects may be found in a facility's information technology area rather than as part of facility operations. However, facility managers must understand, incorporate and utilize technology to support facility operations. Technology can provide critical tools and capabilities to help the facility manager effectively direct and oversee day-to-day infrastructure operations and to integrate new systems, materials and equipment so that they operate as expected and optimize the functionality of the entire facility.

Safety and security systems

In its broadest definition, security in the sense of safety is the freedom from danger, doubt or fear. At this level, a facility manager is expected to operate a facility prepared for threats that include the potential for natural disasters, terrorism, hazardous materials, medical emergencies, power failure, elevator and escalator incidents, civil disorder, workplace violence and more.

From a security standpoint, technology may be applied to prepare for and protect against cyberterrorism techniques. These illegal activities disrupt or damage computer-based information systems to cause fear, injury or economic loss. Less harmful invasions known as "spam" can inhibit productivity. The facility manager has little control over the potential for many of these threats. However, he or she should be prepared to manage if not avoid them so that occupants and visitors can work and reside in the facility safely and with confidence.

Systems and approaches for avoiding threats are varied and can be complex, depending upon the nature of the facility, the value of its purpose, and the perceived threat of its loss. That's why, for example, a bank is more highly protected than a grocery store.

Relative to facility operations, the facility manager prepares for external security threats such as terrorism, civil unrest or any invasion that compromises the integrity of the building. Security systems help protect against direct threats.

The facility manager considers several ranks or rings of security that begin at the outer perimeter of the grounds, at entrance points to buildings, and within buildings.



Layers may include:

- **Perimeter detection and deterrence** such as fences, sensors, electronic barriers, vehicle and traffic restrictions, closed-circuit television (CCTV) cameras, intercom systems, card readers and lighting along with passive landscaping measures.
- **Entrance and structure features** such as locks, doors, revolving doors and turnstiles, remote cameras, card readers, remotely controlled entrances and lights.
- **Building access devices** such as CCTV cameras, electronic locks, card readers and biometric readers (fingerprint, eye scan, voice, face, signature).

Security may include a standoff zone (the area between the facility and the threat) that can extend outward from the facility a few meters/yards to a hundred or so meters/yards. Surveillance and confirmation activities such as remote visual identification, intercom communication, screening deliveries at entrances, and authorized-only access help the facility manager provide security that gives occupants confidence.

In any case, the facility manager develops a security approach that considers many aspects beyond the technical elements of systems themselves. These overarching aspects include:

- Organizational policy and owner expectations.
- Budgetary and operational limits.
- Interactions and relationships with emergency responders.
- Emergency management provisions and crisis management.
- First response practices.
- Emergency rehearsals and tests.
- Recovery and backup preparations.

The facility manager may assess the current state of security, safety and sustainability in a facility using a grid such as that shown in Exhibit 2-12 on the next page. For example, the facility manager could:

- Evaluate the urgency, importance and utility of each safety and security aspect.
- Determine the level of significance (low/medium/high) relative to urgency, importance and utility.
- Select the most significant priorities with the highest urgency.
- Prepare a plan to improve safety and security that resolves the highest priorities first.



Exhibit 2-12: Safety and Security Priority Matrix

Safety/Security Aspects	Urgency			Importance			Utility		
	L	M	H	L	M	H	L	M	H
Fire life safety									
Fire alert									
Noxious gas alert									
Fire suppression									
Smoke/gas exhaust									
Trip/fall hazards									
Standards									
Known requirement									
Code compliance									
Owner expectation									
Security									
Access control									
Preparedness									
Technology									
Communication									
Environmental health									
Energy efficiency									
Sustainability									
Waste/use costs									

Health and safety considerations

There are many concerns about health and safety relative to building systems. Among them are three primary ones that reflect different aspects of building systems: environmental sustainability, workplace safety, and workplace risk assessment.

Environmental sustainability

Sustainable processes and activities ensure that the creation and use of the built environment does not harm the natural environment. Sustainability is supported by policies and practices designed to:

- Reduce the consumption of energy through energy-saving devices and systems.
- Reduce waste production by employing more efficient processes including recycling programs.
- Preserve water through water management programs that include waste, drinking and storm water.
- Protect wetlands and wildlife.
- Protect the natural environment through the use of biodegradable chemicals and materials to clean and preserve surfaces.



By implementing sustainability-focused policies and practices, facility managers and their entire organizations have an opportunity to support the environment while saving unnecessary operating costs by using energy more efficiently. In this way energy management (or energy conservation) is a measure of both sustainability and facility efficiency.

Efficient operations are significant to facility managers for several reasons:

- The high costs associated with energy use increase overall operating expenses, reducing the effectiveness of the facility.
- Any preventable waste adversely affects the facility manager's operating budget by either reducing opportunities for upgrades or limiting choices.
- Misuse or wasteful use of depleting resources adds to both the unit cost of those resources and to higher future energy expenses.
- Overly restricted budgets, possibly caused by excessive energy or materials use, may reduce the facility manager's ability to acquire new materials, systems or equipment that have better long-term advantages.
- Industrial pollution is a primary consideration in the public today, and wasteful use of resources—especially carbon-based energies—is scrutinized for code compliance that ranges from local agencies to international treaties.

High energy costs (particularly for fossil fuels) have caused facility designers and builders to specify better insulated, tighter (meaning less air leakage), more energy-efficient buildings with sealed windows, mechanical ventilation and controlled air exchange. This provides greater control over energy use and loss. It also increases the facility manager's responsibility to maintain adequate ventilation, more closely monitor filtration and exchange systems, and rigorously maintain mechanical system cleanliness so that fungi cannot grow and pollutants are minimized.

Workplace safety

Workplace safety crosses too many areas and issues to fully address in this course. Briefly, it involves accident prevention, fire inspection, health hazards, chemical/biological/high energy hazards, equipment safety, use of personal protective equipment such as hardhats, eyewear, ear protection, fire-resistant clothing, air quality monitors, gloves and tie-offs (for above-ground duties), and much more.



Some statistics on electrical safety in the workplace show how critical this subject is. In the United States alone, the National Safety Council reports 1,000 electrocution deaths and 30,000 nonfatal shock accidents each year. The current required to light a common 75-watt, 120-volt lamp, if passed through the chest, is enough to kill an individual.

The facility manager must strive to embed safety into the culture by incorporating proper understanding, techniques, equipment and resources into O&M activities as well as facility use in general.

For the entire physical infrastructure, it is important to check protocols against local, regional, national and international requirements. This is particularly significant relative to hazardous materials and spills; solid, liquid, gas or infectious agent releases; proper availability, maintenance and use of personal protective equipment; and emergency support, equipment, scene management and cleanup.

The facility manager should also extend safety issues beyond compliance. He or she should promote a culture of safety that supports the mission of the facility, improves productivity, reduces injuries and their costs, considers the environment and helps direct how operations and maintenance activities are conducted.

This responsibility includes an incident management approach that has both leading (pre-emptive) and lagging (follow-up) behaviors so that all personnel, staff, contractors and occupants are aware of and can practice a safe working approach. The facility manager is a champion of workplace safety through actions, assessments and activities that demonstrate its importance.

Workplace risk assessment

Most accidents and incidents in a facility are the result of unsafe acts. Unsafe acts generally occur because of behaviors by an individual or group. Behavior is difficult to change because it can involve personal habits and attitudes. Yet facility managers actively assess risks and strive to reduce them in the workplace. Two examples of risk-reducing approaches are job safety analysis and safety data sheets.

Job safety analysis (JSA) procedures offer effective means to prepare for and perform tasks by giving personnel the opportunity to analyze hazards and potentially unsafe conditions/activities and use tools or equipment to prevent incidents.



This can include:

- Approaches to working in confined spaces.
- Lockout/tagout for energy or high-voltage electrical systems.
- Hot work processes where welding, brazing or soldering occurs and where flammable material is present.
- Handling asbestos coating material, lead, mold, radioactive materials, chemicals and so forth.

A JSA may also be known as a job hazard assessment/analysis (JHA), safe operating procedure or safe work method statement (SWMS). All are designed to explain what's required to complete a job or activity safely, with minimum risk to the individual, others nearby, the property and the environment.

In general, JSAs provide details on an activity while SWMSs provide detail on a task.

Safety data sheets (SDS) generally provide information on the properties, uses, handling and health aspects of chemicals, compounds and other substances that may be found or used in a facility. Countries and suppliers have different protocols and requirements for proper material handling. These protocols may be known under different names or descriptive terms such as material safety data sheets (MSDS), CHIP (chemicals hazard information and packaging), risk and safety statements and so forth.

SDS forms are widely used and universally available to provide the procedures for properly working with a chemical substance but are not generally mandatory for cleaning supplies.

Facility management policies and procedures help influence how people maintain safety in a facility. The facility manager provides the practices, procedures, oversight and diligence to promote and support a safe facility and working environment.

✦ **Topic 4: Assess Interior Furnishings, Fixtures and Equipment**

Typical facility interiors and common deficiencies

Facility interiors generally include all elements within a structure that are not actively supporting the structure. A typical interior is a complex mix of different yet integrated materials, furnishings, fixtures and equipment as well as openings.



Exhibit 2-13 lists some typical elements of facility interiors.

Exhibit 2-13: Interior Elements

Interior Element	May Include . . .
Openings	• Operations and maintenance portals and access openings • Doors and frames of various materials and specialties • Entrances and storefronts • Windows, roof windows and skylights • Glazing • Louvers and vents
Finishes	• Plaster, gypsum board and cement plastering • Tile • Ceiling types (acoustical, specialty, textured) • Flooring (laminated, masonry, wood, resilient, carpeting, etc.) • Wall coverings, carpeting, facings, acoustic treatments • Painting and coatings
Specialties	• Information elements (visual display surfaces such as chalkboards and markerboards) • Display cases, directories and signage • Compartments, cubicles, partitions, screens and service walls • Wall and door protection • Toilet, bath and laundry accessories • Fireplaces and stoves • Safety and emergency cabinets • Lockers, mailboxes and storage shelving
Equipment	• Service equipment (for vehicle, parking, loading dock and pedestrian support) • Security, detention and other commercial equipment • Maintenance equipment storage • Hospitality, office and soft services equipment • Library, laboratory and other scientific equipment • Entertainment, athletic, recreational and health-care equipment • Waste/recycling collection and disposal equipment
Furnishings	• Casework • Countertops • Window treatments • Accessories, lamps, bed, bath and waste receptacles • Rugs and mats • Furniture and multiple seating assemblies • Art, plants and other internal elements

There is great variety within interiors because there are so many different materials, finishes, functional purposes, fixtures, and security and safety



requirements. Among all these options, the facility manager attends to aesthetic and decorative as well as functional O&M concerns. In addition, the facility manager oversees soft (occupant) services (such as custodial) that may influence the condition or quality of a facility's interior.

The scope is so broad that it is good practice to perform assessments and monitor results as part of scheduled maintenance. Exhibit 2-14 shows a sampling of deficiencies that may occur in interiors that could compromise functionality, safety, security or health.

Exhibit 2-14: Deficiencies in Interior Elements

Interior Component	Deficiencies
Interior walls, windows and doors	• Cracks, rips, tears, peeling, flaking • Joint openings, poor seals or poor fit (doors, windows, access panels) • Worn or poor physical condition, deterioration • Water stains, discoloration, other evidence of moisture • Malfunctioning or missing hardware • Abuse, vandalism, security breaches (such as incorrect latching) • Interior doors without appropriate fire-rated door classifications
Ceiling systems	• Settling or sagging • Poor alignment • Stains, discoloration, evidence of moisture • Missing panels or elements • Cracks, surface deterioration, poor adhesion • Poor acoustic quality (high noise levels, distracting traffic noise)
Floors and floor coverings	• Carpet wear, color fading, age, stains, odor from carpet or carpet pad, holes, tears, poor seam conditions • Continuous topping (concrete, terrazzo) cracks, porosity (void spaces in a material), poor joints or seals • Resilient floor broken or loose elements, shrinkage, cupping, cuts, holes, porosity, fading • Wood shrinkage, warping, wear, unevenness, decay, poor finish • Masonry (brick, stone) cracks, stains, porosity, poor joints or seals, deterioration, discoloration • Trip hazards (unsecured electrical cords or cables or items on the floor; uneven or slippery floors)

Many aspects of a building's interior functioning are closely regulated, especially regarding health and safety. Code compliance is critical both for occupant satisfaction and to maintain the building in good standing with regulatory authorities and agencies.



A comprehensive facilities assessment inspection is likely to be an opportune time to systematically identify any functional weaknesses that might hinder free access and movement. Any such barriers may also violate local, national or international requirements. From a business perspective, barriers to mobility might also reduce or inhibit productivity.

The usefulness of any space depends on reliable heating, cooling, electrical systems, water supply and waste removal.

In addition, space usefulness depends upon:

- How the space is allocated.
- How configurations support workflow.
- How management goals are realized.
- How current arrangements permit growth and change.

These strategic and managerial factors affect the ultimate value and effectiveness of O&M. The facility manager recognizes and makes business decisions that support these nonmechanical aspects of building interiors.

Lighting

Let's take a closer look at another element of facility interiors—lighting systems.

A well-integrated indoor lighting system considers the best of both natural and artificial sources to provide adequate illumination for core business functions. It should create a suitable and pleasant visual environment that enables occupants to function visually with efficiency and comfort at a minimum of electrical consumption.

The following concepts are important in the design of lighting systems:

- Reflectance is the amount of light a surface reflects relative to the total amount of light upon the surface.
- Glare is the lighting or brightness reflected from a surface.
- Contrast is the balance of light and surrounding dark areas.
- The color of light relates to the hue that specific artificial lighting produces (usually green, blue, pink and brown).
- Lumens measure the projection of light to an area.
- Brightness or luminance refers to the perception or measurement of the level of intensity in a particular direction.
- Ambient light refers to the available light in an environment or room.
- Task light is a controllable light for a local, well-defined activity.



Effective lighting depends on what lighting levels and light quality the space requires, on ceiling height, and on the reflectance value of the ceiling, walls and floors. Over time, light output from fixtures can drop below initial designed values because:

- Lamps themselves deteriorate with use and age.
- Dust and dirt can collect on lamps, lenses and reflecting surfaces in the room.
- The reflecting and transmitting materials of some lamps gradually change chemically in the presence of air and light.

Good lighting design integrates illumination to support tasks, enhance the architectural image, provide emergency lighting, and reinforce security at night. Lights in unoccupied areas should be able to be dimmed or turned off. Daylight openings should be adjustable using blinds, shutters and reflectors.

Deficiencies in lighting systems can include:

- Lamp breakage, failure or short lamp life.
- Shadow areas or insufficient illumination.
- Radio interference, ballast noise (buzzing, humming or other sounds), lamp flickering, dimming or glowing.
- Insufficient or poor exit and emergency lighting.
- Poor condition or operability of wall switches, cover plates, outlets and grounding (grounded wiring).

Lighting fixtures and reflective surfaces require regular cleaning to preserve efficiency. An area ventilated with forced and filtered air generally stays cleaner than a naturally ventilated space.

Lighting maintenance is now generally accepted on a replace-lamps-as-needed basis rather than replacing entire zones regardless of whether lamps are burned out. An exception to this policy is lamps that are difficult to reach, require lifts or special equipment, or are in limited access areas such as manufacturing lines, hospital operating rooms, building lobbies and so forth. In these instances, facility managers generally have lamps replaced on a set schedule.

Exhibit 2-15 is an example of some of the elements contained in a lighting systems assessment.



Exhibit 2-15: Interior Lighting Assessment

Facility, system and location identification _____

Inspector name and date of assessment _____

Components:
 fixtures, visible motor
 controls, switches,
 cover plates, grounds,
 lighting protection,
 accessible wiring

General causes or concerns**Lighting levels**

- ☐ Excessive, insufficient
☐ Balanced, adequate
☐ Controllable
☐ Aesthetically pleasing

Energy use

- ☐ Controllable lighting
☐ Sufficient wattage
☐ Systematic
 conservation (turn off
 lights, month-over-
 month/year-over-year
 energy use
 comparison)

Deterioration

- ☐ Ballast noise
☐ Flickering/glowing
☐ Dimming
☐ Water/moisture

Maintenance

- ☐ Dust/dirt
☐ Improper installation
☐ Broken/failed lamps
☐ Loose wiring/
 connections
☐ Custodial supervision
☐ Fixture condition

Lamps

- ☐ Energy-efficient
☐ Unexpectedly short life
☐ Lens condition

Safety

- ☐ Proper grounding
☐ Blown fuses/circuits
☐ Overloaded
 circuits/motors
☐ Proper wattage for
 circuits
☐ Dimming
☐ Water/moisture

Compliance

- ☐ Local regulations/codes
☐ Area requirements
☐ National/international
 standards
☐ Required permits

Maintenance history

Component evaluation

Description (type, location, serial number/ID)

Condition (deficiency described)

_____	_____
_____	_____
_____	_____
_____	_____
_____	_____
_____	_____
_____	_____



Health and safety considerations

Facility managers are expected to maintain a safe interior environment. Protecting occupants from unsafe interior conditions includes fire life safety and minimization of hazards. The facility manager must always be alert for conditions that might cause injury, incapacitation or discomfort. Storm preparation is also necessary so that the resources and responses are in place to maintain an open, operating and safe facility during routine storm conditions.

The facility manager is wise to review manufacturer safety data sheets and to be familiar with government-mandated information concerning health and safety. It is especially important that the facility manager be familiar with health and safety compliance, including relevant codes, standards and regulations.

There are aesthetic as well as economic reasons for keeping a facility safe and functional. The facility manager strives to maintain a safe working environment so that occupants and visitors are satisfied with their surroundings and the building provides optimal conditions for productive use. Furthermore, any negative event that endangers life and property could cause an asset to be damaged or lost. This produces a financial penalty that multiplies the cost of any injury or productivity loss.

Health and safety aspects of particular concern regarding interiors include air quality and contaminants and hazardous materials.

Air quality and contaminants

Acceptable indoor air quality (or, more broadly, indoor environmental quality) can be compromised if a building has been constructed so that air quality cannot be controlled and the amount of fresh air and humidity is limited. Biological contaminants such as mold or fungi can grow wherever four essentials are present:

- Viable mold spores
- Acceptable temperature range
- Consistent moisture source
- Nutrient source (paint, paper, wood products, adhesives, plastics, fabrics)

Mold is an invasive and common problem in structures, especially in high humidity environments. Unfortunately it may be hidden, occurring inside walls and ceilings or anywhere moisture collects and cannot be readily observed.



Signs to look for that might indicate mold infestation include:

- Peeling or blistering paint.
- Rot or other damage around windows, doors and air conditioning units that permits water penetration.
- Cracks, breaks or undesired openings in exterior walls and around windows or doors.
- Roof damage that permits water penetration.
- Water pooling in basement window wells.
- Joints, sills and jambs where different claddings meet that could permit water penetration.
- Cracks or damage to basement walls or foundations that may indicate water intrusion.

In addition to mold itself, there are many other potential pollutants in buildings. They can be produced from interior finishes, new furnishings, office supplies, detergents, pesticides, cleaning materials, even emissions from office equipment.

For example, new carpeting may emit gases that contain volatile organic compounds (VOCs) that reduce indoor air quality. To avoid poor air quality, new carpeting may be unrolled outdoors or in a well-ventilated area to reduce VOCs before the carpet is installed in its designated space.

Hazardous materials

Any solid, liquid, gas or infectious agent that can chemically or physically affect life, health or the environment is a concern for the facility manager and should be carefully managed within buildings. A spill that would be unsafe outside of a facility can easily produce a toxic condition inside the structure.

The facility manager must be alert for unsafe conditions such as:

- The presence of hazardous materials (storage, location, handling) that may be part of the facility's purpose.
- The presence of environmental hazards or pollution from materials such as asbestos, lead, PCBs (polychlorinated biphenyls), gasoline, diesel and other chemicals as well as the visible and invisible fumes they may create.
- Grease, oils or slippery surfaces that could not only affect air quality but also cause slips or falls.



Additional concerns include:

- Carbon monoxide or other toxic gases from faulty heating units (which could also be drawn into equipment and air supply systems).
- Refrigerant leaks at condenser or evaporator coils in HVAC units.
- Leaking return ducts in crawl spaces that compromise heating/cooling efficiency or air quality.
- Unclean air handlers that send pollutants and dirt into interior spaces to make cleaning difficult and more expensive.
- Unsafe electrical wiring.
- Poor, inadequate, imbalanced or unreliable fresh air exchange.

Monitors and detectors should be installed and periodically maintained to alert occupants to unhealthy emissions such as carbon monoxide, fuel vapors and other dangerous conditions. Relative to hazardous spills or leaks, the facility manager should have a plan (such as an emergency procedures manual) to manage situations that can appropriately respond with:

- Types of possible releases and their danger levels.
- Safety data sheets and labeling information.
- First-line defense equipment (first aid, fire extinguisher, hydrants and hose, containment supplies).
- Evacuation, communication and transportation management protocols.
- Trained (or emergency) personnel with personal protective equipment and material for support and cleanup.
- Prepared and documented information relating the most stringent or specific (local, regional, national and international) requirements for response, materials management, replacement and review.

Dangerous and hazardous materials are subject to strict environmental codes and regulations around the world. The facility manager must be aware of these restrictions and watchful for these materials within (and near) the facility.

★ Topic 5: Assess Grounds

Typical facility grounds and common deficiencies

Facility grounds include all exterior elements not attributed to separate structures. Examples are listed in Exhibit 2-16.



Exhibit 2-16: Facility Grounds

Type of Facility Grounds	Examples
Earthworks	• Maintenance through site clearing, earth moving, underpinning, retaining walls, berms, dikes, terraces • Soil treatment, rodent and termite control, stabilization • Slope protection, grading • Grouting, expansion joints, maintaining foundations • Tunnels and mining
Exterior maintenance	• Pavement cleaning, rehabilitation, repair • Level changes, bases, ballasts, paving, aggregate surfaces • Curbs, gutters, wheel stops and parking bumpers • Markings, traffic calming, yield/stop signs, pedestrian crossing signs • Fences, gates, walls, bridges • Streetscape furniture (light poles, flag poles, railings, seating areas, reinforced trash receptacles) • Security fences and site access gates, aesthetic illumination, and lighting for way-finding • Sculpture, statues, art, fountains, waterfalls, planter boxes • Steps, stairs, ramps, docks
Wetlands, undeveloped areas, landscaping (restoration and maintenance)	• Planting, seeding • Plant strips, beds and features • Turf, shrub and tree maintenance • Integrated pest management with ecological use of insecticides/pesticides • Water (ponds, reservoirs, cisterns, storm basins) • Irrigation equipment and water runoff systems, culverts, trenches
Utilities and related aspects	• Water, wells, storage, distribution, fire hydrants • Sanitary sewer, storm drainage, piping, mains, and tanks • Manholes, sewer covers, gratings, drain covers, electrical covers • Fuel distribution (natural gas, liquid fuel) • Steam energy • Site utilities, systems and infrastructure (power transformers, energized equipment, gas lines, power lines, water lines, electrical grounding) • Communications utilities (towers, distribution, wireless)
Transportation infrastructure and maintenance	• Roadways, railways, guideways, airports, bridges • Signaling and control equipment • Fare collection equipment
Waterway and marine maintenance	• Waterway preservation, cleaning, deepening (or dredging) • Signaling and control equipment • Shoreline protection, artificial reefs, levees, banks, scour protection, dams

Exhibit 2-17 illustrates an example of grounds and surfaces.



Exhibit 2-17: Grounds



A facility's setting is a particularly important and challenging aspect of O&M, because the grounds:

- Represent a significant long-term financial obligation.
- Provide an immediate impression that reflects on the building owner and occupants.
- Require continuous, labor-intensive maintenance year-round, including turf mowing, tree pruning, hedge/brush trimming, flower care, fertilization, edging, aeration, irrigation, disease and insect control, sweeping and surface cleaning as well as litter, leaf and snow removal.

The smoothly transitioning, visually attractive entrance to a building or complex provides an immediate, positive first impression. This not only influences the surrounding community but also affects occupants and visitors.

Successfully maintained grounds provide:

- Safe, smooth pedestrian paths and vehicle pavements.
- Integrated pest management.
- Barrier-free access for individuals with limited mobility.
- Well-drained sites free of standing water.
- Adequate snow and ice removal.



- Adequate irrigation and xeriscaping (use of native vegetation).
- Intuitive, easily understood travel direction day and night.
- Security and emotional comfort day and night.
- Shade in summer.
- Shelter from wind, rain and inclement weather.
- Seating areas.
- Noise reduction.
- Visual screening.
- Habitat for wildlife.

Integrated property management is an approach to landscape design and maintenance that may be linked to life-cycle costing. This method helps recognize what resource-conserving options support a lasting and healthy landscape at a lower total cost of ownership.

Maintaining grounds is a challenge to preserve existing features and incrementally enhance the exterior environment.

Facility managers may use staff or contractors for grounds O&M. Many select contractors to maintain grounds because these activities are not usually significant to the core purpose of the facility or connected to the functions of the occupants. These services may also be contracted because they are seasonal. The level of care and aesthetics given to grounds is usually dictated by the budget provided and the general impression the owners or the organization desires to maintain.

Roads

Pavements and finished surfaces generally require high and expensive levels of maintenance. Road and walking surfaces are critical for way-finding, stable footing in inclement conditions, visual integration with buildings and grounds, efficient traffic circulation and other purposes.

Exterior paved surfaces can be subject to heavy use as well as the extremes of temperature and weather. Yet they should continually meet aesthetic, directional, safety and security expectations and should be regularly assessed and substandard conditions corrected.

Exhibit 2-18 is an example of some of the elements that could be contained in an assessment of roadways and paved surfaces.



Exhibit 2-18: Road Assessment

[illegible]

Safety and security considerations

Grounds are typically designed to promote a pleasant aesthetic without sacrificing pedestrian or building security. For the purposes of O&M, several security aspects relate to facility management and passive landscaping of grounds.

Streetscape elements are used to deny vehicle access and maintain standoff (security) distance. These elements can include fountains, reflective pools, concrete planters, barricades, walls, reinforced or anchored benches, boulders, trees and light poles. Grounds should be assessed to determine what is being done to prevent or frustrate man-made threats and prepare against natural ones.

Examples of security concerns to look for on facility grounds include:

- Lack of control over physical access (flow patterns, passive barriers, building entry scrutiny and command, locks and locking procedures).
- Hiding places (overgrown vegetation, blind spots, blocked sight lines).
- Inadequate lighting, areas in shadow.
- Unsecure or unmonitored site or building perimeter.
- Lack of or poorly functioning alarms and surveillance systems.
- Ignorance of nearby potentially hazardous facilities (such as fuel storage, chemical labs).
- Ignorance of nearby potentially hazardous natural features (hillsides for potential mudslides, seismic activity, water levels that could flood, trees that could fall).

The facility manager would be wise to prepare a security maintenance checklist that regularly reviews security practices and corrects any weaknesses. It should include contact information for critical nearby fire, police and hospital resources as well as equipment manufacturers' operations, service and emergency shutdown procedures.



Progress Check Questions

Directions: Read each question and respond in the space provided. Answers and page references follow the questions.

1. Which of the following is an element of a building foundation structure?
☐ a. Drilled piers and shafts
☐ b. Roofs
☐ c. Active fire protection
☐ d. HVAC, plumbing and similar systems
2. Which of the following is a common deficiency to look for in a building's structure?
☐ a. Vertical cracks in foundation walls
☐ b. Separation of corners at windows, doors or joints
☐ c. Poor acoustic quality
☐ d. Overgrown shrubs, trees or foliage next to the structure
3. Which of these is the most important reason to recognize and correct structural deficiencies?
☐ a. They are unsightly and bad for morale.
☐ b. They can be the primary cause of poor air quality.
☐ c. They are inexpensive if caught early.
☐ d. They could seriously affect health and safety.
4. Which of these approaches is regularly used to remediate or prevent foundation deficiencies?
☐ a. Position trees near to the building to divert rainfall effects.
☐ b. Grade slopes and position roof downspouts to drain water away from foundations.
☐ c. Provide a level grade out from foundations to support water percolation.
☐ d. Add shutters, awnings or other coverings to deflect rainfall.
5. Which of these would not typically be considered a building system?
☐ a. Stairway
☐ b. Fire suppression/life safety
☐ c. HVAC
☐ d. Electrical
6. Active fire protection measures
☐ a. are generally fixed-in-place walls, shaft enclosures and so forth.
☐ b. require on-premises firefighting personnel.
☐ c. include sprinkler and smoke control systems.
☐ d. are seldom necessary or required.



7. Which of the following is typically not a component in a plumbing system?
- ☐ a. Distribution piping and pumps
 - ☐ b. Filters and vents
 - ☐ c. Backflow preventers and valves
 - ☐ d. Fans, ducts and plenums
8. JSA and SDS are designed for what purpose?
- ☐ a. To reduce the risk of accidents and incidents
 - ☐ b. To catalog equipment and systems maintenance activities
 - ☐ c. To support long-range planning
 - ☐ d. To maintain manufacturer warranty requirements
9. At a minimum, mold or fungi can grow wherever there are viable mold spores, an acceptable temperature range, a consistent moisture source and
- ☐ a. fresh air.
 - ☐ b. pollutants.
 - ☐ c. nutrients (paint, paper, adhesives, plastics).
 - ☐ d. radon gases.



Progress check answers

1. a (p. 15)
2. b (p. 16)
3. d (p. 19)
4. b (p. 17)
5. a (p. 28)
6. c (p. 30)
7. d (p. 33)
8. a (p. 45)
9. c (p. 52)



Chapter 3: Manage/Oversee Operations and Maintenance of Building, Systems and Equipment

After completing this chapter, students will be able to:

- Describe how acquisition begins an asset's life cycle and what factors are used in deciding on an acquisition.
 - Explain the elements of installation and commissioning of systems, materials and equipment.
 - Describe typical preventive maintenance activities.
 - Explain how reliability-centered maintenance is used to balance maintenance activities effectively with analysis and management tools.
 - List the elements of a typical maintenance management workflow process.
 - Describe the operation of building systems such as building automation, energy management, health and safety elements and emergency preparation.
 - Describe typical conditions and measurements that signal it is time to replace an asset.
-

★ Topic 1: Acquire Systems, Materials and Equipment

From a broad viewpoint, operations and maintenance are required for systems/equipment and to support occupant needs. As a way to differentiate these areas, facility managers may refer to O&M for facility systems as hard services and O&M for occupant needs as soft services. Hard and soft services share several common aspects:

- Both require operations and maintenance.
- Both have similar work plan processes by which O&M is accomplished.
- Both may be performed by employed staff or hired contractors.

Here in Chapter 3 the focus is on operating and maintaining facility systems and equipment. Chapter 4 focuses on occupant services.

A facility's hard systems (such as the building structure, plumbing, electrical, HVAC, production equipment and so forth) can easily represent a major portion of the entire organization's physical assets. It is the facility manager's responsibility to operate and maintain those assets. That responsibility generally begins when an asset is acquired. Acquisition may refer to everything from the act of purchasing desired equipment or systems to the entire decision-making process used to determine what to obtain prior to its installation.



Increasingly, facility managers are becoming involved in the original specification, design and construction phases of an infrastructure, including its equipment and systems. Most systems are new when a building is constructed; however, the life of the building is usually longer than the life of many of its components and systems. When replacements occur, the facility manager should be involved.

Acquisitions may be broad and encompassing, such as during a renovation, or individual, such as when replacing a pump. These acquisitions may occur because of changing functional needs, updates, renovations or the functional failure of equipment.

New equipment or systems are also acquired to improve performance rather than just as a replacement for maintenance reasons. A long-term capital renewal plan is important to fund and make these acquisitions possible.

Asset management may begin with specification and design that is value engineered. Value engineering should not be viewed as a cost-cutting approach. Rather, it compares cost-adding aspects with value-adding aspects as a way to assess and analyze the user's requirements for a new asset, ensuring that those requirements are met but not exceeded.

Once the asset is installed, it is the facility manager's responsibility to achieve optimal performance and sustain that performance for the asset's useful life. To accomplish this, the facility manager reviews performance and inputs from ongoing operations, maintenance and related sources.

Acquisition begins the life-cycle phases

Acquisition initiates the life cycle of a physical asset. The life cycle is the useable life span of a product, process, facility, tool, system, technology, natural resource and the like. It is based on the presumption that all things go through a continuous cycle beginning with creation, moving to use and replacement/ disposal, and then ideally starting all over again. Once the asset is acquired, installed and commissioned in compliance with design specifications, it becomes part of the inventory/register. The asset is tracked for the remainder of its useful life as it is operated and maintained (and repaired, if needed). At the end of the item's useful life (or installed purpose), it is removed from service and recycled, scrapped, resold or reused for another purpose.

Tracking an asset once it is acquired provides important information relative to O&M activities. This helps give the facility manager the means to recognize an appropriate time to replace the asset.



It is important to track assets because:

- Assets have different life spans.
- Similar assets have similar life cycles.

Assets have different life spans

All hard components or physical assets have a built-in useful life that is different for different components. For example, the life cycle of a light bulb or lamp is certainly much shorter than that of an elevator. Yet, a commercial elevator may have a useful life that is shorter than the life of the building in which it was originally placed.

In this way, “useful life” refers to both functionality and performance. A properly *functioning* component (such as an elevator) may at some point be no longer considered an adequately *performing* one because changing occupant expectations, new product design, or new technology have made the current system out-of-date or undesirable.

Similar assets have similar life cycles

New acquisitions not only signify the start of an asset’s use but also represent due diligence (the investigation and analysis of a potential purchase to minimize risk and possible loss). The facility manager gathers information and studies, compares and selects among similar assets to help determine what acquisition to make and when.

Some of this knowledge comes from existing assets of a similar kind that are tracked through O&M reports. These reports provide information on frequency and costs for maintaining or repairing existing material, equipment or systems.

The facility manager tracks functionality, benefit and useful life of all installed assets. This helps determine when it appears that the total cost for continuing to keep and maintain the asset will exceed replacement or upgrade costs. At that point, a replacement asset is acquired, and the life cycle and tracking of the replacement begins.

Considerations in deciding on an acquisition

Every hard component has a built-in useful life. The facility manager needs to know the life cycle of the item to maintain it properly and determine when to end maintenance and either repair or acquire new. Several aspects are considered when deciding to acquire new or to continue maintenance, including life-cycle cost, reliability and maintainability, performance and energy use, and sustainability.



Life-cycle cost

Life-cycle cost (LCC) is the total cost of acquisition, operation, maintenance and support of an item throughout its useful life, including the cost of removal. Life-cycle cost reflects the total cost of ownership over the life of the asset, not just its capital investment and operating costs. It includes a valuation or measure of environmental impacts produced by the asset's creation.

A thorough discussion of life-cycle costing can be found in the *Finance and Business* course; however, for the purposes of O&M, there are several significant life-cycle cost considerations:

- First costs (acquisition, capital investment, design, construction)
- Recurring costs (operation, maintenance and repair costs; energy and resource, or sustainability, costs; alteration, improvement and replacement costs)
- Potential added value (any value of deferred maintenance; any realized benefits, such as improved productivity, safety, aesthetics; measurable residual, salvage or resale value upon disposal)

The facility manager also considers other strategic factors and space requirements when preparing to acquire systems, equipment or other significant physical assets. These other considerations may include:

- Owner/management policy—what is required and culturally expected.
- Goals—what the acquisition is expected to accomplish.
- Allocations—how the activities and equipment related to space are proportioned.
- Growth—what potential expansions or changes may require new rounds of acquisition or replacement.

The facility manager must be able to contrast and compare all of these factors. This analysis helps provide the most accurate and complete information to determine the best possible acquisition among qualified alternatives.

Reliability and maintainability

By tracking the cost of maintaining existing systems and equipment, the facility manager has a means to recognize more accurately when it might be time to replace rather than maintain or repair infrastructure assets. Confidence in the reliability of critical systems also plays a role in the decision to acquire an asset. For example, it may be better to replace a critical piece of production equipment before the chance it may fail becomes too great. In this way, criticality as well as life-cycle costing helps inform decision making.

The facility manager can find information on the potential operational reliability and maintenance aspects of an acquisition under consideration from



manufacturer specifications, industry standards and associations, and other facility managers.

It is important to know where to go and whom to contact for objective, professional help. When considering a new acquisition of materials, equipment or systems, the facility manager may need to seek advice from design, architectural and engineering services or knowledgeable contractors. The facility manager may also request vendors to provide references who may be asked for information and insight.

All of these inputs help provide a more accurate understanding of the costs to expect for operating and maintaining an acquired component or system over its useful life. This not only permits cost comparisons among competitive candidate items but also provides information on service and maintenance frequency, possible productivity benefits and other life-cycle cost expectations.

Performance and energy use

Changing needs may also require that new materials, systems or equipment be acquired. A facility may add new functions, different occupants or capacity. Any of these changes could require new materials, systems or equipment. New technology such as energy-efficient lamps, equipment or other performance improvements could provide savings that merit purchasing them, even when existing lamps or equipment are still functioning.

Aesthetics, rather than functional factors, may create the need for new carpet or fresh paint. In these cases, performance is measured in perceptions as much as productive savings. These are some of the aspects a facility manager tracks and considers as part of life-cycle costing and the total cost of ownership.

Sustainability considerations

Both public policy and business perspectives play a role in defining and promoting sustainability. They recognize a common concern to meet short-term basic social and economic needs without undermining future resources or environmental quality. Countries and internationally recognized groups have developed or adopted codes and standards relative to environmental concerns and the sustainable use of resources. The facility manager takes direction from these types of associations and also examines local cultural and commercial expectations. This provides the wide, critical view to help the facility manager understand the long-term effect that a new material, piece of equipment or system has relative to sustainability.

In general, sustainability considerations may be summarized by areas of concern, rankings of sustainability, approach for decision making and factors of influence.



Areas of concern

The following is a brief list of some of the regulated areas the facility manager considers when deciding upon a new acquisition:

- Air pollution (both indoor air quality and exhaust releases)
- Emissions (including those from newly acquired materials that may contain volatile organic compounds)
- Asbestos, lead and other toxic elements
- Water use (waste disposal, pollution, reuse)
- Recycling, reduction and biodegradability (breakdown) potentials

Rankings of sustainability

When acquiring new elements, the facility manager should favor resources that rank higher in relative sustainability. These can be ranked as follows in order of most sustainable to least sustainable:

1. **Perpetual resources** such as solar, wind, geothermal and tidal energy represent the most sustainable and environmentally positive option.
2. **Renewable resources** such as timber, soil and grass represent the next most sustainable option.
 - Biodegradable plant materials such as soy, cork and straw might be considered a more desirable material subset among renewable resources.
 - Biomass products such as mulch, compost, low-cement content, plastic lumber/wood and building insulation are also preferred over less recyclable or renewable resources.

It is important to note that renewable materials are natural, which can also make them more susceptible to rot or decay. Some products may contain added chemicals such as arsenic, asbestos or formaldehyde that could prohibit their use. The facility manager must consider all of these factors.

3. **Recyclable resources** such as aluminum, paper, glass or plastics can be reprocessed and used to make new products but at added (and perhaps not sustainable) expense.
4. With **reducible resources** such as packaging or energy use, changes in how they are handled, maintained or originated can reduce environmental impact.
5. **Nonrenewable resources** such as coal, oil and natural gas, which cannot be reused and may not be replaced as fast as they are being used, represent the least sustainable option.



Approach for decision making

The facility manager may make decisions or support others who make decisions about acquisitions to determine the most sustainable, environmentally friendly option. These considerations could:

- Identify material categories.
- Select the most sustainable options.
- Gather and review technical information; seek expert input.
- Assess materials and potential acquisitions for the greatest safety.
- Compare costs.
- Identify the best choice.

Factors of influence

Sustainability factors play a significant role not only in systems but also in service level determinations because sustainability represents both human and machine aspects, as represented in Exhibit 3-1.

Exhibit 3-1: Sustainability Influencers



Each sustainability factor may affect any of the others. All represent a dynamic system in which the facility manager recognizes these relationships in order to select assets and develop work plans that support more sustainable (energy-efficient, resource-conserving) facility management.



✦ Topic 2: Install Systems, Materials and Equipment

Installation defined

Installed equipment is generally considered that equipment affixed to the owners' buildings and maintained by the facility manager (not the landlord, the functional operator or the line manager). The process of installation puts the material, component, system or piece of equipment in place in preparation for its use.

Installation of equipment and systems generally requires commissioning, integration and documentation.

Commissioning

Installed equipment should be commissioned. Commissioning is the process of evaluating, verifying and documenting the performance of facility systems, subsystems, components and operations (and maintenance procedures) to ensure that they function to the intent of the design and efficiently as a system.

Commissioning may be considered to be the transition between installation and O&M in the physical asset life cycle. It generally consists of:

- Documenting requirements, activities and decisions relative to the acquisition of equipment and its specified purpose.
- Testing for functional performance.
- Establishing baseline performance.
- Training operators.

Commissioning helps make the acquisition ready for useful operation and marks the beginning of its maintenance schedule. However, commissioning does not always occur at the initial installation. It can occur whenever equipment or systems are installed or replaced.

For example, predictive testing and inspection methods such as vibration analysis and motor circuit analysis help not only to ensure efficient performance but also to help avoid future maintenance issues, early failures or other reliability problems. Vibration analysis and MCA are discussed more in Topic 3 of this chapter.

Commissioning may occur at different times and in different ways. The following are a few examples.

- **Recommissioning** or **retrocommissioning** is a systematic process to investigate, improve and optimize systems as a way to support high performance and ensure functionality. Recommissioning or retro-



commissioning should not be considered a substitute for major repairs. Instead, it is most often separate from maintenance procedures and is used to reach a higher standard in energy efficiency, occupant satisfaction or some other performance goal.

- **Continuous commissioning** is an ongoing approach to maintain systems at optimal performance at all times. This process is often applied to energy use, management and conservation in existing buildings. It may also be utilized when portions of a system are replaced to produce assurances that the new equipment or parts support overall system performance and the capabilities of existing equipment or parts.

Commissioning activities may be accomplished by staff, contractors or independent third parties, depending upon the expertise or compliance regulations required. An independent party may be hired to commission complex environmental or health and safety equipment while staff or contractors may operate and maintain it after commissioning.

Integration

The facility manager oversees installation in a timely, efficient and effective manner. In general, this means that:

- Installation is integrated with the ongoing operations and maintenance schedule.
- The schedule is developed from a systematic work plan.
- The work plan is specified through a work order for installation.
- Documentation of O&M activities begins.

Relatively few operations or maintenance activities within a facility should occur in isolation. Independent action may be required to manage an emergency, respond to a breakdown, or react to an unexpected event. However, anticipated installation activities should usually be contained in the O&M routine and occur as part of the schedule.

It is important to schedule installation so that it minimally disrupts occupants, customers and tenants. It is also important to coordinate the installation schedule with other work, including maintenance, repair, alterations and other capital projects. For example, coordinating activities helps ensure that a wall will not be freshly painted by a maintenance team right before it is removed by a renovation team.

New acquisitions that involve utilities (electricity, water, fuel, communication wiring) should be selected and installed with particular care



because they become invisible once installed and are noticed only when they fail to function. Occupants are particularly sensitive to occasions when these highly critical systems malfunction.

Core systems such as plumbing, electrical, heating and cooling can be complex and difficult to inspect once they are behind walls, floors and ceilings. Careful installation and complete documentation are important so that as a building ages and utilities deteriorate, the facility manager can maintain systems for adequate safety, reliability and efficiency.

Once installation is complete, the asset becomes part of the facility manager's O&M schedule. This means that the asset is integrated into the work plans and work orders of facility management. This workflow process is discussed in the next topic of this chapter.

Documentation

Installation marks the start of the useful life of an item or system. Documenting the new installation is important not only because it officially captures the start of the installed element's use (and thus the beginning of its life cycle) but also because it is a convenient source when information is needed in the future. The decisions that led to the selection of the item may be supported or challenged by the recorded and cumulative results of its ongoing operations and maintenance.

Complete documentation of the installation provides the orderly, organized start that the facility manager needs for timely, complete and effective O&M. This documentation is typically organized in an equipment O&M manual. Each documented item should be compiled in the inventory/register to be stored in a central file or conveniently near the system itself. The location is also noted in the work order log so that anyone working on the installed element knows where to find its documentation.

The inventory/register documentation contains important information that:

- Designates location (building, floor, room, placement of item).
- Describes type of equipment or system.
- Locates related specifications or drawings.
- Specifies manufacturer, model number, serial number, inventory/register ID number.
- Specifies any permits, compliance codes or other regulatory requirements.



- Records original cost and installation date.
- Documents service warranty requirements and the responsible service supplier.
- Indicates intended useful life, maintenance plan and recommended intervals.

A facility manager may not have a documented list of assets if, for example, a building is purchased and the new owner is not given an inventory or if the register is incomplete or nonexistent. In this instance, the facility manager may seek other information sources, including:

- Operations and maintenance manuals.
- Drawings and plans.
- Other files, notes or documents available through purchasing, accounting or other departments.

When there is no inventory/register or if it is incomplete or determined to be inaccurate, the facility manager may decide to perform an assessment so that a record may be established.

In typical facility management operations, the documentation (along with work requests) is incorporated into a central, automated system that tracks schedules, priorities, work approach and personnel. This may be known as a computerized maintenance management system (CMMS) or integrated work management system (IWMS).

The information in a CMMS/IWMS may include:

- Manufacturer specifications, parts inventory suggestions.
- Warranty and service schedule requirements.
- Maintenance history (planned and unplanned activities).
- Maintenance attributes such as filter sizes, belt sizes, capacity, horsepower, voltage, amperage, pressures and so forth.
- Cumulative costs of maintenance efforts (labor, downtime, lost production).
- Planning for budget and personnel.

These automated systems generally contain both operations and maintenance data for facilitywide management. The system tracks costs to help the facility manager make informed decisions about optimum maintenance schedules, operating results, and repair/replace thresholds.



Education and training

Education and training are at the core of understanding systems and the impact they have on the organization. Education consists of three interrelated components: research, communication and training.

- **Research.** Facility managers can access a wide variety of information on automated systems to support training programs. However, when trying to educate people on technology and processes, a facility manager may want to include a larger quantity of operations-related information. One way to do this is by finding case studies that demonstrate best practices or lessons learned by other organizations. Another way to research operations and maintenance ideas and processes is by participating online in blogs and forums. Performing research online is an excellent way to stay informed on current trends and events.

What other sort of information could be researched? When implementing a new policy or technology, include information relevant to the initiative. The latest in sustainable technology options is certainly a topic to be researched prior to purchasing. Policies and regulations pertinent to the organization's geographic location, governmental regulations or specific products are also good areas to research.

- **Communication.** The next component of education and training is communication. Sharing the right information with the right individuals at the right time is important to raise awareness and build support for a new operations and maintenance activity. There are a number of options for communicating program policies and new procedures to those involved with the organization, both internally and externally. If the facility manager has the responsibility for any external communication, the company website can be used to communicate to those individuals, such as customers and the public. The primary communication focus is those within the organization, such as the tenants/occupants and FM staff. The facility manager can use a corporate intranet site to communicate with this group.

A newsletter is another way to communicate both internally and externally. Two separate newsletters may be needed in order to customize the information provided to each of the audiences; what is important to one stakeholder may differ from that for another. A facility manager may want to develop a repository for information and tools he or she can use in communications.



- **Training.** Expand the capacity of the staff through training, access to information and transfer of successful practices, procedures and technologies. Investing in training and systems to share successful practices helps ensure the success of the action plan by building overall organizational capacity. An informed staff leads to an increased efficiency of equipment, accurate monitoring and training, preventive and predictive maintenance, and an increased success of initiatives through feelings of ownership and an understanding of synergies. Many organizations have found that informed employees are more likely to contribute ideas, operate equipment properly and follow procedures, helping to guarantee that capital investments in sustainability improvements will realize their potential. The facility manager should understand the steps to designing, developing and delivering effective training:
 - Identify the type of training (online, in-class, on the job).
 - Define metrics for measuring the success of the training.
 - Develop training materials for participants and trainers.
 - Schedule the appropriate amount of time to deliver training.

✦ Topic 3: Maintain Systems, Materials and Equipment

Maintenance defined

Maintenance is the work necessary to maintain the original, anticipated useful life of a fixed asset and the upkeep of property and equipment. Operational maintenance of buildings, systems, and equipment generally relates to activities that keep the facility functioning as intended so that occupants are productive.

Over time, operational costs will far exceed a facility's design and construction costs (three times more according to D. G. Cotts in *The Facility Management Handbook*). Proper maintenance can significantly affect the total cost of ownership.

General operational maintenance costs are usually different from capital-funded costs.

- Capital-funded projects usually focus on large-expenditure repairs or replacements of systems or building components as well as remodeling or other major changes designed to, for example, anticipate property improvements or expand productivity. The funding for these activities usually accumulates as reserves over several years.
- Funding for O&M activities that maintain the building, manage its efficient use, and support current systems and equipment is usually an annual budget item that is based on tracked and estimated expenses.



Types of maintenance

In terms of buildings, systems and equipment, the tasks of operational maintenance can generally be divided into two major types: planned and unplanned.

Planned maintenance

As noted in Chapter 1, planned (proactive) maintenance is any maintenance activity for which a predetermined job procedure has been documented and for which all labor, materials, tools and equipment required to carry out the task have been estimated and their availability assured before commencement of the task. Routine building inspections, tours and watches are part of planned maintenance as well as corrective work identified during its two subcategories—preventive maintenance and predictive maintenance.

Preventive maintenance

Preventive maintenance is an equipment maintenance strategy based on replacing, overhauling or remanufacturing an item at a fixed interval, regardless of its condition at the time. Preventive maintenance brings value to occupants because it is likely to be significant to the function, dependability and capability of building operations.

Typical preventive maintenance activities include the following:

- **Periodic or occasional inspection** is any task undertaken to determine the condition of equipment and/or the tools, labor, materials and equipment required to repair an item. Special technologies or equipment may be used to aid in the inspection.
- **Adjustment** refers to any activity performed on materials, equipment, systems and so on to adjust, recalibrate, adapt or otherwise change the circumstances of that asset to match a standard or meet a requirement. As new assets are added to a facility, changes may be needed to either meet predetermined specifications, perform more effectively within the infrastructure, or adapt to conditions caused by other changes to the system.
- **Lubrication**, in its most generic meaning, means to make smooth or slippery to reduce friction and prevent damage by abrasion. Lubrication can also dissipate heat caused by the friction of moving parts making contact.

For maintenance purposes, lubrication may involve a lubricating oil analysis. This analysis is performed on in-service machines to monitor



emerging conditions, confirm problems identified through other means (such as vibration), and troubleshoot known problems.

- **Cleaning** (noncustodial) can improve operational productivity. Dust buildup; air, oil and steam condensate on filters; grease accumulation on machinery; and dirty atmospheric conditions in general can bind equipment, accumulate on surfaces and affect occupants, and in other ways inhibit how the facility efficiently conducts its mission.

Noncustodial or maintenance cleaning can be established on daily, weekly and monthly schedules. The facility manager must initiate and monitor the cleaning of surfaces, metals, fabrics, systems and components with the same diligence as other O&M activities.

Noncustodial cleaning may involve:

- Steam cleaning or sandblasting as well as traditional rag and brush methods.
- Housekeeping around and inside HVAC, electrical and plumbing equipment such as cleaning vents, filters, hoods, fittings, lines and so forth.
- Checklists and work orders clearly describing what, when and how to clean specific assets.

Periodic maintenance cleaning serves to preserve and improve the facility's appearance and promote health. These requirements also promote greater respect for and attention to traditional concepts of housekeeping functions.

- **Painting** is a general term to describe the act of coating any surface with any liquid for a utilitarian purpose. Maintenance tasks may include applying coatings to protect against corrosion, weathering or deterioration; to promote visibility or durability; or to provide color and aesthetics as when painting furnishings, moldings, walls and the like.
- **Replacement** of parts is the act of replacing an item of permanent investment or plant equipment. It is the exchange or substitution of one fixed asset for another having the capacity to perform the same function. The replacement may arise from obsolescence, wear and tear, or destruction. In general, replacement (as distinguished from repair) involves a complete, identifiable item.



- **Minor repairs** are generally ascribed to maintenance activities that do not exceed one or two workdays per task and do not noticeably prolong service life or add to value.
- **Major repairs** are generally those that exceed two workdays or cannot be performed by staff. They may prolong service life but not increase the value of equipment/systems.

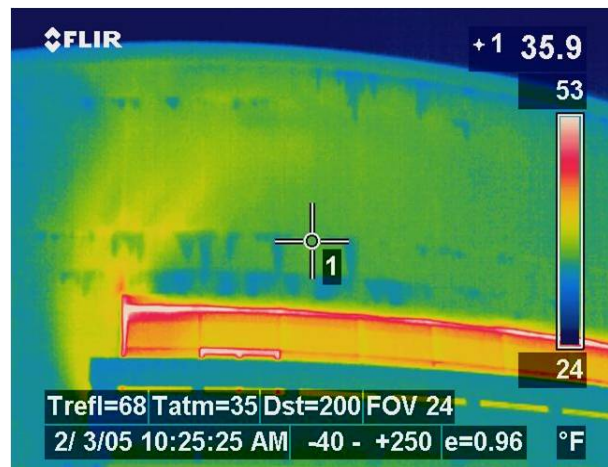
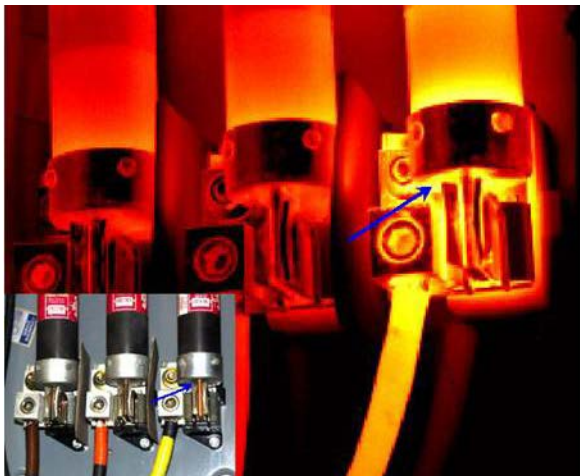
Predictive maintenance

Predictive maintenance (PdM) is a type of planned maintenance that utilizes technology, which allows the forecasting of failures through monitoring and analysis of the condition of the equipment. The analysis is generally conducted through some form of trending of a parameter such as vibration, temperature or flow. PdM is also known as condition monitoring, condition-based maintenance or predictive testing and inspection.

Predictive maintenance anticipates the need for repairs by identifying faults as early as possible. It allows equipment to be repaired at times that do not interfere with production schedules. The following list of predictive maintenance technologies or instruments demonstrates the range and purposes of PdM.

- **Vibration monitoring and analysis** is used to establish a scientific and objective measurement of the running condition of machinery.
- **Infrared (IR) imaging/thermography** is a thermal (temperature or heat) analysis tool that has become widely used for preventive maintenance on mechanical and electrical systems. See Exhibit 3-2.

Exhibit 3-2: Infrared (IR) Imaging/Thermography



- **Oil analysis** is the routine assessment of lubricant properties and suspended contaminants to monitor and report timely, meaningful and accurate information on the condition of the lubricant and the machine.
- **Airborne ultrasonics** use equipment that senses high frequency sounds produced by air or gas leaks, electrical emissions and mechanical operations.
- **Motor circuit analysis (MCA)** is a sophisticated testing and analysis technique to determine the condition of and detect hidden irregularities in a motor power circuit before catastrophic motor failure can occur.
- **Laser shaft alignment** is a technology that is used to determine the collinear condition or alignment of pump and motor shafts.
- **Laser sheave alignment** is the process for magnetically mounting a tool against the inside of a sheave or pulley that measures against the face of the opposite sheave to maintain proper belt alignment.
- **Tong tests** use a current clamp or probe that fastens around an electrical conductor to measure the electrical current in the conductor without requiring the conductor to be disconnected.

Facility managers refer to and use reliability data available from outside resources. See the IMFA FMP Credential Program online Resource Center for lists of some of the many national and international associations and agencies that supply reliability data.

Predictive maintenance can provide several advantages, including:

- Less equipment downtime, catastrophic failure or additional damage.
- Fewer emergencies.
- Better control over maintenance schedule.
- Advanced notice to prepare personnel, materials and equipment.
- A cost-effective maintenance approach.

PdM activities may also be applied as part of the initial phases of facility construction or repair (such as during design, procurement, installation and operational aspects). This could include applying techniques such as root cause and failure analysis, age exploration, and precision alignment and balancing during the life cycle of the facility. These aspects will be discussed later in this topic.

Both predictive and preventive maintenance generally follow a routine based on calendar days, usage (meter readings or run time), or both. An illustration of what might be included in a basic preventive maintenance template is shown in Exhibit 3-3.



Exhibit 3-3: Basic Preventive Maintenance Template

Equipment/asset ID <u>Air compressor</u> Facility location _____ Routing _____	Job # _____ Department # _____ Account # _____																						
<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 60%;">Maintenance</td> <td style="width: 40%; text-align: right;">Next</td> </tr> <tr> <td>Frequency: Days <u>Semiannual</u></td> <td style="text-align: right;">_____</td> </tr> <tr> <td>(units) Meter _____</td> <td style="text-align: right;">_____</td> </tr> <tr> <td><u>Run time</u></td> <td></td> </tr> <tr> <td>Fixed [] Floating []</td> <td></td> </tr> <tr> <td colspan="2">Work group/trade (list)</td> </tr> <tr> <td colspan="2">HVAC shop</td> </tr> <tr> <td colspan="2">Special tools (list)</td> </tr> <tr> <td colspan="2"> 1. Standard tools—basic 2. Belts 3. Lubricants 4. Fin comb 5. Vacuum cleaner 6. Test gauge </td> </tr> <tr> <td colspan="2">Parts (list)</td> </tr> </table>	Maintenance	Next	Frequency: Days <u>Semiannual</u>	_____	(units) Meter _____	_____	<u>Run time</u>		Fixed [] Floating []		Work group/trade (list)		HVAC shop		Special tools (list)		1. Standard tools—basic 2. Belts 3. Lubricants 4. Fin comb 5. Vacuum cleaner 6. Test gauge		Parts (list)		<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td>Job description</td> </tr> <tr> <td> 1. Perform normal checks and operation. 2. Change compressor crankcase oil. 3. Clean air intake filter. 4. Check air dryer. 5. Check condensate drains. 6. Inspect air tank for proper operation. 7. Inspect belt alignment and condition. 8. Check accuracy of gauges with calibrated test gauge. 9. On two-stage compressor, check intermediate pressure. 10. Test relief valves; replace if leaking. 11. Complete preventive maintenance checklist. </td> </tr> </table>	Job description	1. Perform normal checks and operation. 2. Change compressor crankcase oil. 3. Clean air intake filter. 4. Check air dryer. 5. Check condensate drains. 6. Inspect air tank for proper operation. 7. Inspect belt alignment and condition. 8. Check accuracy of gauges with calibrated test gauge. 9. On two-stage compressor, check intermediate pressure. 10. Test relief valves; replace if leaking. 11. Complete preventive maintenance checklist.
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Work results/uncovered issues Work performed by _____ Work checked by _____ Work authorized by _____	Remarks/special instructions 1. Review manufacturer's instructions 2. Coordinate motor preventive maintenance on an annual basis. See Guide M-3. 3. Tank should be inspected and tested by qualified inspector. 4. Secure the electrical service.																						

By measuring planned maintenance, the facility manager can monitor the amount of work being accomplished as a way to anticipate and prevent failures. This also helps the facility manager identify defects that could lead to failure. To maximize the benefits of avoiding failure, some facility managers



promote a high level of planned maintenance work balanced against total cost of ownership; this is also discussed in more detail later in this topic.

Unplanned maintenance

Unplanned (reactive) maintenance, or unscheduled maintenance, refers to any maintenance work that was not on an approved maintenance schedule before it began.

There are three subcategories of unplanned maintenance: corrective, run-to-failure and emergency.

- **Corrective maintenance.** Corrective maintenance is any activity that is required to correct an unanticipated component failure that has occurred or is in the process of occurring, for example, a machine breakdown because of bearing failure. This activity may consist of repair, restoration or replacement of components.
- **Run-to-failure.** Run-to-failure is a strategy in which no routine maintenance tasks are performed on the equipment. The only maintenance performed on the equipment is corrective maintenance, and then only after the equipment has suffered a failure. This is also described as a nonscheduled maintenance strategy. An example would be replacing a small quarter-horsepower pump when it fails in a noncritical system. The cost to maintain the pump would be more than the cost to replace it. Other examples of items that run to failure are door knobs and relamping.
- **Emergency maintenance.** Emergency maintenance is any activity that requires immediate repair because of impending danger to the occupants, their business processes, the building or a building system. For example, this could be repair of a broken door or exterior window or a loosened stair railing.

Typical maintenance activities

Exhibit 3-4 illustrates where different maintenance activities—planned and unplanned—may occur.

Exhibit 3-4: Typical Maintenance Activities

Planned Maintenance		Unplanned Maintenance	
Preventive	Predictive	Corrective	Emergency
• Inspect • Adjust • Lubricate • Clean traps • Paint • Check for noise, odors, vibration • Tighten connections	• Vibration analysis • Ultrasonic • MCA • IR thermography • Laser alignment • Oil analysis	• Hot/cold call • Lights out • Broken fixture • Clogged toilet • Spill/stain • Damaged furniture • Broken lock • Vandalism	• Leak/flood • Power outage • Gas leak • Safety/hazard issue • Chemical spill



Balancing maintenance activities—reliability-centered maintenance

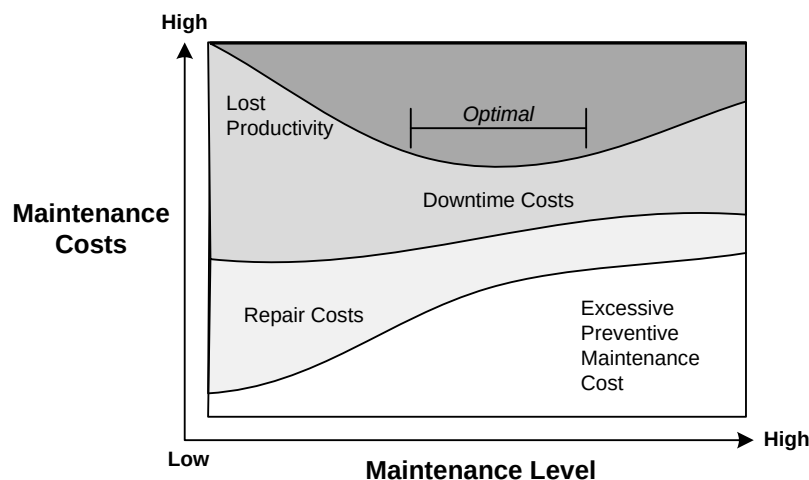
Prioritizing all the options relative to operational maintenance is generally determined by the criticality of the system or equipment.

Ideally, an organization should have more planned maintenance activities than unplanned. In production or manufacturing facilities with large equipment and complex processes, plant engineers recognize that the breakdown of a piece of assembly equipment can be costly not only in terms of labor hours lost and repair expenses but even more in the revenue lost because goods are not produced. Preventive and predictive procedures are critical to avoiding such breakdowns.

On the other hand, in nearly all cases it is not prudent for a facility manager to strive for 100 percent planned maintenance. Labor hours spent in operational maintenance generate an opposing cost, and there is a point of diminishing returns between the cost of total preventive maintenance to produce ideal operating conditions and low cost from minimum maintenance that increases the potential for expensive breakdowns. In addition, unplanned maintenance is not necessarily the result of improperly conducted or inadequately planned activities. The facility and its management are dynamic. The operational status of materials, equipment, systems and even the building itself can change without warning.

The facility manager needs to balance planned and unplanned maintenance. The ratio between the two may vary, depending upon the mission, age, requirements and resources of the facility/owners. The ideal balance occurs by finding the optimal timing and method of maintenance so that planned maintenance costs are at their lowest while keeping the potential or realized equipment downtime also at its lowest. This relationship optimizes maintenance by comparing it to the cost to replace or restore losses due to equipment failure. It may be referred to as a reliability-centered maintenance (RCM) approach and is illustrated in Exhibit 3-5.

Exhibit 3-5: Reliability-Centered Maintenance Approach



Reliability-centered maintenance determines the best mix of preventive maintenance, predictive maintenance, proactive repairs, and run-to-failure approaches. These different maintenance and repair options are integrated and employed in combination to produce the greatest probability that a machine or component functions as intended over its design life cycle balanced against the proper level of maintenance.

Thus, the overall goal is to provide for the stated function of the facility with the required reliability and availability at the lowest cost. Maintenance decisions are then based on requirements supported by sound technical and economic justification.

Many paths or processes can lead to that final RCM goal, especially since the consequences of failure can vary significantly based on the nature of the facility or industry. For example, RCM analysis is rigorously applied by the aircraft and nuclear industries, where functional failures can potentially produce large losses or catastrophic environmental impact. Facility managers in less critical industries or missions may employ a streamlined or more intuitive RCM process because the high analysis cost of the more rigorous approach does not supply equally high results in system reliability savings. The streamlined approach is based on the same principles but may not recognize or analyze all possible failure modes.

The facility manager may apply a number of analysis tools to maximize reliability and performance at the lowest total cost. These tools can include failure modes and effect analysis, root cause analysis, and age exploration.

Failure modes and effects analysis

Reliability-centered maintenance may include a detailed failure modes and effects analysis (FMEA) to help determine failure probabilities and system reliability calculations. This analysis helps the facility manager determine what maintenance approach and tasks to use in addressing each identified potential failure and its consequences.

The facility manager (supported by the system that processes maintenance activities) screens and evaluates work relative to how critical it is to occupants and the facility. Physics of failure (PoF) and criticality are examples of approaches used to support FMEA.

PoF is an approach to incorporate reliability into product design using materials, structures and technology that support reliability assessments.



Criticality can be ranked as values most critical to least critical to the facility mission and strategic plan, such as:

- Protect life.
 - Protect property.
 - Protect operations.
 - Comply with codes.
 - Protect productivity.
 - Protect energy loss or cost.
- Most Critical



Least Critical

Other rankings of criticality or importance may include:

- Dollar value, such as budget and financial policies that determine which job may be assigned for maintenance first.
- Work complexity, for example, if work is complicated enough to require training, advanced preparation, or new inventory order.
- Occupant satisfaction, where social penalties may occur such as lower morale, poor aesthetics, or increased turnover.

Root cause analysis

Root cause analysis (RCA) is the process to discover the systemic cause of system results by finding the underlying or root cause of the problem. Some problems should not be fixed but instead the system or equipment should be replaced. A root cause analysis can help determine this.

RCM is used to identify what *might* cause a problem and is thus a *leading* indicator for deciding what maintenance to perform. RCA is used to identify what *did* cause a problem and is thus a *following* indicator to determine why a system, component or process failed.

Generally, RCA applies five questions (five “whys”) as a technique to reach beyond symptoms to determine the root cause of a problem. Essentially the questions ask why the customer/occupant was dissatisfied in order to reach the basic cause of the problem. The question loops back several times by continuing to ask “why” until there is agreement on what the problem is and how it happened.

Facility managers can use root cause analysis to help understand why systems failed historically and what effects those failures had on the operation or mission of the facility. A root cause analysis is not intended to find fault or blame but to recognize the true cause of equipment and system failures.



Age exploration

As the term implies, age exploration, or reliability analysis, uses historical data to improve reliability or productivity or reduce the cost of maintaining an item. It is similar to FMEA and RCA in that the goal is to understand the condition of the component, equipment or system and the potential/probability of failure. By knowing the age and condition of the item, the facility manager has better information and can decide if maintenance should be made more intensive, if inspections should be more intrusive, or if replacement or repair options should be considered.

Systems, components and machines will deteriorate, break or fail to a level that does not perform to expected standards. It is highly unlikely that any maintenance program can prevent all failures. However, RCA can help provide historical maintenance knowledge, FMEA can help provide proactive maintenance knowledge, and RCM can help the facility manager achieve a balanced maintenance approach to support the productivity and efficiency expectations of the business owner based on lowest total overall cost.

The facility manager should be able to quantitatively justify the balance that is struck between planned maintenance and anticipated breakdowns. Most facility managers would agree that there is a place for demand or corrective maintenance as well as run-to-failure options in a complete maintenance approach.

Maintenance management workflow

Maintaining a facility's assets requires exceptional information input, organization and control. In general, it involves:

- A plan or routine that balances planned maintenance and corrective maintenance.
- An inventory/register organized by location and group.
- An understanding of the maintenance tasks required.
- A prioritized approach to assigning and accomplishing the work.
- Resource allocation to complete the work in a timely, efficient manner.
- Data input once the work is completed.
- A means to capture that feedback as a way to update the status of the inventory/register.

The facility manager uses the directional goals established by the strategic facility plan and the inventory/register to produce a short-range O&M approach (usually up to one year) for maintenance management.



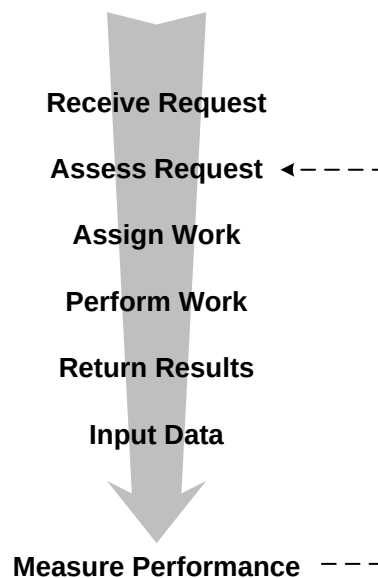
The approach may be different, depending on the mission of the organization, the priorities of the ownership, the state of the facility's equipment/systems, and the facility manager's methods.

All types of maintenance may follow the same general approach. The difference usually occurs at the point when work is requested.

- Requests for planned maintenance generally come automatically based on predefined schedules set for preventive and predictive maintenance.
- Requests for unplanned maintenance can come as telephone calls, faxes, or e-mail messages into a work control or work management center.
- Corrective maintenance is generally added to the schedule based on the criticality, demand and priority of that unplanned maintenance.

A typical maintenance management workflow process consists of the activities shown in Exhibit 3-6.

Exhibit 3-6: Typical Workflow Process



A note on automation

Automation is a tool that helps make the workflow process efficient. This automated relationship is noted by the dotted line in Exhibit 3-6 that connects and encapsulates the steps from when a request is assessed until performance of the work is measured. This forms a continuous loop that becomes the basis for continuous improvement in managing a facility.



Work requests or inputs are gathered to establish all the maintenance work that needs to be done. This information is typically collected in a facility management information system. This is an information technology tool that includes the hardware and software needed to store, manage, distribute and use information that is critical to an organization's mission, such as coordinating operations and maintenance.

In general, the facility management information technology:

- Provides a tool to support the facility's business and maintenance work processes.
- Provides consistent and accurate data and information so that the facility manager can make sound decisions regarding the building and its systems and equipment.
- Provides a tool to measure performance.

These automated systems may be known as computerized maintenance management systems or integrated work management systems.

A **computerized maintenance management system (CMMS)** uses automated technology to assist with the effective and efficient management of maintenance activities. It generally includes elements such as equipment and inventory management, computerized work orders and applications for scheduling planned maintenance tasks as well as numerous other features. In general terms, a computer-aided facility management (CAFM) system may incorporate the functionality of CMMS and add special or geographic information system (GIS) applications. GIS provides a way to reference or map assets and processes relative to location.

With an **integrated work management system (IWMS)**, a facility manager may also have access to software programs for the entire organization. As enterprise-encompassing tools, IWMS can support a comprehensive, controllable approach across the entire infrastructure, its assets and their functionality. It can provide ever-increasing and broadening functionality, which may include features for:

- Project management.
- Capital planning.
- Sustainability.
- Fleet management.
- Environmental health and safety.
- Security.
- Disaster preparedness.



There are several different application approaches for CMMS-type systems, such as the following:

- Client-server-based systems have individual PCs with a facility management program installed.
- Online approaches use an Internet-based wide area network to communicate and maintain facility maintenance records.
- Vendor-hosted approaches use an application service provider or software as a service in which the software is installed at the vendor's hosting facility and the facility has a monthly or annual license agreement that permits connection via the Internet to a secure server.

The cost and complexity of applications can vary greatly. In general, software license costs represent only a fraction of the total costs to implement a CMMS/IWMS.

Facility maintenance software applications can also experience high failure rates. This suggests that many technology implementations fail to meet the needs of their users. Facility management technologies may fail for any of several reasons, including the following:

- Users (owners, executives, facility managers, systems personnel) do not have or recognize a strategic technology plan so they do not clearly express what is expected of a system before it is installed, configured and implemented.
- Data standards are not established and data entry and maintenance are inconsistent, leading to unreliable results and performance measures.
- Inadequate training worsens the situation and can lead to data being incomplete, inaccurate or both, and neither the facility manager nor executing personnel know how to use and respond to the system effectively.

If there is no computer-based system to help determine and organize operations and maintenance schedules or requests, the facility manager uses a manual process to schedule and review operational maintenance activities required for the day, the week, the month and even longer durations.

Whether the requests and work performance data are automated or manually generated, the facility manager's ability to efficiently and effectively maintain building, systems and equipment is only as good as the quality of the record system and the inputs provided.



Typical workflow process

Now let's take a closer look at each of the steps in the typical workflow process as shown in Exhibit 3-6.

Receive request

In this step, requests are collected from authorized sources to recognize all maintenance or repair work potentially needed across the entire infrastructure. A work request is the primary method by which user departments request the initiation of a maintenance task. These work requests may be part of regular, scheduled maintenance, or they may come from occupants or others making requests through a work management center (WMC) or help desk. They may be made by e-mail, telephone, meeting or direct entry into a CMMS. The work request is usually converted to a work order (WO) after it has been authorized for completion.

Service orders, service requests and the like usually represent smaller, service-oriented requests for maintenance that need immediate attention. These are normally measured by the cost and time of the work so that, at some threshold for either, the request is more closely considered and estimated.

Routine maintenance for the building, equipment and systems is also included through work requests that will be turned into work orders. Manufacturer warranty requirements and recommendations largely determine the schedule and frequency of planned maintenance. Routine planned maintenance requests (rather than unplanned maintenance requests) should typically reflect the majority of requests when a facility is managed by a reliability-centered maintenance philosophy.

Assess request

In this step, supervisors or other authorized personnel identify, prioritize, and otherwise sort the work requested. The approach to all maintenance and operations activities, or work plan, is often managed through the work management center—a place or activity where all scheduled and unplanned work is screened and evaluated so that work assignments and activities can be communicated to individuals or teams and trades with specific tasks, expectations of time required to complete the job, and date/time for doing the job.

The WMC may be the exclusive domain of a facility manager or may involve the coordination of many individuals responsible for work reception, preventive maintenance, estimating/planning/budgeting, materials, customer service, and work assignment. Regardless, the WMC is usually the central point that collects and reviews requests for maintenance or repair. At the



WMC typically all work is estimated, planned, coordinated, scheduled, executed, measured and tracked. In a larger operation, an individual (or WMC staff) enters request data into the CMMS/IWMS and routes work orders to a supervisor for review and assignment.

WMC/help desk staff may coordinate many different types of work coming from many different sources. The receipt, assignment, communication and monitoring may include:

- Coordination of activities by individuals and different skills to specific projects.
- Projects, alterations and renovation work directed by the strategic facility plan and the tactical O&M plan.
- Support for operations such as moves, custodial activities, security and other requests (discussed in Chapter 4).
- Planned maintenance including preventive and predictive maintenance, routine building inspections and watches.
- Special or planned events such as reimbursable services, activities coordinated among multiple trades or shops, planned outages and others.
- Previously unscheduled corrective and emergency maintenance activities approved to be done.
- Corrective or demand work that moves to a higher priority.
- Anticipated requests from a backlog that recognizes all work available to be done, including job interruptions.
- Backlog work that has been approved and for which parts are either listed or bought and everything is ready.

Work schedules may also reflect priority handling or special considerations such as the following:

- **Safety.** Is occupant or visitor health, well-being, or security significantly affected positively or negatively?
- **Worker safety.** Are proper work safety procedures in place and being used based on the nature of the maintenance activity, for example, lockout/tagout (to label and keep equipment from accidentally operating), working in a confined space, handling hazardous materials, wearing appropriate personal protective equipment such as hardhat, gloves, eye protection and footwear?
- **Regulations.** Are there potential local code violations or concerns relative to national or international standards?
- **Customer deadline.** Does the occupant require a completion time that is critical to their business purpose, operational productivity or other important need to satisfy?



- **Energy efficiency.** Is there a significant cost benefit in the earliest possible completion?
- **Environmental impact.** Are there pollution, depletion, toxicity or quality factors that would be considerably reduced by the earliest possible completion?
- **Performance improvement.** Is there a significant productivity or commercial benefit in the earliest possible completion?
- **Durability.** Does the operational maintenance add reliability or stability benefits in the earliest possible completion?
- **Economy of work motion.** Is there related or nearby work that could be combined with this maintenance activity to reduce costs?
- **Aesthetics.** Is there added occupant comfort, competitive advantage or other nonspecific benefit to be gained by the earliest possible completion?

Some help desks/WMCs may also be responsible for monitoring building automation systems, fire and life safety system alarms and perhaps security monitoring systems. When this is the case, help desk personnel communicate with facility management leaders, including the facility manager, to ensure that the necessary information is going into the CMMS/IWMS to generate meaningful and accurate performance measures. These performance measures could include:

- Staff productivity.
- Preventive maintenance versus corrective maintenance.
- Work order aging (how long it has been since the WO was prepared).
- Turnaround time (how quickly the WO was completed).
- Emergency response time (how quickly responses occurred).
- Preventive maintenance completion rates (how efficiently it was performed).

Performance measurements and approaches are discussed more fully in Chapter 6.

The information accumulated by the CMMS/IWMS helps to produce a work plan. The work plan proceeds to individual work orders that divide the work plan into detailed tasks.

Assign work

Once a work request is received and assessed, it is assigned to an individual or team usually through a work order (or its equivalent such as a job ticket).



The WO is the prime document for planned and unplanned operational maintenance. It may include information such as:

- Description and type of work required.
- Work location.
- Customer and contact information.
- Work group or service organization assigned to complete the work.
- Repair codes.
- Task priority.
- Job procedures to be followed.
- Parts, materials, tools and equipment required to complete the job.
- Labor hours, costs and materials consumed in completing the task.
- Key information on failure causes.
- Work performed.
- Prior work performed.
- Other details or notes specific to the task and assignments.

In some facilities this information is entered at the work management center and a WO is prepared, or the request is reviewed by a supervisor before the actual WO is prepared. Supervisory-level personnel may review work requests to identify type of work, urgency, skills required, available personnel and other aspects that promote efficiency. In this way the facility manager can determine whether to prepare a WO, place the request on hold or dismiss it.

Work orders can range from simple to complex, depending upon what is required and to what level the work is planned. Typical WOs for maintenance for equipment or systems may require many different tasks and components and may include a list of the trades or skills involved, the job steps for each, and their equipment and material requirements. Usually, work orders do not anticipate or allow for delays, unforeseen events or unusual circumstances.

There are two general types of work orders:

- A **standing work order** is left open either indefinitely or for a predetermined period of time for the purpose of collecting labor hours and/or supplies/materials costs for which it has been decided that individual work orders should not be raised. Examples would include standing work orders raised to collect time spent at safety meetings, or in general housekeeping activities.



- A **standing operating order** represents predictable, routine services that can be planned in advance and occur on a regular basis. These may include inspections and established preventive maintenance. Some facility managers include non-core services such as custodial, waste collection, food services and grounds maintenance as standing operating orders that are automatically included as work reception inputs.

The facility manager uses standing operating orders by:

- Applying them to short-term, repetitive jobs.
- Assigning them to specific jobs that occur frequently.
- Grouping them to reduce random preparation and travel by service people.
- Providing specific instructions or procedures to promote the fastest quality work.

Standing operating orders can reduce individual work orders and paperwork. Using standard, preprinted descriptions fully supports the maintenance activity yet reduces redundant planning, writing and paperwork.

Automated systems such as CMMS can create standard work orders or standing operating orders particularly to support personnel doing short duration (less than one hour) maintenance or repair activities. These quick projects can account for a measurable portion of the total O&M labor hours. Efficiency is important in doing the task and in shortening the interval between activities or locations of activities.

Perform work

Work activities signal the actual execution of maintenance work accomplished by staff or contractors. Personnel execute planned and unplanned maintenance work based on the decisions of the work plan and the directions of the WO.

The work plan and the work order that represents it may include any of several considerations, such as:

- Work to accomplish.
- Any advanced preparation (materials, tools, equipment, or special skills personnel or individual training needed).
- Dependencies among tasks (such as coordinating staff or contractors with different skills or responsibilities so that tasks are efficiently accomplished without unnecessary delay or waiting time).
- Tasks and task sequence in the correct order.



- Coordination and productivity considerations (priorities, relationship with nearby O&M activities, and economic use of time and motion).
- Estimated start and end dates (or work duration).

When jobs are more complicated or involve several different workers or skills, the facility manager (probably assisted by the CMMS) must carefully consider how schedules interact. This can help reduce conflicts and promote more coordinated activities.

Managing activities at one job site may require coordinating several visits by different maintenance personnel at different times. The sequence of work may also prioritize which trades or individuals appear, when they appear, and what they must accomplish for the project to continue in an orderly, efficient manner.

By being aware of potential conflicts among projects (for example, if an individual or crew is needed in two places of equally high priority at the same time), the facility manager may determine alternate project service dates or times or involve more service personnel. Regardless, the facility manager uses this knowledge to assess the conflict and make a thoughtful decision.

An integrated workflow schedule can also help the facility manager assign work crews across projects more efficiently. For example, efficiencies may be lost or gained because activities may:

- Occur near to each other and thus may be done at the same time or in quick succession.
- Require the same tools, parts or skills and may benefit by using the same personnel.
- Be redundant because others are performing the same, similar, or partial work at a different time.
- Be unnecessary because a different, overarching activity negates its purpose (for example, lubricating a machine scheduled to be removed the next day).

Work productivity can easily suffer in O&M because so many aspects need to be recognized and coordinated. Productivity rates can fall when, for example:

- Parts, materials or tools are not available.
- Work groups are not synchronized.
- Emergencies produce conflicts.
- Customers or occupants delay access.
- Poor work descriptions cause confusion.



The facility manager must strive to keep activities efficient, orderly, in the right sequence, on time, within guidelines and fully reported.

Return results

Once the requested work is complete (or at some other designated point), the results of the work are communicated and any additional needs or unanticipated requirements are noted. Staff/contractors return the WO with complete detail on the work performed. This may be transmitted to a supervisor or to the work management center/help desk.

Regarding work activities, the facility manager (or a designate) may also:

- Check and inspect the maintenance work.
- Close completed work orders and perform cost accounting.
- Update activities with reports.
- Solicit feedback from occupants, owners and maintenance personnel to be included as inputs in the ongoing, cyclic management of the maintenance workflow.

Input data

In this step, the work results are collected and organized to regularly update the status of maintenance and the condition of the facility and its equipment and systems. Data from the WO is entered into a facility information management system such as the CMMS/IWMS or equivalent. This data may include work class, repair codes, equipment, time and so forth. Once the data is entered, the WO is completed and closed. If additional or follow-up work is needed, an additional WO is usually created and becomes part of the ongoing work plan.

Work inputs are transferred to a central database to provide the facility manager with current and ongoing information relative to:

- The facility as a logical grouping of equipment and systems.
- Requirements, condition and parts needs of each operating asset (materials, components and equipment across building structure, systems, interiors, exteriors and grounds).
- Maintenance procedures, special instructions and skill requirements for specific projects.
- Sequence and frequency of maintenance procedures relative to all assets scheduled for maintenance.
- History of prior maintenance, results, and out-of-ordinary circumstances.



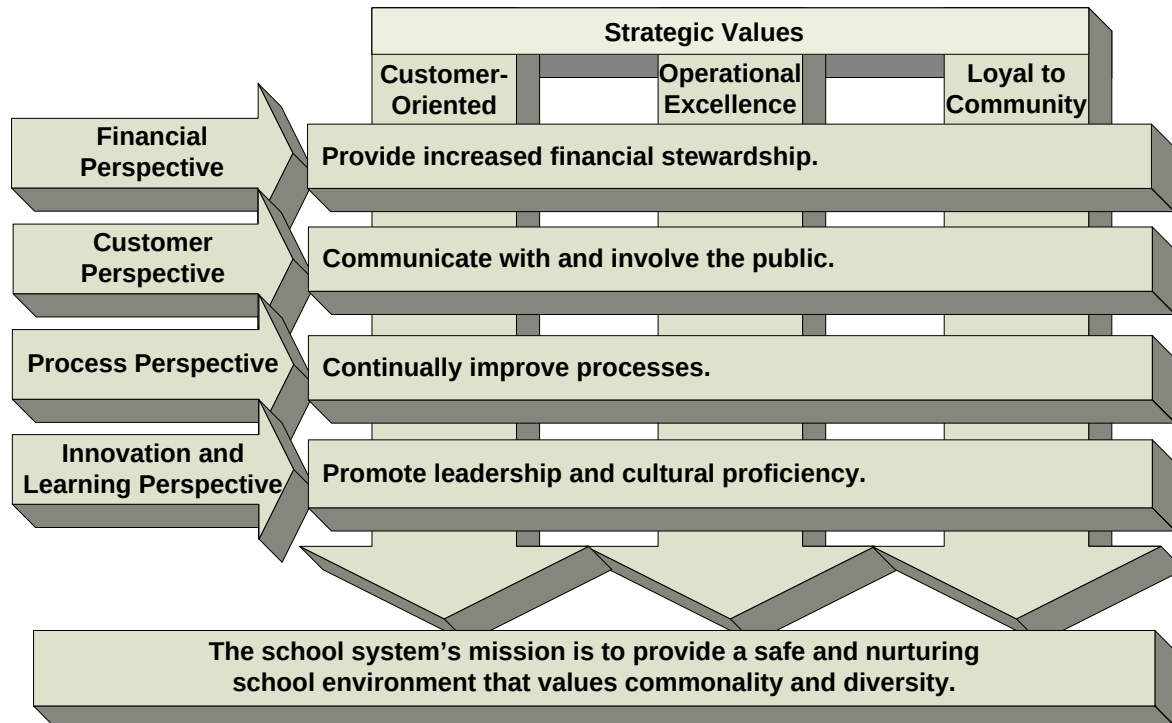
Measure performance

As data is accumulated, the facility manager uses some means (usually automated information technology such as CMMS/IWMS) to compare overall results against predetermined performance expectations. Some best practices relative to a maintenance management policy can also help establish how the CMMS helps in preparing a work plan. The quality of inputs from work orders helps determine the accuracy of information the facility manager uses to support efficient, effective operational maintenance.

When maintenance management is part of CMMS-type automation, the facility manager can benefit from computerization that brings together all business aspects of O&M. This supports not only the execution of maintenance schedules but also the business administration of O&M.

A key performance and measurement approach applied to facilities (as well as processes and the workplace) is known as the balanced scorecard (BSC). A BSC helps the facility manager implement practices that align with the entire organization's mission and strategy. The balanced scorecard approach was first developed in the early 1990s to measure, compare and balance financial performance against employee, business process and customer perspectives. This relationship is illustrated in Exhibit 3-7.

Exhibit 3-7: Example of Balanced Scorecard Strategic Objectives for a School System



By viewing the organization from these perspectives, the facility manager has the means to review results against expectations or performance indicators. It also helps the facility manager organize and present O&M results, requests and objectives in a way that communicates with upper management and facility owners. BSC and performance indicators are covered more completely in Chapter 6.

The CMMS can deliver a reasonable work plan that supports the BSC when the facility manager considers and applies rules such as the following:

- Productivity and results are highest when staff/contractors know what to do, when to do it, how to do it and who (what individual or team) is expected to do it.
- System input and use is more than document management and is a means to inform, direct and implement maintenance activities.
- Relationships and service quality are core requirements whether using staff or contractors.
- Measurement determines control; in general terms what is measured is performed.
- All assets and their condition are known, inventoried, recorded and updated with each service visit in a dynamic, responsive system.
- Authorized personnel at all levels have access to the information needed to do their jobs.
- System access and use ensures privacy, security and integrity.

Facility management requires many other abilities that influence this basic maintenance management approach. These include planning, estimating, budgeting, materials control, payments, contracts, quality measurements, reports and customer service as well as other skills and practices that the facility manager needs in order to deliver competent management. These skills are discussed in other IFMA courses.

Maintenance during peak periods

Managing a facility can be especially stressful during peaks when resources may be strained. This is a time when the facility manager becomes a leader and motivator to help staff and occupants manage demands.

The *Leadership and Strategy* course is a good resource to use in learning more about leading personnel. Concerning O&M, when demand is high, the facility manager demonstrates leadership by handling situations in a manner that sets a proper tone.



Some examples include the following:

- Be present as a facility manager by creating a positive atmosphere and being available to answer questions and resolve conflicts.
- Prepare staff, contractors and occupants for the challenge of higher demands; recognize the heightened emotional state.
- Encourage involvement and ownership so that the focus is on efficiently handling tasks and solving problems and not finding blame or becoming distracted by complaints.
- Help dampen stressful situations and upset individuals; encourage staff as well as occupants to express their anxiety and frustration; encourage patience.
- Support efforts made to respond; reinforce success with rewards, recognition and acknowledgement; share successes; emphasize challenges undertaken rather than deadlines missed.
- If facility technical staff are union members, the facility manager should fully understand and address the related issues, concerns and work rules.

★ Topic 4: Operate Building Systems and Equipment

Operations defined

Operations is about ensuring that the facility's infrastructure and how it is used and managed provides a satisfactory work environment, is in compliance with laws and regulations, meets financial performance goals, and protects the surrounding community and environment.

Guidelines for how systems operations is used and maintained can vary depending upon the facility, its purpose, its level of specialization, its age, the owners' mission and other characteristics. In any case, the infrastructure should provide appropriate and reliable temperature control, lighting, equipment and all the other operating aspects required for effective and efficient building systems, structures, interiors, exteriors and grounds. As technologies and products change and as facilities age, the facility manager must be diligent in maintaining a level of service that supports occupant safety and comfort and energy efficiency.

Energy efficiency, environmental sustainability, and air and water quality are all significant to facility management because they directly relate to issues that concern the facility's productivity, efficiency, safety and code compliance. Facility managers must not just be aware of these issues but must be prepared to deal with them as part of their management responsibilities. Energy use, air quality, and sustainability are business issues that affect operational planning and ultimately the mission of the facility.



Maintaining the operations of systems and equipment may cover several categories, including:

- Systems operation and monitoring.
- Building automation.
- Energy management.
- Health and safety inspection.
- Appearance inspection.
- Emergency planning and preparation.
- Material handling and storage.
- Snow and ice removal.
- Policies, procedures and safety.

This topic briefly summarizes each.

Systems operation and monitoring

Routine, ongoing systems operations and the monitoring of those operations are significant aspects of facility management. Operations that may be continuously monitored include water treatment, cooling systems, heating systems, ventilation systems and so forth.

The operation of systems and equipment can be generally understood as all activities other than their maintenance. These activities can include:

- Startups and shutdowns (such as starting up boilers in cool weather to warm a building before occupants arrive).
- Monitoring and adjusting (such as adjusting thermostats or dampers based on reference measurements and reading pressure and temperature gauges).

Activities such as these are common routines for managing a facility and may or may not be automated (discussed next).

Building automation

Building automation systems (BAS), also known as building management systems (BMS), are integrated, automated systems that control several aspects of building operations such as HVAC, lighting, energy, elevators, fire suppression and security. This integrated automation technology generally uses monitors and computers in a distributed control arrangement to automatically regulate facility systems. Ideally, BAS/BMS technology helps produce what is often called an intelligent or smart building that self-regulates so that the facility operates at peak efficiency, with the greatest cost savings, lowest greenhouse emissions and optimal energy use. In this way, BAS/BMS can be a significant method to produce effective energy management.



Integrated automation technology from a systems standpoint refers to many different equipment and maintenance elements, including:

- Network equipment, conductors, cables and pathways.
- Maintenance schedules and commissioning.
- Devices, gateways, networks, control units and software.
- Terminal devices relative to facility equipment, conveying equipment, fire suppression, plumbing, HVAC, electrical systems, communications, safety and security (and the control sequences for these devices).

While integrated automation may be most common in large buildings, it is coming into wider use because a high-performing BAS/BMS can help provide:

- Comfort and productivity for occupants.
- Greater control over energy consumption and targeted goals.
- Early detection of equipment mechanical problems or other conditions.
- Greater reliability and durability of facility equipment.
- More efficient use of maintenance personnel.

Energy management

Energy management is an operational aspect that spans the entire facility infrastructure: systems, structures, interiors, exteriors and grounds. Facility managers are usually mandated to reduce waste, increase efficiency, conserve use and meet quality requirements. Because energy is used to heat, cool, provide light and run equipment, it is vital to supply and expensive if misused.

As the cost, sustainability and significance of energy use continues to become more important, facility managers must be alert to methods and opportunities that support better use of resources and more reliable delivery. A process for managing energy could include the following elements:

- **Control demand.** Some widely recognized behaviors help control demand such as turning off lights when leaving a room, closing drapes and shades to retain heat in cold seasons and block sunshine in summer. Others involve integrated automation in the form of an energy management system (EMS) to monitor, control and optimize the performance of electrical generation, transmission equipment and related utilities.
- **Develop a baseline.** The facility manager uses invoices, meter readings and all available methods to measure and track energy consumption for each utility (electricity, liquid/gas fuel, water, communications, and so



forth). The use of these energy resources may be further divided by location, building, type of use, age of the system or other factors. The energy manager or facility manager tracks use and uses business returns, life-cycle costing and other measurements to determine where efficiencies may be created through better maintenance, replacement with more efficient alternatives, improved conservation or others in a combination that delivers the lowest total overall cost.

- **Determine a hierarchy of energy management measures.** The facility manager is always alert for approaches that support savings or sustainability or provide other means to reduce cost and resource use. This may include developing and ranking a complete list of possibilities relative to their cost, criticality, capacity to pay for their use, value in occupant satisfaction and other factors.
- **Consider energy management opportunities.** Design, repair, upgrade and renovation decisions should be considered relative to energy conservation. As new technologies and more energy-saving systems and alternatives become available, the facility manager must be aware of these potential solutions and apply sound business principles to select among them.

For energy conservation alone, there are many aspects to consider that go beyond maintaining room temperature for human comfort. Effective operations and maintenance demand the continuous delivery of energy, dependable heating/cooling equipment operation, humidity control and consistent maintenance. The facility manager can provide policies and procedures to support energy conservation such as:

- Provisions for using outside air and energy-saving equipment such as fans during moderate weather.
- Installing ventilation systems that are the most efficient possible in terms of their configuration and design.
- Using and maintaining air filters that remove particulates to levels that comply with best practices, air quality codes and occupant needs.
- Instituting and monitoring a program that encourages turning off lights when and where possible and purchasing energy-conserving lamps in bulk quantities.

The facility manager can develop and apply best practices across the entire infrastructure to monitor use and performance.



Ways to reduce energy consumption could include the following:

- Record and track all energy bills; check bills and readings for accuracy; identify energy use by zones to more accurately measure and compare areas, equipment mix and systems.
- Schedule and perform predictive and preventive maintenance to optimize equipment efficiency as well as support higher reliability and better occupant satisfaction and avoid major breakdowns and their productivity and repair costs.
- Plan non-emergency repairs and renovations to integrate with routine maintenance schedules, minimize disrupting occupant activities, and most effectively use resources (personnel, tools, materials) to produce the maximum work result.

The facility manager supports energy management solutions that produce greater energy efficiency in order to realize cost savings, maintain facility productivity and occupant comfort, and support environmental sustainability.

Exhibit 3-8 on the next page is a partial example to demonstrate some considerations when reviewing a facility's resource use. Assessments such as this could be helpful for different facility O&M aspects such as resource conservation (electrical, fuel, water, chemical), indoor air quality, waste management, grounds and soil, renovation and recycling, protecting plant and animal habitat, and conservation engineering practices to save energy and resources costs.

Health and safety inspection

Part of a facility manager's operational responsibility is to oversee regular inspections for the health and safety of the facility and its environment. This includes not only indoor air quality, discussed in Topic 4 of Chapter 2, but also all the general factors relative to health and safety. These are also discussed in Chapter 2 when assessing the condition of the building structure (Topic 1), exteriors (Topic 2), systems (Topic 3), interiors (Topic 4) and grounds (Topic 5).

In general, health and safety inspections relative to the building and its systems and equipment may review:

- Accident prevention (trips, falls, working around systems and high energy equipment).
- Personal protective equipment (such as hearing and eye protection).
- Hazardous chemicals.



Exhibit 3-8: Resource Conservation Assessment

Facility, system and location identification _____ Inspector name and date of assessment _____		Components: electrical devices, fixtures, systems, building structures
General causes or concerns		
Water system ☐ Leaks, losses ☐ Water bills not tracked or compared ☐ Inadequate electrical pump maintenance ☐ No pipe insulation ☐ Waste not managed, properly disposed or recycled	Interior ☐ Electrical items (computers, photocopiers, room lights) not off when not in use ☐ Thermostat not set or monitored for human comfort ☐ Not replacing incandescent with energy-saving bulbs ☐ Not tracking energy use of appliances and devices ☐ Photoelectric cells not used to automatically turn off lights in vacant spaces such as storerooms, restrooms ☐ No use of solar panels or sunlight for secondary heat ☐ Renovations do not consider separate switching systems and heating zones	Compliance ☐ Local regulations/codes ☐ Area requirements ☐ National/international standards ☐ Required permits ☐ Barrier-free access Fixtures and devices ☐ Energy conserving ☐ Maintained for lowest energy use ☐ Energy-efficient motors, tools, compressors ☐ Security lights reviewed for savings
HVAC systems ☐ Not maintained for optimal use ☐ Energy bills not tracked/compared ☐ Useful life tracked and trends not compared to plan ☐ Radiators impaired by closely set furnishings ☐ Fresh air and door openings not supporting efficiency		
Component evaluation		
Description (type, location, serial number/ID)	Condition (deficiency described for correction)	
_____	_____	
_____	_____	
_____	_____	
_____	_____	
_____	_____	
_____	_____	
_____	_____	



In similar ways, air quality, water quality, recycling and related forms of resource management are the facility manager's responsibility. They are also regulated and monitored for compliance. If a facility or system is shut down or operates unreliably because of a loss of power, poor management of service, unhealthy air quality or some other issue, the economic cost can be significant.

Appearance inspection

As part of maintenance routines and as general rules for staff performing O&M activities, the facility manager may inspect or assess appearance. This may include items such as wall coverings, painting or flooring; cleanliness of restrooms; and general appearance and condition of equipment doors, hinges and hardware. It may be anything that is visible to the observer. These inspections may be formal or informal, separate or part of scheduled maintenance. The general purpose of appearance inspections is to catch forgotten or unattended conditions that might affect the image or functionality of an asset.

Emergency planning and preparation

Emergency preparedness is a state of readiness designed to respond to potential unscheduled events and natural disasters that can affect building occupants, disrupt business operations and potentially impair the physical facility. The facility manager makes preparations for all types of threats whether caused by humans (such as terrorism, workplace violence, medical crises), the environment (such as hazardous spills) or nature (such as storms or power failures) through operation, escalation and publication.

Operation

Operational preparation brings all aspects of the infrastructure—systems, structure, interiors, exteriors and grounds—into a complete plan that assigns responsibilities and develops templates for activities before, during and after an emergency. The facility manager may prepare for a safe response to operations by having a plan that includes:

- Emergency contact list of area responders and support agencies, capabilities and typical response times.
- Security procedures (including building and elevator keys, responses).
- Alert and communication systems such as public address, stairwell intercom, and other emergency warning systems (also called emergency warning information systems).
- Emergency procedures developed and published for operating mechanical, electrical and utility equipment affected by storms.
- Debris and snow removal, deicing, water/flood management.
- Readiness of vehicles, structures and equipment for operation in storms, excessive cold or heavy snow (winterization), excessive heat, unusual crowd conditions.



- Procedures and checklists of required information, communication protocols, recovery and safety elements.
- Emergency communications backups such as battery-operated radios and two-way transmitter/receiver (or walkie-talkie).
- Pathway lighting.
- Emergency equipment and personal protective equipment, locations and assignments.
- Information prepared and conveniently placed describing weather-related or environmental possibilities (such as heavy rain, winds, snow, cold or heat) and potential effects (mudslides, flooding, entrapment, exposure and so forth.).
- Emergency and standby power systems.
- Communication from the work management center.
- Operations checklists relative to emergency actions for heating, cooling, traffic flow (pedestrian and vehicle), air-handling and harmful gas exhaust and so forth.
- Emergency directions for occupants or visitors based on building locations, storm situations and evacuation plans.
- In-place procedures for structures in the event that evacuation is incomplete or delayed.
- Emergency electrical, heating/cooling, water and injury response capabilities.
- Emergency equipment cache such as first aid, portable lights/generators, sandbags, tarpaulins, shovels, axes, lumber/plywood and nails, water, ready-to-eat food, infant supplies, clothing, bedding, sanitation supplies.

Escalation

From an emergency operations standpoint, escalation is the preplanned determination of how to respond to specific conditions with prescribed levels of intensity. An emergency can escalate from an incident with possible loss, to a disruption of normal function, to a dangerous situation, to a crisis in which an unsustainable condition requires immediate, urgent response. The intensity of a response should be appropriate to the intensity of the emergency.

In the case of most emergencies, especially those of greater intensity, response generally proceeds through stages such as:

- Alert.
- First response for safety.
- Operational security.
- Public welfare (occupants and visitors).
- Recovery.



The facility manager prepares for these stages with an escalation process in which greater emergencies produce a faster and more intense response. This preparation would likely include:

- Prescribed indicators that trigger response levels. (For example, a fire in a wastebasket would not require the same response intensity as a fire near fuel storage.)
- Roles and responsibilities assigned to individuals who are trained in their requirements.
- Rules of engagement that provide responsible individuals with clear communication relative to threat level.
- Response approach and activities relative to the nature of the emergency. (For example, the response would be different when detecting a harmful gas release than an approaching hurricane.)

Publications

The facility manager not only prepares for emergencies but also organizes and publishes those preparations so that others in the organization are competent in their roles. This requires including other personnel active in the facility such as administrators, managers, business overseers, floor supervisors, and contract workers or staff who are typically on site such as custodial, food service, waste management and other services.

Publications may include these general instructions:

- Notification and alert descriptions
- Individual responsibilities and expected reactions (including equipment, system and security management)
- Evacuation procedures
- Human safety and public welfare practices
- Needed lifesaving skills and responder capabilities

Publishing emergency plans not only helps the facility manager prepare for contingencies but also involves the entire organization in carrying out a controlled, safe and efficient response.

Material handling and storage

Building systems and equipment as well as other operations may require material handling and storage systems, activities and processes. Some examples of these include:

- Chemicals, spare parts, and related emergency or supplemental equipment or components.
- Waste stream management.
- Loading docks and material management.



- Warehouses and other storage facilities.
- Audits and reconciliation of inventories.

Snow and ice removal

In cold weather climates, facility managers oversee the prompt removal of snow and ice to:

- Provide ready access to the facility for continued productivity and the convenience of occupants and visitors.
- Maintain safe passageways for vehicles and pedestrians free of slippery surfaces.
- Remove potentially damaging buildup of snow and ice on roofs.

Policies, procedures and safety

Facility managers help create, produce and continually update standards and expectations for maintaining the infrastructure as well as occupant services. This information is written down and made available to staff and contractors in a tenant services manual or similar document that is prepared for and provided to facility occupants.

Policy and procedure manuals, safety manuals and other kinds of documents supply clear directions, systematic thinking, proper courses of action and so forth. They exist not only to support proper, efficient O&M but also to represent the foundation from which performance expectations and tasks can be determined and measured.

★ Topic 5: Replace Systems, Materials or Equipment

Replacement defined

Replacement is the exchange or substitution of one fixed asset for another that has the capacity to perform the same function. The replacement may arise from obsolescence, wear and tear, functional failure or destruction of the item being replaced. Replacement involves a complete identifiable item, while, as basic distinctions, repairs are curative and maintenance is preventive. In the sense of operational maintenance, the replacement of a component, piece of equipment or system does not merely mean to remove it, discard it or throw it away. For the facility manager, replacement signifies the end of the asset's intended operational use and a plan for what to do with it next.

Well-planned replacement also represents a systematic approach to sustainability. It consists of several elements or actions:

- **Reuse**, which uses the product more than once such as using maintenance rags rather than paper towels
- **Reduce**, in which the quantity of items or outputs is decreased, such as by using less packaging material or other disposables, energy resources, or single-use products



- **Recycle**, so that a product or item is reprocessed into a new raw material to be used as a new product
- **Salvage**, in which existing materials or equipment are reapplied in a productive way for another purpose without being recycled or otherwise reprocessed
- **Disposal**, in which an item may be scrapped, treated as waste or otherwise discarded

The terms “disposal” and “salvage” may also be understood as methods to organize or order the use of items. This meaning should not be confused with how their descriptions are intended here.

Conditions for replacement

At some point in the life cycle of any component, system or piece of equipment, the facility manager will need to determine whether to continue to maintain that item, replace it or upgrade it as an entirely new installation. The ability to make that determination can begin when the original item is acquired. As part of the life-cycle cost analysis, the facility manager takes into account the estimated useful life of the item and thus recognizes the potential time to replace it.

In general terms, an asset is considered no longer useful for its original purpose when the:

- Wants or needs of occupants and users exceed the capability of the asset to provide for that need.
- Conditions of its installation or intended use change.
- Cost of ongoing maintenance and repair exceeds the cost of replacement.
- Integrity or productivity of other facility elements is compromised by the continued use of the asset.
- Operating potential of a new replacement asset justifies replacing the current asset, for example, new products or technologies produce measurably more effective results, environmental sustainability or occupant satisfaction.

By comparing alternative items, the facility manager may consider:

- Lifetime energy and resource costs.
- Lifetime operations, maintenance and repair costs.
- Replacement costs.
- Residual value.
- Scrap, disposal or handling expenses (for example, fees to manage hazardous elements).
- Health and safety considerations (such as hazardous materials, chemicals and preservatives in the asset’s construction, toxicity at disposal, compliance codes, environmental impact and waste management).



Other business considerations also contribute to determining when to replace an asset, such as:

- Whether to repair versus replace depending upon the estimated costs and anticipated benefits between those choices.
- Annual cost of ownership versus cost to maintain as a long-term life-cycle cost analysis.
- Energy use as an expense for operating utilities with less efficiency or outdated technology.
- Sustainability in which materials, equipment or systems are initially chosen because their creation has minimal long-term effect on the environment. Decisions made when acquiring an asset in the first place also generate better options for the asset's eventual replacement.
- Reliability relative to the need for maintenance, frequency and time that maintenance requires, potential costs of downtime caused by breakdowns and so forth.
- Performance measured in productivity as well as durability of the equipment or system in operation.
- Image or aesthetics in which an outdated but operational system or machine no longer projects the impression ownership desires.

While there are no concrete rules for determining this transition, a general threshold may be that when the annual cost of maintaining an asset exceeds the estimated annual cost of ownership based on life-cycle costing, the facility manager should consider replacement. More information on planning approaches can be found in the *Finance and Business* course.

Managing disposal

Facility managers administer, organize, publicize and monitor sophisticated materials management programs to identify, collect and dispose of assets at the end of their useful life. This includes not only parts, components and materials but also the supplies, chemicals and waste products that are produced by them. In this interpretation, there are two aspects regarding disposal: asset disposal and waste product disposal.

Asset disposal

When it is time to decommission and/or remove an asset from service, the facility manager considers several options:

- Resale to another party
- Reuse in another fashion (for example, salvaging renovation lumber for use in a second application or more directly somewhere else in the organization)
- Recycling, scrapping (paper, plastics, metals, glass, packaging)
- Donation



Some facilities offer warehouse or drop-off areas where replaced items are gathered for occupants and visitors to purchase at vastly reduced prices or even take home free as a benefit of being an occupant. These resales can be an opportunity to dispose of replaced or damaged items in a more popular, useful and environmentally supportive way.

Equipment, furniture, building materials and system components may have some residual value instead of being thrown away as trash. Typical examples of recycling or reselling items could include:

- Computer parts and equipment.
- Copper plumbing, conduit, tubing, etc.
- Electronics.
- Used furniture and fixtures.
- Kitchen equipment and other specialty items.

The goal of replacement is to obtain the best overall value at the end of the component's life cycle through a business return, environmental savings, further adaptability or other accepted measures. It does not necessarily mean achieving the highest monetary return but instead producing the greatest total return.

Waste product disposal

Programs that support removing the byproducts of a work environment generally include these key aspects:

- Convenient collection points (conveniently near to users and clearly labeled)
- Regular collection and removal
- Services and/or approaches for sorting and efficient management
- Regularly solicited feedback on service levels and occupant satisfaction
- Accounting practices to measure and compare costs, savings, returns and sustainability results

Environmentally appropriate waste management programs recognize, define handling guidelines, and properly dispose of anything the facility may use or create, such as:

- Hazardous waste (solid, liquid, gas).
- Industrial waste (chemicals, paints, preservatives, grease, lead, mercury [fluorescent bulbs], refrigerants [CFCs or chlorofluorocarbons], asbestos-containing materials or ACMs).
- Construction debris (brick, concrete, wood, gypsum).
- Furnishings (materials, textiles, fixtures, batteries, tritium exit lights [which contain radioactive material]).
- Equipment waste (electronics, metals, plastics, packaging).
- Biodegradable waste (vegetation, composting, water management).



Progress Check Questions

Directions: Read each question and respond in the space provided. Answers and page references follow the questions.

1. The life cycle of a typical physical asset
 - ☐ a. describes the entire useful life of that asset.
 - ☐ b. is unknown until the asset fails.
 - ☐ c. should not be considered when determining cost of ownership.
 - ☐ d. is between 10 and 15 years.

2. The hierarchy of sustainability ranks recyclable resources above _____ resources.
 - ☐ a. perpetual
 - ☐ b. renewable
 - ☐ c. reducible
 - ☐ d. natural

3. The installation phase of an asset's life cycle
 - ☐ a. may require commissioning.
 - ☐ b. marks the start of that asset's useful life.
 - ☐ c. should be completely documented.
 - ☐ d. All of the above

4. Which of these best describes the process of commissioning?
 - ☐ a. Paying incentives to contractors for superior work
 - ☐ b. Evaluating and documenting performance prior to operation
 - ☐ c. Providing rewards to staff such as special privileges
 - ☐ d. Running equipment to failure before replacement

5. Vibration monitoring, airborne ultrasonics and laser shaft alignment are all examples of
 - ☐ a. air quality concerns.
 - ☐ b. equipment failure.
 - ☐ c. PdM analysis tools.
 - ☐ d. root cause analysis requirements.

6. Periodic inspection, adjustments and lubrication are examples of
 - ☐ a. unscheduled maintenance requirements.
 - ☐ b. poor maintenance practice.
 - ☐ c. mission-critical predictive maintenance.
 - ☐ d. preventive maintenance activities.



Match the following analysis tools with their description.

- | | |
|---------------|--|
| 7. _____ RCM | a. A process to discover the systemic cause or root cause of a problem |
| 8. _____ RCA | b. A detailed analysis to help determine failure probabilities and system reliability calculations |
| 9. _____ FMEA | c. A maintenance philosophy of optimizing reliability and availability at the lowest cost |

10. Which of these activities would be considered unplanned maintenance?

- () a. Corrective maintenance
- () b. Run-to-failure
- () c. Emergency maintenance
- () d. All of the above



Progress check answers

1. a (p. 64)
2. c (p. 68)
3. d (p. 70)
4. b (p. 70)
5. c (p. 78)
6. d (p. 76)
7. c (p. 82)
8. a (p. 84)
9. b (p. 83)
10. d (p. 81)



Chapter 4: Manage/Oversee Occupant Services

After completing this chapter, students will be able to:

- List typical occupant services and describe the process of determining guidelines for occupant service performance expectations.
 - Describe a typical workflow process for occupant services.
 - Explain the key aspects of a successful occupant services plan.
 - Describe the guidelines that support the workflow process.
 - Describe an effective approach to preparing for modifications to occupant services.
-

★ Topic 1: Recognize Required Occupant Services

Occupant services defined

The term “occupant services” refers to anything that facility occupants or visitors might need beyond the operational maintenance of the hard assets of the building and its systems. These occupant (or soft) O&M services are the critical activities through which the facility manager directly affects the users of the building. When competently done, these asset-based activities are largely invisible to occupants, yet the occupant services themselves are highly visible and present the image of the organization to the rest of the world.

Occupant services should align with the organization’s values. Since these services are the main interface with the organization’s own customers, their quality and consistency imply the facility manager’s competence and value. Any failure in service to occupants is immediately apparent and detracts from the organization’s mission as well as its image.

Examples of occupant services

Occupant services may occur facilitywide (such as waste removal or custodial cleaning) or be applied to specific areas (such as information security or child care). Potential occupant services for a facility could include those listed in Exhibit 4-1 on the next page. This partial list serves to demonstrate the variety of services a facility may be expected to provide. O&M for occupant services extends far beyond heating, cooling, water and power.

To manage/oversee these services, the facility manager needs to identify what services facility owners, occupants or guests may require. In addition, the facility manager must determine what staff or contracted party should supply them.



Exhibit 4-1: Occupant Services

- | | |
|---|---|
| <ul style="list-style-type: none"> • Custodial (light cleaning, vacuuming, dusting, trash removal, restroom refreshment) • Cleaning and refurbishing (window washing, floors, fixture replacement) • Waste management (collection, recycling, incineration, hazardous materials, shredding, secure destruction) • Food service (coffee, ready-to-eat, cafeteria, dining rooms, banquet) • Parking (pickup, delivery, vehicle security) • Fleet (vehicle cleaning, maintenance, acquisition, lease, resale) • Travel (reservations, itinerary, transportation) • Concierge (facility help desk, greeting, directing, support for individuals) • Collections or curator (art, sculpture, display management, security) • Mail (pickup, distribution, packaging and shipping, fees, tracking, insurance, messenger) • Shipping and receiving (pickup, packaging, transport, routing, tracking, insurance) • Managing relocations or moves and changes (MAC), which can include packing, shipping, unpacking, sorting, setup and so forth) • Information technology (equipment procurement, maintenance, disposal, information management such as e-mail and server support, information security) | <ul style="list-style-type: none"> • Copying, printing, document management • Records management (media, files, security, disaster recovery) • Finance (accounting, taxes/fees, reports) • Child care (facility, nursery, preschool, before- and after-school support) • Health (nurse services, benefits administration, counseling, fitness) • Human resources (recruiting, staffing, relations, training, compensation) • Engineering and trades (specialty skills such as air conditioning technicians, design/fabrication/repair expertise, inspection) • Landscaping (grounds maintenance, planting, design and so forth) • Security services (personnel, watch, and response separate from the systems themselves) • Communication (line telephones, cellular phones, fax, wire and wireless communication separate from network systems themselves) • Emergency preparedness (fire, medical, disaster, terrorism training, practice/drill, triage) |
|---|---|

★ Topic 2: Develop Usage and Service Level Guidelines

Recognize occupant needs

O&M planning relative to occupant services is the process that produces facilities that occupants find comfortable, safe and effective. While a facility manager may not be responsible for deciding which occupant services are provided, it is important to understand how they are determined. Thus, the facility manager evaluates current conditions to be assured that the desired level of service (LOS) is provided.

Relative to occupant services in facilities, LOS refers to the amount and quality of service provided to meet occupant or customer needs without hindering the



mission or profitability of the facility. Level of service can vary widely based on budget, expectations and other aspects.

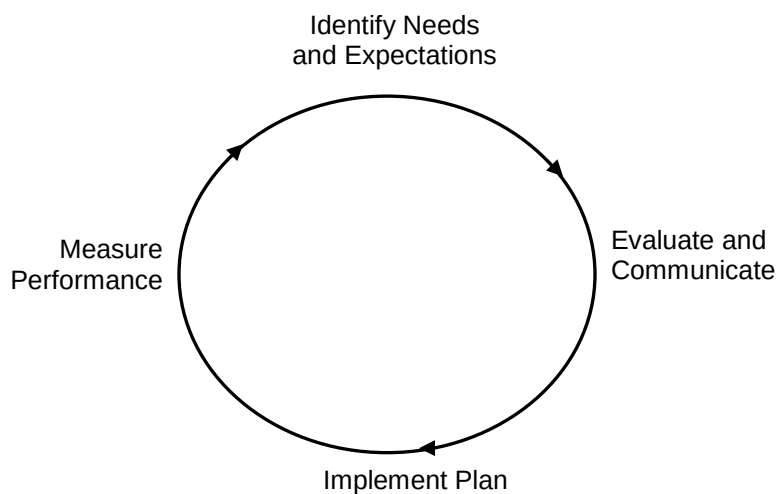
Along with LOS is the mix of services in a facility. The selection of services is usually based on the form of the facility and its functions, economic conditions, the organization's culture and other factors. By recognizing how services are determined, the facility manager has a better understanding of what level of service to provide and how to determine if the service is sufficient.

To determine guidelines for occupant service performance expectations, the facility manager will likely:

- Identify needs and expectations.
- Evaluate and communicate.
- Implement plan.
- Measure performance.

Exhibit 4-2 illustrates these aspects of occupant service planning and operations.

Exhibit 4-2: Typical Occupant Service Planning Approach



Identify needs and expectations

The facility manager needs an accurate accounting of occupant services to be assured that the facility is supplying the right services in the right amount so that occupants are satisfied and the mission of the facility is not compromised. If the facility manager is new to the position or there is not enough information about what occupants expect, he or she must assess the condition.



Occupants can provide an invested and relevant viewpoint on the state of the facility. They are the core contributors to the facility mission. They represent the basic capability of the facility to sustain its mission. This makes their satisfaction with their environment significant to the overall purpose of the facility.

The facility manager must learn to listen to occupant opinions about facility services. By soliciting feedback from occupants and attending to what they say, the facility manager learns about problems in the early stages and does not wait for complaints to accumulate.

As a source of information on the quality, reliability, efficiency and effectiveness of the facility they are in, occupants' opinions can expose gaps in expectations, needs and the current level of service. A rapid, responsive approach helps ensure that service quality meets occupant expectations.

The facility manager must develop and implement plans and policies that reflect occupant expectations while supporting management objectives. To determine those expectations, the facility manager can begin by surveying groups and determining acceptable standards.

Survey for input

Several different groups might be surveyed to develop an understanding of service needs. For example, the facility manager might canvass:

- **Occupants** by directly talking to some or all, preparing and conducting a survey, and/or sampling key occupants and visitors.
- **Owners and executives** by survey or interview to determine how services are meeting or not meeting the current vision or strategic facilities plan.
- **Visitors or guests** with simple response forms, brief on-site interviews, and/or follow-up satisfaction surveys to learn their opinions on the variety, mix and quality of services.

In addition, facility managers must continue to conduct surveys to stay current on occupant opinion. Feedback from occupants on service needs and expectations can be reviewed over time to detect trends or potential adjustments. This information can help the facility manager predict new needs or the impact changes may have on existing building systems, structures, interiors, exteriors and grounds. This is possible only if the facility manager:

- Consistently gathers information and opinion.
- Compares it to prior inputs.
- Looks for gaps and opportunities.



Determine and apply standards

Input from contractors, staff, occupants, visitors, owners and executive management can provide the full range of important data that the facility manager uses to establish standards for the level of service, aesthetic quality, facility efficiency and other operational requirements. From this input the facility manager can develop descriptions of services that clarify what current services exist and what they provide and inquire into what they might or should provide.

These standards become the basis for measuring results and are a way to identify trends that might suggest the need for change. This understanding of needs can be formalized into an occupant service plan discussed later in this topic.

Evaluate and communicate

The delivery of occupant services relates to both the assignment and the performance of occupant services O&M. Scheduling and performance are only mentioned here because they are described more fully later in this chapter as the services workflow process. However, evaluation and communication also relate to how inputs and complaints are handled.

Initial interviews and surveys provide a preliminary level of input. It is wise to respond directly and quickly to any expressed dissatisfaction; the facility manager's request for information may also cause the responding individual to expect further action and feedback.

Direct and straightforward communications can help reduce or eliminate routine complaints. Rapid, focused response also encourages relationships that can help avoid future conflicts or support better understanding.

The facility manager must know and teach maintenance personnel how to respond to an occupant expressing a problem. That complaint may signal the first time a weakness in service is occurring. For example, an occupant might mention a noise coming from a vent that might mean a component has failed in the air conditioning system. This information provides early recognition that can prompt a timely solution before problems escalate or dissatisfaction increases.

Communication not only occurs between occupants and the facility manager but from the facility manager to others responsible for the facility, including owners and executives. The facility manager can use inputs and evaluations to promote and defend the budget for occupant services and earn approval from management based on clearly defined and mutually understood expectations. A well-constructed plan with measurable objectives also gives the facility manager a sound presentation to executive management.



Implement plan

An occupant services plan should formalize service satisfaction requirements so that expectations are clear. This helps to make the delivery of occupant services measurable, consistent and of recognized quality just as with O&M services that sustain the building, systems and equipment.

There are many implementation aspects, such as the following:

- Service requirements are documented and agreed upon.
- Response criteria are perceived as effective and in the occupants' interests.
- Service complaints and feedback are responded to efficiently and in a timely manner as perceived by occupants.
- Dissatisfaction expressed by one or more occupants is quickly addressed, respectfully answered, and managed to avoid escalation.
- Planned moves, changes or other facility disruptions are identified, communicated and executed to minimize interruptions to occupant activities.
- Service requests are reviewed, prioritized and executed with the same care and completeness as other O&M work orders.
- Other technical and quality factors are included to maintain satisfactory results relative to recognized performances. These can include:
 - Total quality management (TQM) or a similar approach that systematically and consistently examines and integrates all organizational functions such as operations, finance, engineering, training and so forth to add value to occupants, visitors and facility personnel by solving problems early or even before they start, if possible.
 - Indoor air quality in which the air inside buildings is satisfactory relative to pollution from dust, mites, mold spores, radon, or noxious gases and chemicals from materials and appliances.
 - Fire life safety in which construction, protection and occupancy features are in place to minimize danger to life from fire, including smoke, fumes or panic.
 - Others as directed by ownership and/or the facility manager.

The value of developing a process to determine and deliver occupant services extends beyond the occupant service flow itself. By determining occupant needs and measuring expectations, the facility manager accrues valuable information concerning the facility itself and how to better manage it.



Measure performance

Just as facility managers monitor O&M infrastructure activities to promote continuous improvement, they also monitor occupant satisfaction to detect deteriorating or unsatisfactory service activities as early as possible. This helps corrective action to be the least invasive, expensive and challenging. By combining and comparing all inputs on services, the facility manager can develop an accurate accounting of how well existing services are performing, what new services might be helpful, and what level of value or priority each may have.

Performance can be evaluated and improved by monitoring parameters, checking progress and correcting variances.

- **Monitor parameters.** By monitoring operational results, including occupant satisfaction, service levels relative to expectations, and other indicators, the facility manager can detect when a change in measurements may indicate a deteriorating condition.
- **Check progress.** Reports, periodic visits, checking back on corrected actions, and other communication can help the facility manager not only maintain improved conditions but also develop insight for future space or service needs. Regular, consistent communication with occupants is important to facility management. For example, by understanding owner/management policy and gathering ongoing opinions from occupants, the facility manager may discover the need to expand or change custodial services. Concerns may come to the facility manager's attention early, giving time to most appropriately respond to quality issues or add new custodial services for a more complete program.
- **Correct variances or variations.** "Variance" can have a technical, mathematical definition, in which it indicates the distribution and relative values of data points around a mean or average value. For the purposes of assessing occupant services, the facility may consider a variance to be the amount of change or departure from an expected result. This implies that the difference is measurable and the change is recognized through some pre-established standard.

For facility occupant services O&M, the service level agreement can set the standard. It is assigned, measured, executed and monitored through the work plan and work logs that should account for how all labor time is spent.



Occupant services workflow process

The facility manager uses these inputs along with life-cycle cost analysis and specific performance indicators to determine how well occupant services goals are achieved.

Delivering satisfactory operations and maintenance to occupant services generally follows a work plan with steps similar to the approach for O&M for hard systems and equipment as described in Topic 3 of Chapter 3. Both work plan approaches are summarized in Exhibit 4-3 to demonstrate that a typical workflow process parallels building, systems and equipment O&M.

Exhibit 4-3: Operations and Maintenance Workflow Process

Occupant Services (Soft)	Building and Equipment (Hard)
• Assess for occupant needs • Schedule services • Assign tasks • Perform work • Return results • Input data • Measure performance	• Assess/devise inventory/register • Schedule maintenance (preventive maintenance, help desk request) • Assess and assign work • Perform work • Return results • Input data • Measure performance

A workflow process such as this provides the basis for effective, efficient delivery of occupant services. The facility manager monitors workflow to be aware of how the work is going. Whether occupant services are contracted or performed by staff, the facility manager needs to know where work is occurring, how the projects are tasked, and what priorities or exceptions exist so that all O&M activities in the facility are coordinated and checked for efficient use of personnel, scheduling conflicts, redundancies and prioritization. In this way, the facility manager can:

- Maintain order.
- Stay abreast of all tasks going on in the facility.
- Exercise control without being invasive.
- Be available to handle exceptions as they arise.

All workflow activities should reflect maintenance management standards such as the following:

- Productivity is enhanced when each individual is assigned definite job parameters and is competent in accomplishing the assignment.
- Measurement should precede control so that what is mutually understood is the basis for how results are measured against goals.



- The work order should be the basis for all facility maintenance, operation, alteration and renovation activities.
- Wherever possible, work orders should describe trades, tools, materials and tasks, preferably by hours.
- Occupant satisfaction should be considered a key measurement for successful facility operations and management.
- Team size is considered optimal when the fewest people can provide the desired speed and quality of results in a safe way within the required time.
- Priorities should be both defined and compared so that work planning provides the most simple, reliable and manageable activity possible.
- Visits or other oversight should be scheduled at control points that occur frequently enough to detect problems yet not so frequent that the work itself is unnecessarily interrupted or delayed.
- Specifications that measure outcomes are the best work appraisal and offer the most neutral, reasonable and acceptable approach to successful work oversight.

With these principles underlying the approach to maintenance and service level standards, the facility manager has both the tools and the discussion points to competently oversee the work as it is performed and accomplished. The basis for these principles should arise from facts, not opinions, determined by quantitative information.

Guidelines support the workflow process

The workflow process provides a way to merge, prioritize and coordinate O&M activities to help the facility manager know where, when and how to maintain physical assets and serve occupant needs. The work plan determines work execution across all maintainable assets for the entire infrastructure in support of occupant services. These guidelines can generally be described from two main aspects: facility environment and facility systems.

Facility environment priorities

The facility manager may determine how to measure for quality by applying guidelines such as the following:

- **Safety.** Safety is usually regulated by outside agencies that establish a minimum level of compliance. The facility manager must be aware of the requirements and restrictions made by local jurisdictions and must be diligent in meeting those minimums.

The facility manager, owner or occupants may require a level of safety higher than the minimums require. For example, because of terrorism or



the threat that a nearby dike or hillside might collapse in bad weather, the facility manager may schedule emergency exit drills several times a year even though the local standard may not require such frequency.

- **Cleanliness.** The cleanliness of a facility is measured by more than the absence of dust or trash. As an environmental aspect, cleanliness relates to the condition of filters, vents, traps and other maintenance aspects that are significant to the efficiency and air quality of the work space. It may also represent custodial requirements that support the needs of the facility, its operation and its reputation. Cleanliness can become a serious issue when, for example, poor cleaning maintenance results in such high levels of disease-carrying bacteria that occupants become sick or debilitated. Then cleanliness is a productivity and safety issue.
- **Comfort.** Reliable room temperature and water and electricity supply, useful furnishings and fixtures, and adequate space for working are all environmental comforts that may become a serious issue if occupants voice dissatisfaction or facility efficiency is compromised.
- **Accessibility.** Barrier-free access is generally regulated by code requirements. However, beyond required minimums the need for accessibility in the facility can reflect positive, productive workflow so that work areas are not overly cramped, aisles and hallways are convenient, and storage and service areas are easy to reach and use.
- **Productivity.** Interior design and the reliability of utilities and systems are all considered by the facility manager. Decisions about these relative to the working environment will depend upon what productivity factors the owner, facility manager and occupants consider significant and should be included in the work plan.
- **Energy savings.** A facility mission determines the concern and attention given to reduce energy use or develop structures, systems and methods that meet strict requirements. Some facilities may put high priority on producing the highest possible sustainability through environmentally suitable construction. This interest could make the energy-saving environmental factor a top priority and thus change the priorities of other factors in relationship to it.
- **Aesthetics.** Use of color, light, design, quality materials and many other aspects may be an important consideration for a facility. If aesthetics are a



priority, it may also mean that factors such as comfort, cleanliness, and safety would also take on higher significance in the relationships among factors.

- **Replacement/renovation.** A facility that is undergoing physical changes because of obsolescence, deterioration, new occupants, new mission, or changed productivity and comfort priorities may decide renovation is a high priority. This decision can affect other relationships to the building environment such as raising the priority of energy savings, aesthetics or comfort.

The facility manager is tasked to conserve resources and provide a quality environment at the lowest possible cost. To do this requires the insight gained by relating all of the factors just discussed and then devising an O&M work plan that helps the facility manager execute, monitor, assess and make changes when necessary to meet performance expectations.

It is unusual for a facility manager to have the luxury of a virtually unlimited budget and the ability to place all environmental considerations at the highest, critical level. More often, the facility manager may use relational tools to plan work activities that support the relationships deemed most important without sacrificing those deemed less important.

Facility systems priorities

When devising the work plan, the facility manager also considers factors in comparison with processes, systems and occupant expectations. This adds another layer of information and capability. The facility manager uses these comparisons to decide how to detect and correct inadequacies in systems within the total infrastructure in ways that best support the facility mission.

There are several different kinds of systems considerations that are all possible approaches to work plan development. The facility manager must review and compare them to determine the best mix to reflect in the work plan:

- **Reliability.** Equipment and systems that support occupant work should certainly be reliable. The facility manager may prioritize different assets for reliability to most productively support the mission.
- **Productivity.** The productivity of the facility can be a highly significant factor based on how systems influence the environment. For example, an electrical service outage caused by a frozen bearing that overloads a circuit may have been prevented, but instead a production line is shut down, resulting in lost productivity.



- **Maintainability.** Access to service points, availability of tools and checklists, and a precise schedule of maintenance all play a role in how well systems are functioning.
- **Redundancy.** Backup capability or the ability to operate or compensate during failures may be a high priority if the facility environment has a life-critical mission or if productivity is important and downtime is costly.
- **Instructions.** Clear, measurable and complete maintenance instructions are important to keep systems operating at maximum effectiveness. The inventory/register of all infrastructure assets should be current and specific if the work plan itself is to be efficient and effective.

Systems considerations such as these are part of what the facility manager considers, prioritizes and relates to environmental factors. They may be applied to virtually any O&M responsibility, whether it is repairs, parts replacement, routine service or other activities.

For example, custodial cleaning is a prevalent occupant service that is an expensive and ongoing challenge. In addition, cleaning or custodial services typically produce high injury rates relative to other occupant services, which aggravates the issue. A facility manager may be overly concerned about costs and which cleaner to use, yet only marginally consider performance requirements and if they are being met. This represents poor prioritization. The following discussion expands on custodial services as an example.

Custodial cleaning example

The facility manager must develop and incorporate cleaning functions as diligently as other O&M tasks to help produce an environment that encourages productivity, is safe, satisfies occupants, meets building regulations and derives as much value as possible from the facility as the organization accomplishes its mission. This requires considering a broad range of aspects in a proficient custodial cleaning program, including the following:

- **Cleaning benchmarks.** The facility manager establishes a clear understanding of what is required so that personnel know when the work is properly executed. It involves developing benchmark expectations that are specific and measurable.
- **Cleaning levels of service.** The level of service defines expectations or possibly benchmarks that represent minimum requirements relative to response, availability, performance or other attributes of service, such as



occupant satisfaction or communication. These conditions allow everyone (contractor, staff, occupants, facility manager, others) to understand what service is necessary and how it will be measured. For example, APPA (The Association of Higher Education Facilities Officers) has established custodial staffing guidelines that define five levels of what is clean as a way to assess facility appearance. These guidelines are also adopted by some industry associations.

- **Staffing levels.** The APPA Custodial Staffing Guidelines are an example of a benchmark resource that the facility manager can apply to help decide how many custodial personnel with what types of skills are needed. These criteria may be based on the facility's mission, occupant mix, size, level of service expectations and other associated characteristics.
- **Performance contracts.** Developed within a service level agreement or as a separate set of requirements, contracts hold personnel accountable to specific levels of service, cleanliness, hygiene or other conditions. These contracts may include rewards, penalties and other incentives based on performance. A neutral or third party may be used to audit performance so that both the facility manager and the service provider are treated equally in the assessment.
- **Team versus zone cleaning.** The facility manager can play a role in determining whether custodial work is accomplished by a team or by an individual working a zone.
 - When cleaning as a team, each individual is assigned one task (such as light duty dusting/spot cleaning, vacuuming, or restrooms) and that is their responsibility for the facility, the floor or the area. Team cleaning provides a systematic approach in which individuals are proficient at their assigned task and the work is expeditious.
 - Zone cleaning is essentially the reverse of team cleaning. In zone cleaning, one person performs all the required tasks in a specified cleaning area. Zone cleaning gives the individual a sense of ownership and pride in the quality of their area.
- **Green cleaning.** Green cleaning refers to the practice of using cleaning methods, products and ingredients that preserve health and the environment. A green cleaning approach avoids using chemically reactive and toxic cleaning products or those that emit volatile organic compounds. Green cleaning also refers to cleaning products themselves that are



manufactured, packaged and distributed in more sustainable and environmentally supportive ways.

Green cleaning should not be confused with green washing, a term that is meant to criticize misrepresenting a product as having green benefits. For example, green washing would refer to a product or approach presented as a way to reduce using resources but misrepresented in order to justify cutting costs or eliminating steps.

- **Handling materials.** A broad aspect of a typical custodial service includes how to safely handle cleaning products, how to handle hazardous materials, and how to manage waste. This is a major topic that is too encompassing for this O&M discussion; however, these are important considerations for the facility manager.

Custodial personnel should be well trained on how to recognize and handle hazardous materials and properly dispose of waste. In most areas there are strict standards for managing, recycling and disposing all materials.

✦ Topic 3: Prepare and Execute Modifications to Occupant Services

Preparing for modifications

Facilities may need to be modified for many reasons, such as:

- Turnover of occupants, which refers to the movement and relocation of occupants within an organization.
- Expansion or contraction of facility equipment or space needs.
- Changes in facility mission involving a reordering of priorities, systems, spaces, personnel and more.
- Moves to different facilities or floors or expansions/contractions of current occupants.
- Newly defined needs because of rearrangement of current facility spaces, travel paths, communications and so forth.
- Legislative or regulatory changes that require changes in facility systems, layout, service arrangements, etc.

The facility manager uses the information gathered from periodic conversations with occupants, surveys and interviews to maintain an ongoing understanding of occupants' wants, needs and expectations. This also helps the facility manager establish relationships with occupants as a means to



communicate anticipated modifications that may arise if there are changes in management plans or a facility's mission.

Approach

To prepare for modifications to occupant services, the facility manager may use an approach similar to the way in which routine O&M activities are prepared for, scheduled and performed. The approach applies key aspects that allow the facility manager to do the following:

- **Identify modifications required** by assessing feedback from occupants, owner/management and staff to determine what the modifications are and any special requirements, such as occupant schedule, critical activities or other factors, that are vital to the facility mission and cannot be compromised.
- **Determine assets needed** such as equipment, systems and components; materials, tools and trades; or special expertise such as structural engineering, architectural drawings or equipment schematics.
- **Identify resources needed** including team sizes, numbers, and skills, specialties or trades.
- **Prepare the work plan** that assigns work to specific individuals/teams, with clearly defined tasks sequenced so that the right skills and personnel are accomplishing work in the right order.
- **Communicate the schedule** to all service providers (staff and contract) as well as all occupants affected by the modification. Provide feedback systems for input and questions before, during and after the modification.
- **Perform the modification** within assigned specifications and mindful of other O&M schedules to avoid delays, interruptions or unnecessary conflicts and costs.
- **Document** all assets, specifications, manufacturer warranties and service schedules as part of the total infrastructure inventory/register.
- **Monitor results** for routine, ongoing occupant feedback and continuous improvement.

With routine occupant service modifications as well as systems/equipment modifications, the facility manager is focused on providing responsive, high-



quality results. This includes capabilities and competence in areas such as the following:

- Well-defined procedures, specifications and organizational structure.
- Clear lines of communication to occupants, staff and contractors.
- A centralized, organized WMC that communicates a work plan that aligns skills, tools, materials and equipment with the modification schedule and with ongoing operations, maintenance and alterations activities.
- Follow-up on the completed work to ensure that it is checked, inspected, accounted against budget, analyzed and measured.
- Feedback from all involved parties (occupants, staff, contractor, owner, and so forth) to:
 - Determine the success of the modification.
 - Uncover gaps.
 - Resolve disputes.
 - Maintain quality management principles for continuous improvement.

Alterations

Alterations refer to the work required to change the interior arrangements or other physical characteristics of an existing facility or of installed equipment so that it may be more effectively utilized for its currently designed purpose or adapted to a changed use as a result of a programmatic requirement. Alterations may include work referred to as improvement, conversion, rehabilitation, remodeling or modernization.

Alterations may occur almost continually in many facilities because of moves, upgrades, aesthetic improvements and mission changes. Funds for alterations are generally kept separate from those for maintenance so that budgets are discrete and accountable.

Move management

Alterations are sometimes included within a larger concept known as move management, in which alterations may not be required yet facilities undergo changes because occupants move or leave or work areas change or grow.

This concept may be measured as churn. “Churn” is a term widely used in facility management to describe the amount of movement and relocation of occupants within an organization within a specified period. It is defined as the ratio of the number of employees moved annually compared to the total number of employees in an organization.

Higher churn rates (greater than 50 percent) generally indicate rapid growth and/or significant change. Rates this high may notably affect the facility



manager's operating and project budgets as well as require excessive time to oversee activity and manage resources.

Moves can range from simple to complicated and may require boxing and transporting items, moving furnishings or moving entirely to accommodate construction or renovation. There are too many aspects of move management to cover in this discussion; however, critical stages of the process include the following:

- Defining scope
 - Why the relocation/alteration is important to the facility/owner
 - Who is moving
 - Who will fill that space
 - What will be moved
 - Where items will be moved
 - When moved items are needed
- Establishing the relocation/alteration team (including staff, contractors, administration, purchasing, security and so on)
- Determining the budget
- Determining the schedule
- Collecting data on current conditions (coding and identifying items, descriptions, condition, location, assignment)
- Performing the move
- Evaluating results (including commissioning, performance and occupant satisfaction determinations)



Progress Check Questions

Directions: Read each question and respond in the space provided. Answers and page references follow the questions.

1. Which of the following is not typically considered an example of a soft or occupant service?
☐ a. Custodial cleaning, window washing, waste management
☐ b. Food service, records management, human resources
☐ c. Parking, shipping and receiving, security
☐ d. Inspection, testing, commissioning

2. A typical occupant service planning approach identifies needs, evaluates, communicates, implements a plan and
☐ a. measures performance.
☐ b. assigns staff rather than contractors.
☐ c. processes all occupant service requests first.
☐ d. All of the above

3. To determine the right mix of occupant services, the facility manager considers
☐ a. the opinions of occupants and owners.
☐ b. only lowest-cost and highly competitive contractors.
☐ c. low-bid requests for services.
☐ d. staff over contract.

4. The workflow process for occupant services is _____ the workflow process for building systems and equipment.
☐ a. completely different from
☐ b. exactly the same as
☐ c. similar in approach to
☐ d. not considered as important as

5. In general, a facility manager should conserve resources and provide a quality environment
☐ a. at maximum sustainability.
☐ b. that always exceeds minimum standards.
☐ c. at the lowest possible cost.
☐ d. without using outside contractors.

6. Zone cleaning refers to
☐ a. a systematic approach by a team to an area.
☐ b. deferred maintenance-based priorities rather than schedules.
☐ c. comparing the work of one team or individual against another.
☐ d. having one individual perform all tasks in a specified cleaning area.



7. Occupant turnover, changes in facility mission and even legislative changes would most likely stimulate which of these facility management activities?
- ☐ a. Modifications to occupant services
 - ☐ b. Elimination of O&M budgets
 - ☐ c. Replacing contract workers with staff
 - ☐ d. Complete revision of performance contracts, SLAs or similar
8. The movement and relocation of occupants within an organization is commonly called
- ☐ a. downsizing.
 - ☐ b. turnover or churn.
 - ☐ c. workflow process.
 - ☐ d. productivity management.
9. Green washing is generally understood to mean
- ☐ a. an inappropriate practice of misrepresenting green benefits of a product.
 - ☐ b. an open work order that is not done when scheduled.
 - ☐ c. environmentally supportive cleaning.
 - ☐ d. use of low-phosphate cleaning products.



Progress check answers

1. d (p. 115)
2. a (p. 116)
3. a (p. 116)
4. c (p. 121)
5. c (p. 124)
6. d (p. 126)
7. a (p. 127)
8. b (p. 129)
9. a (p. 127)



Chapter 5: Select the Best Resources

After completing this chapter, students will be able to:

- Discuss the elements, value and use of service level agreements.
 - Describe the process to follow when determining whether to use staff or contractors for each service or operation.
 - Discuss the process of comparing and selecting among outsourcing candidates for O&M services.
 - Explain the negotiation process for a typical service level agreement.
 - Describe the principles of measuring competency, monitoring performance, evaluating outcomes for work/service, and developing/promoting best practices.
 - Describe key factors and attitudes relative to contract disputes and their resolution.
-

✦ Topic 1: Develop Maintenance and Occupant Service Specifications

Service level agreements defined

Service specifications and expectations should be detailed and defined. They can be prepared for both maintenance and occupant services tasks and used with both contractors and staff.

A service level agreement (SLA) is the part of a service contract where the service expectations are formally defined. SLAs are terms negotiated between the service provider and the facility representative (such as the facility manager). The service provider may be internal staff or outside contractors who perform any delivered service such as operations and maintenance of the facility and its systems as well as occupant services such as custodial, food service, waste management and so forth.

In the SLA, the service provider commits to an agreed-upon level of service (LOS). That LOS is usually described in quantitative measures that detail how the provider will support service through availability, serviceability, performance, operation or other attributes. The LOS may also be described in the contract by a listing of specific tasks the supplier is to perform. This is known as a statement of work (SOW).



The relationship may be represented in this way:

Service level agreement = Contract specifications + Statement of work

A basic SLA usually contains these elements:

- SOW that specifies tasks to be done
- LOS that defines expected minimums indicating acceptable performance
- Measurements or metrics (relative to key performance indicators [KPIs]) that compare results to mutually agreed-upon goals
- Oversight measures that describe how performance to LOS is reviewed
- Incentives for achieving goals beyond minimums (There may also be special aspects such as performance contracting in which future savings from more efficient systems or service delivery become part of the terms of the contract or are leveraged into the cost of the contract.)
- Work schedules that define when and at what frequency work tasks will occur
- Standard operating procedures (SOPs) that specify how tasks will be accomplished as general work and safety rules

A typical SLA is designed to help the facility manager (and the service provider) recognize what level of service quality is required and has been delivered. As such, an SLA's usefulness lies in how completely and accurately the LOS is described. Both the provider and the facility manager understand and agree upon what the minimum level of service is.

A service level agreement and the level of service and performance indicators that measure it are vital for high-quality occupant services. Whether an SLA is devised for staff or contractors, the expected (and defined) service levels need to be clear, realistic and mutually understood. When the contract is specific and clear, the facility manager's duty to oversee service providers is likely to be less stressful and be recognized as reasoned, not arbitrary.

In this way, the contract essentially defines and documents the relationship between the facility manager and the service provider. Then, as service activities are conducted, the SLA may determine the quality and tone of how the facility manager and providers (whether staff or contractors) relate to and experience each other.



Service level agreement specifics

In greater detail, a service level agreement generally includes the items listed and described in Exhibit 5-1.

Exhibit 5-1: Service Level Agreement Elements

Element	Description
Introduction	- Purpose and objectives (prioritized) - Parties to the agreement (supplier and customer) - Starting date and contract duration - Definitions
Statement of work describing the scope of services to be performed	- Descriptions of work to perform (describes the work plan including checklists, if available) - Specified standards for acceptable work product - Labor time standards or engineering performance standards (or some basis for measuring and estimating/calibrating work time) - Workforce needs (crews, labor hours, trades, mix) - Equipment, materials, fabrication requirements - Experience, skill, training, certification requirements - Rules of performance compliance (specifications on acceptable versus unacceptable work) - Work prioritization (service priority categories and rankings by criticality, dollar value, complexity) - Subcontracting provisions
Quality and performance-related expectations	- Supervision - Tracking (including methods by which service providers record complications or extra work) - Monitoring performance measures and performance - Reporting
Safety	
Communication	- Meetings - Procedures for revising the SLA
Exception (or problem) management	- Problem definition - Escalation guidelines - Approach to disputes and dispute resolution
Pricing (or compensation) fees and payment terms, incentives and penalties	
Legal compliance	- Confidential information - Intellectual property rights - Security - Default - Transfer of responsibility
Reasons for contract termination	
Signatures of both parties (confirming agreement to the SLA)	



The *Finance and Business* and *Leadership and Strategy* courses provide insight into contract preparation and relationship management.

Value of service level agreements

Regardless of the manner in which or detail to which SLA specifications are determined, they are vital because they help the facility manager:

- Align SLAs with owner/management mission and goals.
- Determine and understand the work to be done.
- Describe how that work should proceed to agreed-upon standards, work expectations, and service level goals or targeted expectations.
- Review how the work has been accomplished.
- Measure results against specific, written standards.
- Evaluate if work performance meets expectations.
- Reduce conflict if there are disputes.
- Support resolution by recognizing and adhering to previously agreed-upon conditions, performance measures and individual responsibilities.

The facility manager may also require or support having some or all activities (tasks, schedules, results and so forth) entered into the CMMS/IWMS or other computer-aided facility management (CAFM) system so that all aspects of service are available for review and control. Automated capabilities help the facility manager administer the contract from the time it is awarded (for contractors) or established (with staff) until it is closed out or replaced by another contract.

The facility manager is usually responsible for administering the contract and determining that the intent, requirements, terms and conditions of the contract are met. However, the SLA should contain all information that the facility manager and service provider require.

A contractor typically submits their standard operating procedures and schedules as part of the proposal. These are included to show how the statement or scope of work, level of service and other performance metrics, including key performance indicators, will be met.

LOS, SOW, and KPIs help provide the standards by which the service provider and the facility manager can evaluate the conduct and outcome of the work.



Service level agreement benefits

Service level agreements are important to the facility manager because they can produce several benefits, such as the following:

- **Priority response.** SLAs can be written so that support level, service performance, response times, reliability, other measures of responsiveness, and accountability can be predetermined.
- **Focus on outcomes.** Performance standards can be specified so that results are measurable and service providers are rewarded (or sanctioned) depending upon how well they deliver.
- **Independent expertise.** For some services (such as engineering or information management), the use of a contractor may bring added technical experience or special knowledge or training that the facility may need but is beyond the training or capabilities of staff.

On the other side, an SLA can give the facility manager control over services that may not require high-level skills but are labor-intensive (such as lawn care services).

- **Availability of added services.** An SLA can support the facility manager when outside contractors with special or extended services are needed in emergencies, at critical periods, or when temporary, added support enhances the mission.
- **Maintain focus on core competency.** Facility managers are expected to support and enhance the core purpose of the facility and its occupants. With SLAs, the facility manager has a written agreement on what will be done to provide services and how that supports the facility's mission and strategic plan.
- **Control.** A well-crafted SLA provides the direction, means and measures to give the facility manager control over resources and deliver quality results at predictable, negotiated costs.

When building systems and equipment O&M are contracted, an SLA can help the facility manager negotiate a fixed fee for planned maintenance or for labor and materials for services on request such as corrective maintenance.



✦ Topic 2: Select Competent Service Providers (Staff or Contract)

Determining what to contract

Virtually any systems/equipment O&M or occupant service could be performed by any competent provider. Thus, the question of whether to use staff or contract work is not an us-or-them consideration but a choice based on sound reasoning.

Contracting competent providers that are not staff may also be known as outsourcing, which means to hire an outside firm to supply services, or out-tasking, which means to hire an outside firm or individual to perform a specific task within a function, such as cleaning or HVAC maintenance.

The facility manager may look at a contractor rather than staff because they may offer capabilities that staff:

- Does not have.
- Cannot justify because of the nature of the work.
- Does not concentrate on because the work activities are less relevant to the core purpose of the business.
- Needs as additional support to supplement staff activities.

For example:

- Work activities that generally require low expertise and are of low relevance to the facility mission (such as custodial cleaning or grounds maintenance) may be candidates for outside contracting.
- Work activities that require special or extensive expertise not relevant to the core mission (such as security or information technology) may be considered for contracting because the experience level and training requirements make a staff capability unaffordable.

On the other hand, the facility manager may determine that specific activities should be handled by staff. For example:

- Work activities that require low experience but are highly important to the mission of the facility (such as maintenance cleaning of critical equipment) may represent good options for staff assignment.
- Work activities that generally require high expertise and are critical to the facility mission (such as managing occupant moves or strategic planning) may or may not be developed as staff responsibilities based on the frequency of the activity, the level of special expertise needed, or the budget available.



There can generally be no specifically stated rules for making these contractor/staff decisions, because there can be as many exceptions as there are standards. For example, the stringent requirements for security for a building might require a security service, while the requirements for guarding a manufacturing parking lot could be quite adequately handled with staff.

The availability of contractor services or of competitive candidates for a particular service may also be factors the facility manager considers. For example, if outside record-keeping services are in short supply, the facility manager may suggest developing that expertise with existing or added staff.

In addition, a facility manager may wish to hire contractors and their expertise as a way to share the risk in complex, sensitive, highly technical or otherwise higher-risk service activities. A contractor who is willing to take on responsibility suggests self-confidence; use of such a contractor can reduce risk or penalty for the facility should something not go according to plan.

In general, the facility manager considers several factors when assessing whether to contract services or manage them internally using staff. These may include:

- Relevance to the facility mission.
- Level of control.
- Service specialty or uniqueness.
- Speed or service response.
- Productivity or oversight requirements.
- Total cost of service.

Relevance to facility mission

Facility managers generally focus staff efforts to support the primary mission of the organization. For example, if a facility's mission is education, then it might contract repairs or landscaping. If a facility's mission is to manufacture a product, it might outsource waste management or cafeteria services. The less relevant the service is to the core mission of the facility and its ownership, the more likely that service may be considered for contracting.

In general, this is because the service itself may be outside the experience or current focus of facility management. A facility manager might, for example, be immersed in staff activities that maintain hospital cleanliness and hygienic standards but contract for services to maintain a sterile research lab.

The business measurements and budgets contracted through SLAs can also provide the facility manager with a foundation for measuring the possible cost



for adding systems, structural elements and other infrastructural assets. By monitoring costs compared to results, the facility manager can:

- Make a business case relative to contracting or staffing.
- Develop cost analyses to determine the impact of altering or acquiring new infrastructure.
- Expand/alter services or remain with the status quo.

Level of control

Depending upon the facility mission and the facility manager's capabilities, the decision to use staff or contract for services may be determined based on how much control is wanted or needed. For example, a facility manager (or owners) may prefer that certain critical activities be closely monitored by the facility manager through inspection, tightly written statements of work, precisely expressed levels of service and rigorous oversight of key performance indicators.

A facility manager usually has more direct control and authority over staff but can still maintain excellent control over contractors through the SLA. In most cases, however, contracted people report first to their supervisors. It is unusual for a facility manager to directly intervene with contracted staff, nor would the manager want to since that is the responsibility of the contracted supervisor to make sure that SLA requirements are being met.

Service specialty or uniqueness

Contract services may be the alternative of choice because skills or capabilities are needed beyond the scope of staff or daily needs of occupants. For example, contractors may be preferred for:

- Highly technical services that require special education, unique tools or distinctive skills such as engineering, architectural design, accounting or maintaining high-energy equipment.
- Labor-intensive tasks or tasks that have seasonal periods of high demand such as lawn care, landscaping, custodial services, or moves and relocations.
- Specialty services that may be difficult to manage, irregular or seldom required such as window washing, master carpentry or floor installation.
- Extensions of capabilities such as waste management resources that can also handle recycling and environmental disposal as well as custodial cleaning.
- Services that are shared or represent greater economy when contracted as part of a group, such as health insurance management or security monitoring services that benefit from larger customer pools.
- Services involving expert oversight or critical health and safety management issues that exceed the facility manager's expertise or available time. For example, contracted specialists could supply anything from vaccinations and fitness programs to marketing and customer satisfaction surveys.



Speed or service response

Circumstances that require an unusually rapid response or significant increase of resources may be better served through contracts. For example, few facilities have their own fire response team or on-premises paramedic unit because the continuing expense in personnel, training and equipment exceeds the potential risks of depending on the services of an outside source. When needed, the quick response and availability of a fire-rescue service also helps lower the total potential damage and thus helps the facility manager control risk more capably.

The facility manager may determine the need to contract services where flexibility or service response is critical. For example, contract services may be selected to:

- Manage, maintain and upgrade information technology systems, communications devices and other specialized services.
- Sort, package and distribute products or materials through fulfillment services or moving companies.
- Dedicate availability and personal attention to occupants and visitors (such as through a help desk or lobby information service).

As was previously mentioned, there are no definite rules for selecting between contractors or staff that work every time. For example, a facility manager may decide to contract services to respond to emergency power outages, critical equipment breakdowns, or severe IT malfunctions where speed and numbers are vital. However, another facility manager may choose to do the opposite and decide to use staff because responders are already on site and instantly available and not subject to a conflicts as a contractor might in responding to other conditions.

Productivity or oversight requirements

Regulatory constraints that require certifications, code compliance or licensed workers may also call for services to be contracted. Strict job requirements and regulations may preclude training a staff person to perform the work to specification that could be readily accomplished by, for example, certified electricians, elevator maintenance experts or hazardous materials contractors.

Where the standard for quality or level of compliance is markedly higher than in routine operations and maintenance, the facility manager may wisely hire outside experts, for example, to maintain on-site security, to test and charge fire sprinkler systems and fire extinguishers, or to perform work in high, exposed areas or small, confined spaces.



Total cost of service

The facility manager builds a business case to help determine whether a service should be handled by staff or contracted. A primary consideration is the balance between the controllable costs of contracting versus the estimated costs in staff hours as well as what services those potential staff hours might better serve.

In addition, the facility manager assembles and analyzes costs that account for:

- Types of systems, activities or tasks to be done by describing the work required and performance expectations.
- Frequency of the occurrence to determine staff requirements compared to economy through contracting.
- Crew size and mix to anticipate total range of capabilities, personnel and resources required.
- Total labor hours, materials, tools, equipment and other costs to maintain staff competency.

Each of these factors helps form the basis for a business case that accurately depicts what can be expected and what is required of staff and contractors. In many cases, the business case is self-evident. For example, a moving company may offer greater expertise, packing materials, transportation and labor services than a facility could provide using staff. Even with facilities that experience frequent occupant moves and relocations, a moving contractor may still be preferred based on cost, experience, risk reduction and productivity benefits.

Determine contract or staff assignments

The facility manager decides when to rely on the capabilities of experts and when to use staff or contractors in operations, replacements, renovations and occupant services. Much of this depends upon the facility and its use, staff, location and other factors. But it is wise to be able to recognize when a task or activity can be accomplished better by an outside expert and to know where to go to find that expertise.

A common approach is to use economic principles to determine whether to develop staff capability or buy contractor expertise. For example, it may be more cost-effective to hire someone to install a raised floor or carpeting than to assign such tasks to a staff carpenter.

The process of determining whether to use staff or contractors requires a competent assessment of facts and includes the following:

- Determine the estimated hours it would take for staff to acquire the knowledge, materials, equipment and capability.
- Determine any aspects that may add to cost or increase concern, such as engineering questions; compliance requirements; quality of the work;



acquisition costs; losses or delays to other tasks; downtime costs if staff cannot perform the work as quickly, completely or within minimum time constraints; chance for error or unacceptable quality; and redo costs.

- Compare all recognized costs and concerns against the responsibilities taken on by a contractor at a negotiated fee. The numbers should speak for themselves.

A process such as this might indicate that it is more realistic and economical for the facility manager and internal staff to create their own capability rather than seek a contractor. It may be that because the maintenance, operation or renovation requirement is so specialized, because there is no available contractor, or because incoming bids are so high that it makes better sense to accomplish the task internally.

Considerations for contracting

From an operations and maintenance standpoint, contractors and staff form an interdependent and mutually beneficial relationship. This relationship should be based on a foundation of trust, respect and consensus on performance. The facility manager must also be aware that internal staff may feel threatened by contractor support and should manage those perceptions intelligently.

The facility manager may typically use several factors to compare and contrast candidates as well as to determine whether staff or contracted services can provide what is best for the facility and occupants. The facility manager also considers what will receive the support of senior management.

These factors include comparing advantages and disadvantages, rating candidates, ranking candidates and determining candidate compliance.

Comparing advantages and disadvantages

By acknowledging possible advantages and disadvantages of contracting, the facility manager can better understand and prepare for potential issues with contractors or staff. The facility manager generally decides on whether to use contractors or staff resources based on which can best achieve the highest possible quality, with the greatest efficiency, at the lowest total expense.

For example, a facility manager may favor a staff response to a critical equipment failure because, all things being equal, the contractor is too distant to react as quickly as on-site personnel. Or the facility manager may contract for traditionally recognized staff functions because it better serves the facility's mission and thus everyone in the organization. When relationships develop based on positive criteria rather than competitive distrust, both staff and contractors recognize that they are needed and valued.



Exhibit 5-2 outlines some of the aspects by which a facility manager may make critical comparisons. These perceptions may be real or imagined. Regardless, the facility manager is aware of these potentials so that all personnel feel respected, acknowledged and understood.

Exhibit 5-2: Advantages and Disadvantages of Contracting

Advantages	Disadvantages
• Cost control • Possible improvement in service quality • Less liability (insurance, legal) • Fresh ideas, best practices • Less involvement in internal politics • Gain in expertise • Better operating efficiency • Flexible sizing to meet each need • Latest technology and tools • Service can be terminated • Greater flexibility of staffing • In-depth specialization • Minimum on-space site needs • Negotiated, dependable cost • Fast, complete response, potentially unlimited worker availability • Accountability to SLA specifications • Contractor responsible for labor relations • Reduced staff work burden • Predictable, controllable cost • Reduced operational expenses (extends staff capabilities without adding to staff/benefit costs)	• Lack of loyalty • Loss of immediate control • No ownership • Different agenda (money for contractor) • Unfamiliar with business culture • Contract restrictions • Performance gaps not visible • Poor service unless measured • Training curve costs • Less control over personnel assigned • Turnover of personnel • Security issues • Minimum on-site empowerment • Future cost escalations • Unavailable for non-contracted response • Less flexibility in out-of-specification areas • Potential union/nonunion conflict • Potential for poor staff morale • Possibly higher cost than staff • May require different supervisory skills than used for internal staffing

Rating candidates

To measure the relative business effectiveness of candidates, the facility manager may look at important factors such as:

- Personnel expertise, training, work reputation, turnover.
- Personnel supervisory quality, flexibility assigning or changing workers.
- Service reputation among other facility managers.
- Licenses and certifications (such as whether the candidate can provide inspectors for boiler certifications, qualified elevator inspector certifications, testing for fire alarm and suppression systems and backflow preventers, fall protection, and so forth).



- Business experience, financial stability, pricing policy.
- Local presence for support, response, capacity to react to new conditions or requirements.
- Speed of response to service calls, support for emergency needs.
- Available tools, equipment, facilities and materials.
- Capacity to provide aligned or extensions of services.
- Communication, culture, relationships with facility managers and occupants in performance of services.

Whether the facility manager is considering staff or a contractor as the service provider, the process for determining needs should be clear and effective. A summary of guidelines is shown in Exhibit 5-3.

Exhibit 5-3: Typical Service Approach

Approach to Contractors	Approach to Staff
Define services needed. Prepare documents. • Request for proposal or invitation to tender. • Service specifications. • Service level agreement. • Contract conditions. Determine quality and cost. • Comparison to staff cost. • Quality and performance measures. • Terms, conditions and communications criteria.	Identify inputs and gather information. • Owner/management. • Occupants. • Visitors. • Staff. Define services. • Combine inputs. • Rank (prioritize). • Describe requirements as measurable specifications. Establish staff capabilities. • Training, experience. • Available hours. • Equipment, materials, resources. Review service execution. • Monitor cost and quality. • Continuously improve.

Ranking candidates The facility manager may consider the suitability of candidates by giving each a high, medium or low ranking based on the competency area. These ratings can then be summarized in a scorecard or grid such as the one shown as Exhibit 5-4. This allows the facility manager to conveniently and impartially compare the overall suitability, strengths and weaknesses of different service providers.



Exhibit 5-4: Candidate Comparison

Competency Area	Candidate A	Candidate B	Candidate C
Personnel training, expertise			
Supervisory quality, awareness			
Service reputation			
Licenses and certifications			
Business competency, stability			
Local presence, availability			
Service response, flexibility			
Tools, material, equipment support			
Capability to expand services			
Communications, relationships			
Candidate Rank (total/column)			

Scores per competency area: Low—1–3; Medium—4–6; High—7–9

Determining candidate compliance

When reviewing or selecting contractors, it is also important to determine that they comply with:

- The facility ownership and facility manager’s mission and vision.
- Legal guidelines such as codes, permits and accepted standards of practice.
- Insurance, bonding and licensing requirements.
- Worker safety and prescribed methods for handling hazardous or noxious materials.
- Sustainability, environmental protection and related quality concerns.
- Provisions for barrier-free access, equal employment opportunity and related human resources concerns.

Another good practice when selecting among candidates is to assess the level of technology and management systems the contractor has available. If a location is remote or the contractor has limited information technology access or capabilities, the skill level of personnel may still be the more significant factor. However, considerations on how a contractor can support data exchanges with the facility manager’s system can be useful for decision making.

The facility manager may use all of these factors, along with price, to compare rival contractors (and estimated staff costs) as a way to arrive at the most competent candidate.



The facility manager may use quality measurements such as those found in the total quality management approach. These measures of effectiveness may be applied to staff as well as contractors. A typical quality management approach includes the following:

- Focus on the customer (occupants, visitors) to understand their needs.
- Give purpose and direction to the organization and involve people to encourage them to contribute their abilities to benefit the organization.
- Use processes to install quality considerations as early as possible. For example, quality controls could begin with the materials that form a component to ensure greater reliability and consistent performance as it operates within a system.
- Improve efficiency by anticipating problems to avoid them if at all possible before they can start. Then if a problem appears, recognize and mediate it as efficiently as possible to maintain the facility mission. Finally, if a problem persists, resolve it before complaints occur.
- Continually improve overall performance.
- Make decisions based on analyzed data and information.
- Recognize the interdependence of mutually beneficial supplier relationships.

In the contract preparation, bidding and selection process, the facility manager:

- Assesses candidates using specific and measurable criteria that fully and accurately compare their qualifications.
- Uses service level agreement information, known best practices, service response requirements, key performance indicators, and other measurable qualities to arrive at the best candidate and make the selection based on concrete and justifiable factors.
- Carefully documents the selection process. Today's business and legal environment has made it important that the facility manager prepare and retain all documentation to be able to demonstrate the basis for selection and its justification if requested or needed at some later time.

To select or compare contractors or staff, the facility manager typically applies the performance criteria defined in the SLA (or similar service contract). Once candidates are selected, the facility manager can also use these performance criteria to monitor activities and hold service personnel accountable for their work.

A complete discussion of contracts, contracting and competent approaches useful to the facility manager is contained in the *Finance and Business* course.



✦ Topic 3: Negotiate Service Level Agreements

Negotiation defined

“A discussion intended to produce an agreement”—this is the most general description of a negotiation as a means to a contract. A contract that is dictated by one party is not an agreement; it is an order. This implies that negotiating a contractor agreement:

- Involves discussion between the awarding and performing parties.
- Implies cooperation between the parties.
- Has an agreement as the mutual goal.

Negotiation should begin before the contract is written. The goal of negotiation is to:

- Arrive at a statement of work (the part of the contract that lists the specific tasks to be performed by the contractor).
- Develop requirements agreed upon by both parties that are clear and quantitative so that they are measurable.
- Define expectations using specification terms that describe essential technical requirements for materials and processes rather than vague words like satisfactory, functional or acceptable.
- Define insurance requirements, compensation, timing of that compensation, method of contractor invoicing, and method of payment by the organization to the contractor.

Written service level expectations prepared by the facility manager or a designate can provide the basis from which candidates (in this case, typically contractors, not staff) make presentations. This helps both contractor candidates and the facility manager be more aware of, synchronous with and confident in the process.

When sending an invitation to tender, it has become common to include a sample contract. This helps invited contractors recognize where they may not comply so that these factors may be discussed during negotiations.

Common negotiation characteristics

Typical negotiation strategies contain some common characteristics:

- Negotiation implies compromise.
- Negotiation permits expectations on both sides so that, once an agreement is made, both feel they have made concessions and have cooperated to achieve the outcome.
- Negotiations must support and sustain legal, business and regulatory requirements.



- Negotiations should meet corporate objectives (of the facility owners as well as the service providers).
- Negotiation uses accepted fiscal and maintenance constructs to clearly define performance criteria measures through:
 - Return on investment, which measures how efficiently management is employing financial resources invested in the firm.
 - Hourly task time or historic data.
 - Standard books that reflect typical work times by task.
 - Other similar means.
- Negotiation should be straightforward and confident, gracious and professional.
- Negotiation should be flexible within minimum (or non-negotiable) expectations.
- Negotiation should not be threatening or exploitative.
- Negotiation should not be intimidating in the tone, location or manner in which it is conducted.

Generally, the facility manager or a designate prepares a request for proposal (RFP), which is a request for bids to deliver a specific product or service. The contractor's response to the RFP begins the dialogue and negotiations.

Negotiated terms

Virtually anything can be negotiated, depending on the requirements of the facility and its manager and the capabilities of the contractor. Examples of negotiated terms may include:

- Statement of work.
- Level of service and performance minimums.
- Schedules and standard operating procedures.
- Monitoring, oversight and reporting methods.
- Exceptions, change orders, minimum and maximum labor levels.
- Supervision approach, handling of disputes.

A typical process for negotiating a public-sector contract is likely to be prescribed by law, regulation or approved administrative procedures. A typical process for contract negotiations in the private sector can be as unique as the individuals and organizations engaged.

Relative to negotiations themselves, two significant areas requiring diligence are service level factors and documenting determinations.



Service level factors

Work planning generally includes prioritization, coordination and flow requirements. The work plan is central to the negotiation because it specifies the expectations, approach and limits concerning the O&M service to be done. The service specification should quantify the minimal acceptable service standard that fulfills the facility mission, meets statutory requirements, and accomplishes the performance outcome.

Service specifications, as part of service level agreements themselves, are tools the facility manager can use to negotiate many aspects, including:

- Scope of work and control over change.
- Acceptable quality performance level.
- Service level minimums and maximums (or when work is done).
- What constitutes work beyond specification or agreed-upon response.
- Value or pricing of services provided; fee changes for added or reduced services.
- Consequences of failure to meet service level requirements (fines, fees, loss of incentives, contract reduction, termination).
- Rights and obligations of both awarding and service-providing parties.

The facility manager regards service specifications not as fixed statements of requirements but as foundation descriptions for continuous improvement.

Documenting determinations

All aspects of the agreement should be documented so that the contract reflects the understanding of both parties. This supports service activities as actual O&M work and occupant services proceed.

Clear, precise service level descriptions represent the agreed-upon core of service requirements and thus the understanding between both parties. The negotiated, documented agreement is the basis for the relationship between the facility manager and the contractor and the source for measuring performance and outcomes.

Methods and incentives to influence continuous improvement and inhibit inferior work help both parties maintain contract expectations with less conflict, misunderstanding or communication challenges.

★ Topic 4: Monitor Work/Service Performance

Measures of competency

A facility manager is usually responsible for managing/overseeing all facility O&M, which includes the services provided to occupants or tenants as well as the operations and maintenance of building equipment and systems. This is



true whether services are performed by staff or outsourced contractors. How effectively and efficiently the facility manager does this is not only a measure of competency but may also determine how competitive, attractive and efficient the facility is for its occupants and the mission. This makes it critical that services be properly executed.

Typically, the facility manager judges that performance by comparing actual results with agreed-upon requirements as they are stated in the service contract. By specifying performance criteria in SLAs, the facility manager uses them to measure work performance in a way that is impartial, less stressful and most professional.

The facility manager needs to have an effective and realistic oversight plan that clearly shows whether contractors and staff are meeting agreed-upon requirements. This ability is critical so that the facility manager can review performance expectations in an accurate and timely manner.

The source for analyzing effectiveness should be the SLA with its statement of work, level of service requirements, and key performance indicators that set goals. It is important to emphasize its importance by restating it here. With a complete, quantitative, agreed-upon set of performance requirements and activities, the facility manager has the means to not only measure effectiveness but also to support change to develop effectiveness where gaps occur.

Monitor to manage

A common saying regarding management is that “what is measured gets done.” This statement implies that O&M activities and the conduct of occupant services are inspected and monitored. Monitoring work to determine that measurements are achieved is necessary and useful to the facility manager for several reasons:

- Comprehensive monitoring provides assurance that contract terms are met.
- Regular, ongoing inspection alerts the facility manager early to eroding conditions, unmet expectations, potential misunderstandings or other service delivery questions before small exceptions become large challenges.
- Routine monitoring (and the contractor knowing that inspections will regularly occur) is one way to prevent or reduce fraud as well as ensure that maintenance work is being properly completed.
- Inspections not only alert the facility manager to changing conditions or possible service delivery problems but also support the relationship and avoid conflicts when misunderstandings or unexpected requirements arise.



- Diligent and thorough inspections encourage regular communication between the facility manager and the service provider.
- Ongoing monitoring signifies that the facility manager is serious about the maintenance work and supports what the service provider is accomplishing; it reinforces the partnership.

Inspection should not imply adhering blindly to work specifications or over-monitoring to find fault. Contract work inspection is a practical and realistic means to keep all parties informed and in agreement about what is being done compared to what is expected. It is useful way to maintain communications after the contract is negotiated and signed.

Monitor through SLAs

As has been mentioned, the service level agreement is the facility manager's reference point for monitoring the delivery of services. Properly written SLAs clearly describe the scope of work, defining the agreement among parties about what is to be done, when and to what extent. This makes an SLA a useful tool the facility manager can follow to monitor work and identify discrepancies.

Service level agreements (or similar documents that identify and clearly describe the work to do and how results are measured) can be useful whether the work is performed by internal staff or contracted services. SLA-type contracts may be developed with staff, but they may be applicable as part of the responsibility of human resources or administration of the facility and not part of a facility manager's duties.

The SLA can provide guidance that both contractors and staff find useful as a means to accomplish the work and to know what is measured to determine that results are satisfactory. A well-written scope of work document helps give all parties—facility manager, staff, contractors, occupants, and executives—the confidence that the work is performed well and efficiently so that it is accomplished properly the first time. A comprehensive review for administering and maintaining contracts is found in the *Finance and Business* course.

The facility manager must recognize the importance of having satisfied occupants and must strive to make O&M comply not only with physical, functional expectations but also across a range of other aspects. Occupant opinions may take what began as a functional specification and expand it to include aesthetic and social impacts. This can make it a more encompassing idea of the results they expect to be achieved.



Agree on performance specifications

Work quality or performance should be measured by clear-cut standards, not vague references that can lead to different interpretations and misunderstanding. When work order expectations, service level agreements, and other contracts are clear, precise and quantitative, the work is more likely to meet expectations without controversy. Performance measures (including those determined by life-cycle cost analysis or total cost of ownership) not only indicate the goals to be achieved but show both the service provider and the facility manager when performance is achieved and when it is not achieved.

A few general principles relative to performance specifications may be described within three overall elements:

- Agree on specifics.
- Identify known outcomes.
- Evaluate results.

Agree on specifics

The facility manager uses the work order to communicate all work whether for maintenance, repairs, upgrades, ongoing or delayed work, activities involving occupant services, or emergencies. The expected results of the WO should be clearly described and defined. Outcomes should be specific and measurable.

Relative to occupant service activities (whether custodial services, records management, child care or security, for example), the desired results should reflect realistic and agreed-upon standards that all parties view the same. Agreements should not be left to individual judgment but should reflect mutual consent. For example, some specifics relative to custodial services could include specifications such as:

- Wipe until dust- and smudge-free, and polish all unpainted metal surfaces within reach; use accepted safety precautions to clean in the same manner all other visible metal surfaces.
- Replace fluorescent lamps semiannually and ballasts upon failure; clean light diffusers annually so that no residue, dirt or other foreign items are present.
- Vacuum all hallway carpeting edge-to-edge before 7:00 a.m. to avoid interfering with regular occupant arrival.
- Do not allow unsightly conditions such as dirt in corners, mop splashes on base boards, or floor wax applied over dirt spots.

The contractor or staff assigned to the work should be involved in developing the standards expected of the work. This helps make everyone aware of the expectations as well as participants in their creation. It encourages agreement before work begins and is an excellent starting point if inconsistencies or misunderstandings occur.



Just as engineering specifications are precise and complete, expectations written for occupant services and the people who provide them should be unambiguous and thorough. By involving workers in the development of specifications, the facility manager makes use of their experience to help make the expected work approach realistic, understood, concrete and in everyone's best interest. When details are precise, the worker knows what is expected and can feel confident about doing the task right the first time.

Identify known outcomes

When the approach to the work is clear, the results are also likely to meet expectations. Certainly the outcomes are different for sanitizing a hospital surgery than they are for cleaning an office hallway. However, both should have written descriptions that accurately state what should be done and what to achieve.

To do this, the facility manager strives to identify known outcomes or expectations, budgets or reasoned estimates of specifics so that work plans are practical and settled among all parties. Work plans can include, for example, estimates of:

- Crew size, mix of trades.
- Anticipated work hours.
- Travel and setup time.
- Work duration.
- Work tasks (including checklists).
- Skills, tools, equipment and preparation needed (including special training or personal protective equipment).

The more complete, specific and mutual the work plan is, the more efficient and expected will be its outcome.

Evaluate results and variations

The facility manager can apply the clear, measurable expectations to evaluate results and recognize variations. This helps the facility manager focus on exceptions such as:

- Out-of-specification results that did not meet the agreed-upon contract requirements and need to be corrected.
- Moves, adds and changes that altered the original work plan or in some way caused specifications or expectations to change.
- Service orders carried over that were incomplete so that work results could not be evaluated and perhaps delayed or interrupted work plans that were to follow.



In addition, the facility manager collects results over time to reveal trends, search for gaps and recognize weaknesses. For example, the facility manager may use the CMMS to create reports that summarize and indicate weaknesses:

- Carry-overs or unfinished jobs by month and year-to-date may, for example, expose unrealistic timelines or inefficiencies.
- Orders received compared to orders completed may, for example, indicate workload challenges, inadequate estimates of work required, or other unforeseen changes that may reflect a pattern.
- Work completed compared to work not completed may, for example, reveal expectations that are too high or not high enough.

By monitoring work results and comparing them to service level agreement specifications for that work, the facility manager may recognize discrepancies and determine where performance does not meet agreed-upon outcomes. Further, with clear specifications and workflow descriptions, the facility manager has the information to support corrective action.

Factual review is also made possible when work plans and outcomes are continually updated in the CMMS so that actual performance may be compared to service level agreements. This not only provides clear evidence for negotiations, specific work corrections, and awarding of incentives but also provides the facility manager with the data necessary to determine if contracts should be renewed, rewritten or terminated.

Value of incentives

Incentives are useful tools to reward and direct work output and results. They may be used to avoid a strike or encourage a specific activity. They may also help resolve disputes and avoid the potential cost of litigation or disruption of operations. However, the use of incentives must never be perceived as a payoff or corruption or give the appearance of bribery or inappropriate influence.

An incentive can be financial or nonfinancial to influence a particular action, behavior or result. Typical incentives include:

- Financial incentives such as rewards, bonuses, monetary prizes and contract extensions.
- Moral incentives such as promoting the right thing to do or encouraging team effort.
- Social incentives through recognition such as personal letters, awards, added responsibility, special status and supporting the contractor's reputation.



Contracts can be reward-based and may offer incentives for meeting work requirements, exceeding goals or simply consistently meeting specifications. Different types of incentive-based contracts include the following:

- **Fixed-price-incentive** contracts provide added remuneration or other incentives if the contractor meets or stays beneath a targeted cost yet delivers quality or performance at or above the targeted minimum.
- **Cost-plus-award** contracts hold the contractor to an agreement on maximum expenses and rewards with a bonus when costs remain at or below those fixed expenses.
- **Award-term** contracts award contractors who consistently meet specified goals by extending or adding more years or expanding tasks to the current contract.

Incentives and rewards may be made spontaneously to recognize positive activity in the moment or announced in advance to emphasize work toward a specified goal. Incentives can also be used to deter unwanted action, behavior or results. For example, a bonus may be lost because of below-standard quality because a deadline or other specified goal is not met.

When incentives are included in the scope of work for contractors or staff, the facility manager uses precisely understood and measurable specifications to determine work performance. These performance measures should be carefully prepared so that whether incentives are awarded or withheld, the affected personnel recognize that the decision was correct according to the agreed-upon specifications.

Evaluate through best practices

Best practices are techniques or approaches that consistently provide the desired results across an industry or within an organization. In facility management some general best practices are driven to produce the following:

- **Occupant satisfaction.** How do occupants perceive their environment and the quality of services and infrastructure?
- **Financial best value.** How do budgets compare to costs, and what is the trend relative to costs versus benefits?
- **Emphasis on facility core mission.** How competently is O&M supporting the facility's core mission and satisfying occupants?
- **Efficient and effective operation.** How is the facility doing in terms of delivering to the purpose intended and to the expectations of the owner/executive management?



Best practices in facility management help the facility manager monitor O&M work so that it continues to meet specified expectations. These best practices may be developed as performance management measures that provide evaluations the facility manager can use. Examples of best practices derived as performance measures or benchmarks include:

- Year-on-year comparisons and quarterly reports that show relative progress over time or against earlier times.
- Comparing activities such as priorities met compared to priorities set, appropriate assignment and use of skill sets, actual hours compared to estimated hours, compliance with codes and regulations.
- Efficiency and effectiveness performance management measurements that support performing the activity with the minimum wasted effort and the maximum relative value.

Both efficiency and effectiveness are necessary for adequate performance measurement. Taken together, efficient and effective performance evaluations help the facility manager measure, monitor and discuss a contractor's or staff person's work. More examples are shown in Exhibit 5-5.

Exhibit 5-5: Efficiency/Effectiveness Measures

Efficiency-Based Performance	Effectiveness-Based Performance
• Work time and total time per work order • Percentage of total work completed within a given time and compared to previous periods • Standard hours divided by actual hours (not counting delays) • Total maintenance cost as a percentage of replacement cost • Number of work orders completed as scheduled or within a given time compared to previous periods • Number of checklists and work aids compared to previous periods • Hours or percentage of time spent in preparation, travel or setup compared to previous periods • Crew sizes and percentage of lower to upper skill mix compared to previous periods • Funding of maintenance as a proportion of the entire infrastructure budget compared period to period or to average • Percentage of contractors working from service level agreements	• Level of customer satisfaction as a percentage • Change in percentage of deferred maintenance or number/hours of delays • Percentage or number of change orders compared against previous periods • Change in percentage of preventive maintenance versus corrective maintenance • Percentage of work plan executed compared to assigned • Percentage of predictive maintenance work orders compared to preventive maintenance as a proportion of all work orders • Percentage of equipment failures or run time to failure against previous periods • Percentage of callbacks or returning to maintenance job because of incomplete or incorrect work • Percentage of occupant complaints or accolades compared to previous periods • Quality, sustainability and conservation programs and performance compared to previous periods



Communicate through feedback

Communication provides feedback. Feedback helps each party know how the other party feels about the way in which work is proceeding and how each party is living up to their responsibilities. Frequent communication helps maintain a balance and comfort level between parties so that issues can be raised and resolved with minimum conflict or tension.

Once a service provider (contract or staff) is selected, the facility manager clearly communicates what has been established that describes successful performance. The objective is to publish and follow agreed-upon criteria so that the facility manager and the service provider concur on what maintenance or service is to be done, how it will be done and how that performance will be measured. These criteria hold whether the work is operational or asset maintenance or supplying occupant services such as custodial cleaning, food services or mail distribution.

This clarity can be particularly important when working with contractors who are responsible for maintaining an asset. In general, it is easier for contractors to maintain new equipment and have the facility owner pay for replacement. Facility managers trust their contractors yet retain the ability to independently and objectively verify their recommendations and agree on any capital expenditure plan.

When repair and replacement decisions must be made, contention can occur between the facility manager/owner and the service provider.

Questions must be resolved regarding:

- Who pays for replacing the asset?
- When does it need to be replaced?
- Why does it need to be replaced? (Was it the end of its useful life or was it improperly maintained?)

The facility manager communicates regularly to keep contractors and staff alert to the performance requirements that are already agreed upon. These discussions should seldom contain surprises because they start with the stipulations of the work agreement.

They may come under dispute about what one side or the other sees as exceptions to the agreed-upon contract principles. However, with a well-defined and mutual SLA and scope of work agreement, the discussion can be about interpretation rather than blame.



✦ Topic 5: Resolve Contract Disputes

Start with the SLA

It is in the best interest of the organization to have contractors and staff who are satisfied with their work environment, focused on outcomes and concentrating on tasks. This is how building maintenance and occupant service work gets done, occupants are served and the facility operates for maximum results. Contract disputes do not serve any of those issues, and the facility manager must determine ways to avoid and resolve contract disputes.

The most effective way to resolve contract disputes is to have the means to address them in the first place. The facility manager must focus on specifics to support the relationship and encourage discussion. A concrete and mutually understood SLA gives both the supplier and the facility manager the basis for discussing a conflict.

Complete, accurate, measurable and specific work descriptions make it far easier to recognize, discuss and resolve disputes. A well-crafted SLA with a clear statement of work and complete terms and conditions offers several benefits.

- It establishes standards that likely relate to an issue in question.
- It provides a basis for the perspective and range for viewing an issue.
- It contains the protocols or rules for how to approach the other party with an issue in question.
- It supports pragmatic and measurable methods to share ideas, progress to understanding and resolve the issue in question.

The quantitative requirements of the SLA are an excellent starting point because they provide a basis that both parties are familiar with, understand and agree on (at least when the contract was initially signed). Those measures may come to be disputed for any number of reasons, such as:

- Objectives or expectations are considered to be too high or too low.
- Conditions have changed, making performance objectives unachievable or unrealistic.
- New conditions relative to expectations have changed the capability to achieve or measure original goals.
- Either party identifies a gap that changes requirements and their outcome.

The facility manager must regularly and at least annually review the balanced scorecard (discussed in Chapter 6) to validate assumptions, check against the strategic facility plan, and be confident that work descriptions and outcomes reflect current, mutually agreed-upon parameters.



The value of existing measures remains important, even if their validity has changed. The original standards can still be the initial basis for reviewing what has changed, how to approach it and how both parties can support a resolution. This can more easily happen when the dispute or new expectations are clearly described so that differences of opinion can be clearly understood and compromises can be devised.

A positive result can be more readily achieved when:

- A previously agreed-upon process is already in place and respectfully followed by all parties for resolving disputes.
- The issue in question is documented and that documentation is shared among all parties.
- The atmosphere surrounding discussions remains cordial and focused on specifics.

It is unlikely that an SLA can anticipate every possible contract dispute and thus avoid all conflict. However, the facility manager can use a well-drafted service level agreement to begin the conversation positively and to develop a mutually satisfying result. Even if no basis can be found in the SLA, the discussion can still be treated as a negotiation between parties to find an equitable solution in the same way that the initial contract was negotiated.

Maintain a positive attitude

To maintain dialog during disputes, the facility manager must maintain a positive attitude. This approach concentrates on finding the solution by analyzing the problem, not the party involved. The facility manager demonstrates this attitude in many ways, including:

- Leading by example.
- Not taking criticism personally.
- Maintaining a patient, professional manner with a positive tone.
- Supporting the opinions of others by listening fully, without interrupting.
- Assisting by discussing information and options while managing expectations.
- Working for resolution without offending those in dispute.
- Finding common agreement in an atmosphere of accord.
- Summarizing agreement as discussions proceed.
- Checking with everyone concerned to ensure that the final agreement is understood and acceptable to all.
- Thanking and affirming all participants.
- Following up (documenting) discussions in writing.



Facility managers must concentrate on understanding the nature of the relationship and working with the service provider before focusing on the contract. But the contract itself should clearly reflect what is expected of the service provider and how those expectations are measured. Without clearly stated measurements, discussions may be less productive and more adversarial.

Respect, flexibility, willingness to listen and an unbiased attitude are qualities the facility manager must bring to discussions. They not only calm tense situations but also help produce an atmosphere for productive resolution and an even stronger future relationship between the facility manager and the staff or contractor who provides the service.



Progress Check Questions

Directions: Read each question and respond in the space provided. Answers and page references follow the questions.

1. What is the most accurate description of the difference between a service level agreement and a statement of work?
☐ a. There is no difference; they are identical.
☐ b. A statement of work can be part of a service level agreement.
☐ c. A statement of work can be negotiated; an SLA cannot.
☐ d. A statement of work is for staff; a service level agreement is for contractors.
2. The typical SLA for contracting outside services would
☐ a. contain all information both parties require.
☐ b. not stipulate how disputes will be handled.
☐ c. provide the facility manager with negotiating advantages.
☐ d. discuss incentives only in the event of disputes.
3. Standard operating procedures are typically submitted by
☐ a. owner/executive management.
☐ b. facility manager or designate.
☐ c. contractor or candidate service provider.
☐ d. legal staff or administrators.
4. Outside contractors may be hired because they offer competencies that facility staff
☐ a. do not have.
☐ b. cannot justify.
☐ c. do not concentrate on because they are not relevant to the core mission.
☐ d. All of the above
5. An activity that has high mission relevance and requires low expertise would
☐ a. generally be performed by a contracted service.
☐ b. seldom require an operations and maintenance focus.
☐ c. most often require expert services beyond the level of staff.
☐ d. most likely represent a good option for staff assignment.
6. Outsourcing or out-tasking is the process of
☐ a. rewarding high performers.
☐ b. scheduling staff or contractors for maximum productivity.
☐ c. comparing results of staff against contractors.
☐ d. contracting for services, not using staff.



7. Why does a facility manager weigh advantages and disadvantages of contracting?
- ☐ a. To better understand and prepare for potential issues with contractors or staff
 - ☐ b. To foster an us-versus-them competition for greater productivity
 - ☐ c. To favor staff over contractors wherever possible
 - ☐ d. To better justify the use of staff over contractors
8. Which of these would not be considered an appropriate characteristic of negotiations?
- ☐ a. Discussion implied with compromise
 - ☐ b. Meet objectives of service providers and facility owners
 - ☐ c. Designed to give the facility manager an advantage
 - ☐ d. Specific performance measures and non-negotiable minimums
9. The most effective way to resolve contract disputes is by
- ☐ a. using incentives and rewards.
 - ☐ b. establishing concrete and mutually understood SLAs.
 - ☐ c. using flexible scheduling and providing access to human resources.
 - ☐ d. maintaining a positive attitude.



Progress check answers

1. b (p. 135)
2. a (p. 137)
3. c (p. 137)
4. d (p. 139)
5. d (p. 139)
6. d (p. 139)
7. a (p. 144)
8. c (p. 149)
9. b (p. 160)



Chapter 6: Measure Operations and Maintenance Performance

After completing this chapter, students will be able to:

- Discuss life-cycle cost analysis relative to total cost of ownership.
 - Use appropriate measuring and data-gathering tools such as reliability-centered maintenance, design for reliability, energy modeling and risk assessment.
 - Describe how to use the workflow process, specific performance measurements and key performance indicators to characterize performance using a balanced scorecard.
 - Explain how to determine key performance indicators that represent the strategic facility plan.
 - Discuss techniques for monitoring and improving customer satisfaction.
-

★ Topic 1: Determine Life-Cycle Cost

Life-cycle cost analysis

An asset's life cycle, life-cycle phases, and life-cycle cost (LCC) were introduced in Chapter 3 as part of the considerations when acquiring a new asset. In this manner, life-cycle cost analysis begins in the planning/design stage and is useful in comparing options in equipment, systems, materials or components. LCC analysis can also be helpful when choosing between contractors or staff to source work. Once an asset is placed in operation, the facility manager uses life-cycle cost analysis and related factors to help determine whether to service, refurbish or replace that asset later on in its life cycle. To accurately identify and track life-cycle costs, the facility manager should capture all time expended and materials used to maintain facilities because this:

- Factually supports and helps justify repair versus replacement decisions.
- Provides input to evaluate and calibrate or estimate task completion times.
- Produces more complete and accurate information on work activities to support facility management decisions.
- Helps allocate resources properly and justify those allocations.
- Provides data to measure against key performance indicators to assess compliance, challenges and potential changes.
- Supports a focused approach to a strategic facility plan for effective use of the facility as it accomplishes its mission.



Other fiscal factors and estimates that relate to LCC include total cost of ownership, economic analysis sources, forecasting and sampling.

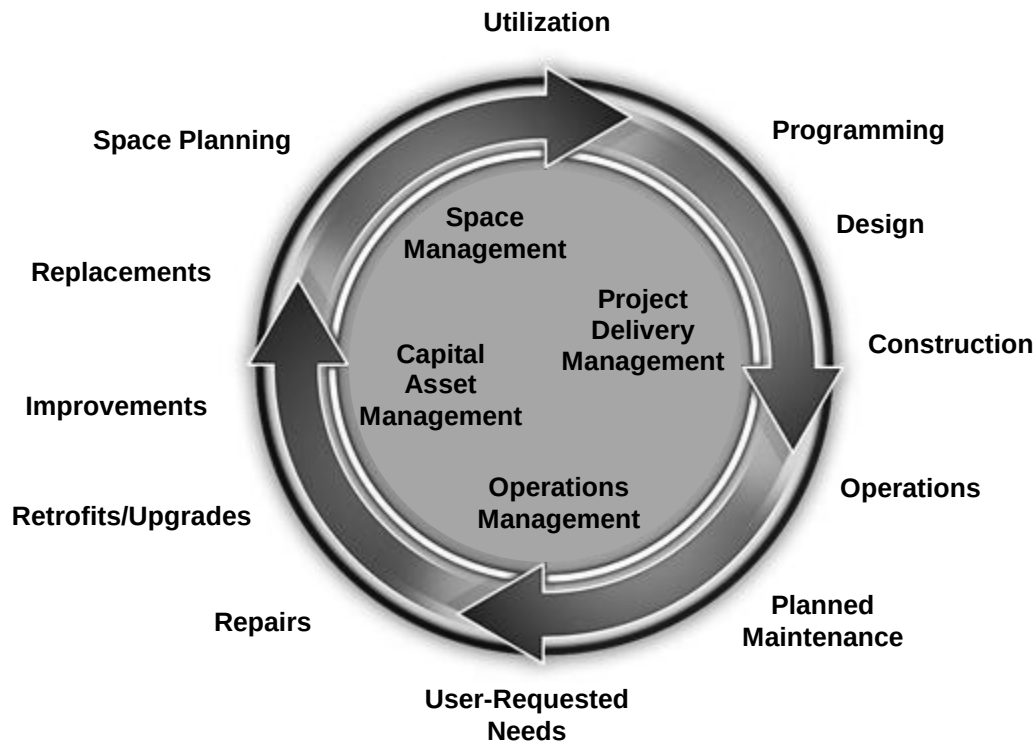
Total cost of ownership (TCO)

Total cost of ownership is a financial management strategy that accounts for the complete life-cycle (cradle-to-grave) measurement and management of a physical asset's useful life. This includes both direct costs that are specifically related to the item or asset as well as indirect costs shared by several grouped assets.

The life cycle of a building can generally be divided into planning, design and construction; operations and maintenance; and capital renewal and renovation through to disposal. All activities except initial planning/design/construction are usually in the realm of the facility manager. They usually represent 60 to 80 percent of the total cost of ownership and, thus, a significant responsibility. Facility managers must understand and apply life-cycle cost analysis to develop budgets for, determine actual costs of, and make decisions about whether to service, refurbish or replace elements.

A model of factors related to total cost of ownership is shown in Exhibit 6-1.

Exhibit 6-1: Asset Life-Cycle Model for Total Cost of Ownership



As a financial estimate, TCO helps the facility manager (and others who have a need or requirement) calculate known and anticipated costs of an asset. These costs may include the following:

- **Initial purchase costs.** This includes all initial money outlays (or capital) to acquire, transport, design, construct, install and commission the asset.
- **Operating expenses.** These recurring costs include utility (energy/water) use; floor space requirements; and estimated maintenance, repair and downtime costs for the useful life of the asset.
- **Hidden factors.** These can include environmental sustainability and negative social costs such as pollution that affect the neighborhood or society. There may also be other hidden factors that are difficult to measure. For example, the facility manager can consider competitive positioning, aesthetic improvement, community standing and similar social benefits.
- **Carrying charges.** Carrying charges refer to any fees associated with having the asset, such as interest payments on the loan balance, taxes or licenses.
- **Replacement and decommissioning costs.** These include not only the costs for alterations or improvement but also recycling, waste transportation and related disposition expenses.
- **Residual value.** Any resale, salvage, reuse or other savings realized when the asset is removed from service are also considered and may reduce the TCO estimate.

The facility manager may also separate operational expenses from capital replacement/renewal budgets. Yet, while these are often divided into separate budgets, they are also considered together. For example, the capital replacement cost for replacing a chiller may also be supported because it could significantly reduce operating costs through lower utility bills.

In this way, the facility manager incorporates all factors relevant to the asset as a way to summarize and analyze what that asset will cost throughout its lifetime of use. This calculation helps indicate when the



continuing cost of maintaining or repairing the asset may not be justified so that replacement or renovation is a better business alternative.

These considerations are useful for small upgrades as well as acquisitions and major renovations. The TCO analysis is useful to indicate the payback period (the length of time it will take to recoup the initial investment cost). This may indicate the value (and wisdom) of acquiring a specific asset in the first place. For example, if the TCO analysis indicates that the payback period exceeds the useful life of the upgrade or the initial asset, the facility manager may consider less expensive alternatives.

Economic analysis sources

The facility manager also refers to other sources to support the LCC economic analysis. These references may include:

- Studies presenting experience with similar components or systems.
- Maintenance experience gathered from contractors, staff or other facility managers familiar with the component or system.
- Published data by the equipment/system manufacturer on maintenance cycles, specifications and so forth.
- Information or requirements available from local, regional or national regulatory or planning boards.

Each of these resources may provide information that the facility manager includes in a complete analysis that anticipates life-cycle costs for a material, component, equipment or system. All may be considered by the facility manager.

Forecasting

The facility manager may also forecast the costs for maintaining an asset at a predetermined service level. Forecasting helps the facility manager develop long-term capital budgets. It also gives the facility manager the means to assess operations and maintenance plans relative to life-cycle costs. This input can be projected to a budget that reflects an understanding of:

- What is anticipated to happen among the various assets of the entire infrastructure.
- What total costs are associated with O&M across the entire infrastructure.

The level of detail and accuracy of a forecast depends upon the expertise, time constraints and information available to the facility manager. The forecast may be intuitive and approximate on some aspects, use generic



standards for others, and have comprehensive data on still others. The final result will be as dependable as the quality and accuracy of its business and financial inputs.

Sampling

A facility manager may decide to include sampling (also known as parametrics) as a way to increase accuracy or confidence in a life-cycle costing estimate. Sampling is the practice of gathering a subset of the total data available from a process or a population. For example, the facility manager may estimate utility costs for one area by looking at specifications for energy use by all devices in that area and then applying that same measurement to similar areas in the facility. By preparing a sample cost for one area, the facility manager can more reasonably and accurately anticipate similar costs for other, similar areas.

Potential areas or aspects a facility manager might sample could include:

- Personnel costs for particular tasks.
- Utilities.
- Maintenance.
- Breakdown emergencies (productivity losses).
- Repair/alteration projects.
- Furnishings and fixtures.
- Grounds (roads, signs, traffic management and so forth).

These sampled calculations can provide insights into key areas where replacing an asset might be a better alternative than keeping it, even if its operating life is not over. For example, the facility manager may look at utilities costs and loss of productivity during breakdowns and determine that a specific system or item of equipment should be replaced because a more energy-efficient and reliable alternative should produce an overall savings against utility, personnel and breakdown losses.

Importance of LCC

Life-cycle costing is important to the facility manager because it can indicate general guidelines for when replacement may be more suitable than repair. Properly prepared LCC information is made possible with accurate, complete work log records and maintenance reports. These records and reports should include the following information:

- The age of the component, equipment or system compared to factory specifications on durability or estimated useful life
- Reliability records, including all service activities, results and costs (which can be projected to show a trend against manufacturer or facility manager estimates of typical costs for the asset)



- Total cost of owning the asset compared for best value against alternate assets considered to fulfill the same function
- Relative value of operational and hidden costs/benefits to improve productivity, support occupant needs and promote sustainability
- Predictive measures of expenses to avoid such as emergency or after-hours repairs, temporary relocations, lost business or productivity as a threshold number that would support replacement or renovation over repair or maintenance of a deteriorating asset

This type of information helps the facility manager anticipate the life cycle of an asset and the point at which replacement is prudent. This is most relevant because an estimated useful life is fairly predictable for most building systems and components.

Recognizing an asset's useful life

Useful life refers to the generally expected time that a component can be left in service before it will start to experience a rapidly increasing probability of failure.

There is no standard rule to predict useful life across all the different materials, components, equipment or systems that may be found in a facility. However, some general principles for all of these assets can be suggested:

- All assets deteriorate over time.
- Assets deteriorate at different rates (for example, a lamp fails before the electrical system wiring fails).
- Routine or preventive maintenance may not extend service life beyond expectations, but it can avoid rapid deterioration that may shorten life compared to expectations.

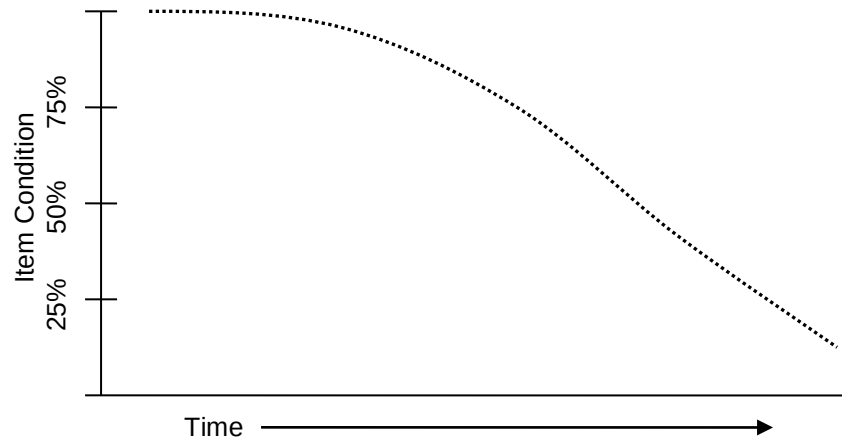
Two useful approaches to determine an asset's useful life are degradation curves and data-measuring tools.

Degradation curves

The facility manager may acquire and use or even develop degradation curves, which estimate the rate and probability at which a component, piece of equipment or system may continue to meet or achieve its intended function over time. Degradation curves may be developed for virtually any known element, including structures, roofs, roadways, planned maintenance and so forth. Estimates generally relate to asset functionality or condition measured over time and assume proper planned maintenance. Equipment manufacturers may also provide statistics and even warranties on typical useful operating life for their products. Exhibit 6-2 illustrates a generic degradation curve.



Exhibit 6-2: Degradation Curve



An actual curve would indicate expected product condition or capability across a course of time such as years. Depending upon the purpose of the asset, the information might estimate its value, functionality, reliability, maintenance cost, efficiency, or other factors over time.

For example, a sample review of the condition of carpeting or paint may reveal that it should be replaced or repainted because of aesthetic rather than functional reasons. A parametric sampling (discussed previously) of one building or a few floors could represent the measurement of several buildings and become part of a degradation curve estimate as well as a forecasting approach.

Forecasting may incorporate degradation curves as part of the information to project future performance based on historical data and measures of current conditions.

Data-measuring tools

A range of possible measuring and data-gathering tools are also available to the facility manager. Several have already been mentioned to measure total cost of ownership and life-cycle costs. These tools could include the following:

- **Reliability-centered maintenance** (which was introduced in Chapter 3) is an ongoing process that determines the mix of reactive, preventive and proactive maintenance practices to provide the required reliability at the minimum cost. It can use diagnostic tools and measurements to assess when a component is near failure and should be replaced. The purpose is to eliminate more costly unscheduled maintenance and minimize preventive maintenance.



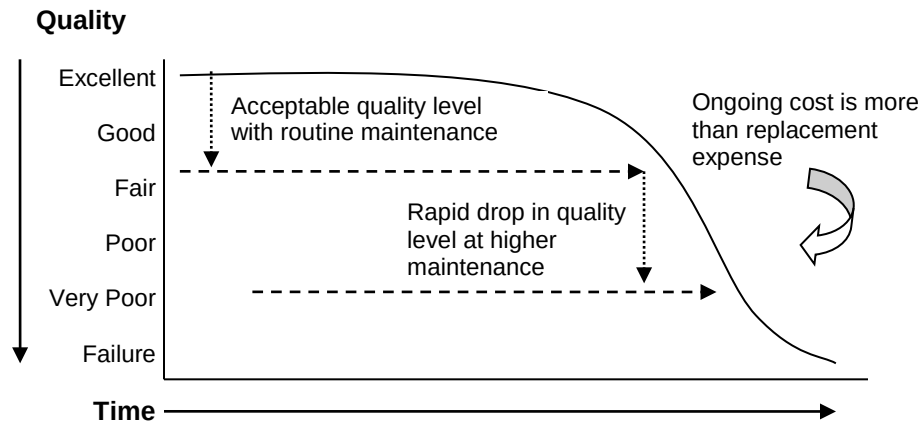
- **Total productive maintenance** is an approach that emphasizes having the equipment operator be responsible for and knowledgeable about the equipment's performance and maintaining that performance.
- **Design for maintainability** is a quality effort in which the product is designed with greater maintenance ease built in, for example, by making grease fittings prominent and accessible on the front of a machine.
- **Design for reliability** is a quality effort in which the dependability of the product is produced by design rather than through inspection.
- **Energy modeling** compares specific materials, systems and equipment relative to their expected annual energy cost, lifetime performance and any efficiencies that might enhance them.
- **Probabilistic risk assessment** is a top-down approach used to apportion risk to individual areas of plant and equipment and possibly to individual assets so as to achieve an overall target level of risk for a plant, site or organization. These levels of risk are then used in risk-based techniques, such as reliability-centered maintenance.
- **Hazard and operability (HAZOP) studies** help develop appropriate equipment maintenance strategies that identify required equipment modifications. A HAZOP study is generally done by experienced teams who examine engineering and process standards to uncover potential operational hazards or malfunctions and how they might affect the facility and surrounding area.

Whether acquiring or tracking assets by using approaches such as these, the facility manager can strike a balance for the lowest risk, the lowest cost, and the highest benefit for a given asset.

Exhibit 6-3 illustrates that, with maintenance, an asset should generally provide excellent or good service to service level standards for a predictable period. The exhibit is shown for illustration purposes only and does not represent an actual asset or relationship to any group of assets. What it shows is that, at some point, the cost of maintaining the asset at a required level of quality exceeds the value the asset provides when compared to a replacement.



Exhibit 6-3: Life-Cycle Costs and Estimated Useful Life



The point at which natural deterioration makes continued maintenance impractical can be recognized as a trend in maintenance costs and reliability concerns reported in work order logs. If the facility manager waits too long to make appropriate repairs or replacement, the cost to bring the component or system back to serviceable condition may be much more than the cost of making the repair or replacement.

✦ Topic 2: Monitor Use and Performance of Facilities

Facility managers recognize that the performance of virtually any facility would most likely benefit from improvement. This places several expectations on the facility manager:

- **Tell the facility management story.** Promote the work and achievements of O&M across equipment, systems and occupant services. Be prepared with information—results, needs, ideas and accomplishments—so that executives and occupants alike are aware of what facility management is doing and how it supports the strategic mission.
- **Justify appropriate resources.** Develop the means and measurements to factually represent what the facility needs, including staff and contracted resources, capital and maintenance budgets, and the plans for managing everything wisely.



- **Support accountability and transparency.** Develop metrics and data-gathering approaches that offer reasonable and factual measurements so that executives and all interested parties understand how facility management is proceeding and how it is doing.
- **Provide credible performance measures.** Use data-gathering tools, performance goals and measurement approaches that deliver accurate information that decision makers understand with confidence and that sustain the facility mission.

Facility managers must not be content with simply doing a good job; they must also be aware of the need to measure, justify and clearly communicate what is being accomplished. It is important to promote what facility management is doing and what's needed so that this vital activity is integrated with and enhances the purpose of the facility.

Respond to the plan

The facility manager is responsible for overseeing the O&M of the facility and its systems and the proper execution of occupant services. Service level agreements, statements of work and standard operating procedures describe what and how it is done; maintenance schedules and workflow processes describe when and where it is done.

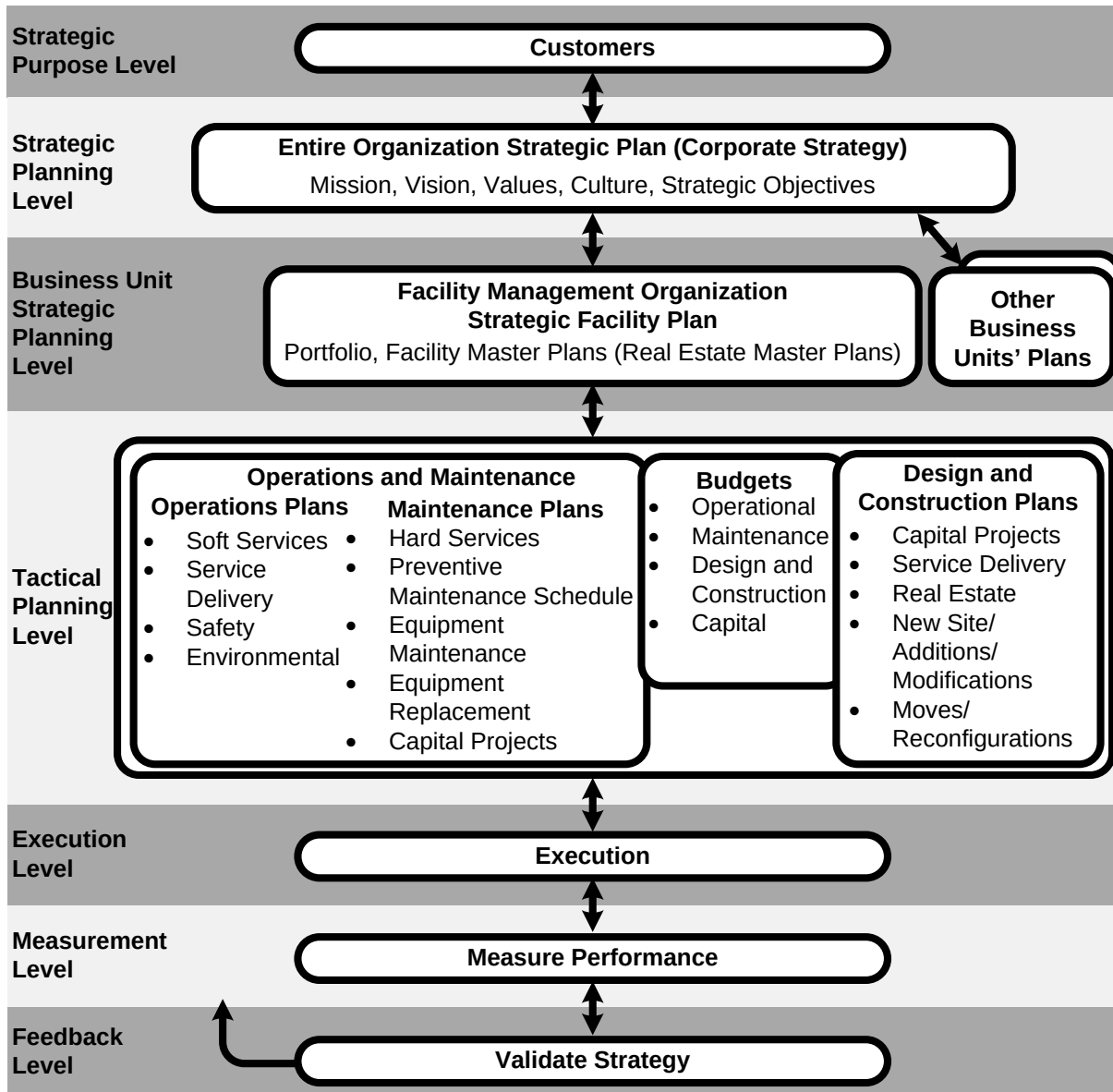
The facility manager oversees the conduct of these activities to support the mission of the facility and its strategic facility plan and the O&M approach that is developed from it. In this way, the emphasis of O&M is on outcome, not merely on activity.

As first introduced in Chapter 1, the activities that occur with O&M (as well as paralleling tactical planning areas such as budgeting and design/construction) are driven by the business strategy for the facility and the entire organization to the benefit of customers.

This relationship was first shown in Chapter 1 (Exhibit 1-1: Strategic Planning Overview Model). Exhibit-6-4 extends that model by adding details to the tactical planning level and completing the model by including execution, measurement and feedback phases. The model thus outlines the approach a facility manager may take to take the organization's strategy and turn it into a sound facilities plan and an O&M program with measured results.



Exhibit 6-4: Strategic Planning Overview Model Extended

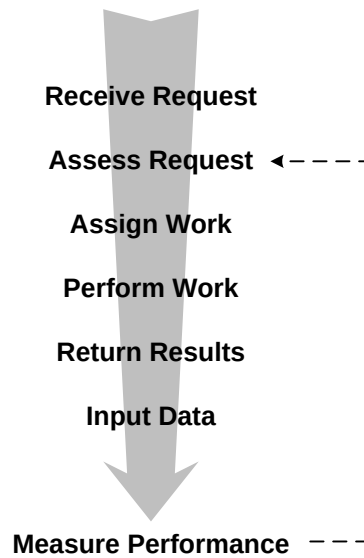


The facility manager takes direction from the strategic facility plan and interprets its goals into the specific activities, assignments and outcomes that generate the work of O&M for building systems and occupant services.

Whether the workflow process represents planned or unplanned maintenance or occupant services, the approach can be generally the same. It was first described in Chapter 3 and is repeated in Exhibit-6-5: Typical Workflow Process.



Exhibit 6-5: Typical Workflow Process



The facility manager measures actual performance against realistic, quantitative goals. These goals may be described as performance requirements, and performance requirements may be summarized into a few, mission-critical ones called key performance indicators (KPIs).

By defining outcomes to achieve specific goals, the facility manager has the means to develop appropriate work plans, oversee their accomplishment and compare actual results to desired performance. This is the basis for professional facility management that can provide building owners and executives with useful information on how well O&M outcomes are supporting strategic business goals.

In short, the strategy of the facility plan determines tactics and the delivery of the workflow process executes that plan. This capability is built around knowing:

- What to measure (performance indicators).
- How they compare (benchmarks that represent a point of reference).
- How to bring it together (balanced scorecard).
- How to track and record it (work log results).

By applying these techniques, the facility manager has an accurate and meaningful presentation that justifies what activities were undertaken, what work plan decisions were made, and how facility operations supported the



building's purpose and its occupants. It is not only the basis for how things have been accomplished but is an organized approach for continuous improvement.

Determine performance measurements

In determining what performance to measure, the facility manager considers the facility mission. For example, if it is a production facility, a performance measurement could be minimal anticipated downtime. If it is a hospital or apartments, a performance measurement could be customer satisfaction.

Specific performance indicators or outcomes can be determined to monitor the results of virtually any activity, task or outcome that is considered important to the purpose of a facility. They measure performance relative to the service level agreement, statement of work, or other written requirements developed in the workflow process.

Key performance indicators are business metrics that stipulate outcomes crucial to the success of the organization. They are identified so that O&M activities can be measured against expected results. KPIs usually summarize or collect lower-level performance measurements into representative critical outcomes.

The facility manager compares actual results to KPIs to evaluate how successful a maintenance activity in the statement of work or SLA/SOW has been accomplished relative to KPI expectations. Key performance indicators may be generally described using the SMART approach, varied types, and quantitative and qualitative measures.

SMART approach

The SMART approach is a way to develop feasible key performance indicators (or any performance outcomes) that represent a critical facility management goal. SMART is an acronym that represents a test of feasibility with a KPI that is:

- **Specific** and clear to avoid misinterpretation. The KPI or performance measurement should represent a single element of the performance outcome and be written so that any stakeholder with a basic understanding of the facility's purpose can understand it.
- **Measurable** to be compared to other data. The KPI should define how to measure whether it is achieved; this is usually a quantitative measurement that represents a specific, understood outcome.



- **Achievable** (attainable) with a reasonable expectation to be reached under the conditions expected. This means that the goal can be met using the skills, assets and capabilities of the intended staff/contractor and work plan. It also assumes that requirements do not conflict with each other.
- **Relevant** in that it fits within the organization's current capabilities, pertains to the task at hand and is cost-effective. It must be feasible and agreed upon by the responsible parties (those who do the work as well as those who oversee the work).
- **Time-bound** in that it should be accomplished within the given duration and is appropriate compared to other priorities or capabilities of O&M and the facility mission. Accurately estimated deadlines promote efficiency and provide a foundation for reporting and control.

There are many ways to develop key performance indicators and thus develop or determine best practices. More common stages for developing a KPI may include the following:

- Identify potential area.
- Determine what to measure (hours, activities, work completed, planned versus unplanned maintenance and so forth).
- Set performance expectation based on known standards or historic results.
- Collect information/data on current activity.
- Compare findings against expectation to determine performance gap and use as a benchmark.
- Set improvement goals based on corrections or work approach changes.
- Monitor progress.

SMART goals are further discussed in the *Project Management* course.

Varied types

KPIs can be as varied as facilities and facility managers themselves. There are likely hundreds of potential metrics relative to facility management. The goal is to select the right mix of metrics to develop as key performance indicators that set goals and measure achievement. They may be developed to enhance facility management from different perspectives, such as:

- Operations decision making with a goal of reducing operating costs and risks and maximizing output.
- Reliability monitoring to maximize operating time and preserve asset integrity.
- Workflow or maintenance management to minimize corrective maintenance or restore asset condition.



- Health and safety to reliably provide expected and code-compliant electrical systems; air, water and waste management quality; security and emergency preparedness; and a safe working environment free from hazardous conditions.
- No downtime (such as for a data center or productivity on a manufacturing line).

Quantitative and qualitative measures

KPIs are measurable; however, some may apply qualitative factors that are descriptive of kind, type or direction as opposed to size, magnitude or degree. Others will apply quantitative factors involving measurements and numbers. In both instances, KPIs should reflect standards that are understood, communicated and agreed upon by the facility manager, executive management, staff, contractors, interested occupants and any other relevant parties.

KPIs for operations and maintenance may include any or all of these quantitative and qualitative aspects:

- Quantitatively measured energy and resource costs (to help determine if a more energy-saving alternative will pay for itself in long-term savings)
- Quantitatively reviewed cost cycles that identify when the cost of maintaining an asset becomes more than the life-cycle cost of replacing it (This measure can be drawn as an item from the cost of operation, which provides the total costs associated with the daily operation of a facility. It includes all maintenance and repair costs (both fixed and variable), administrative costs [clerical, time-keeping, general supervision], labor costs, custodial, housekeeping and other cleaning costs, utility costs and indirect costs [i.e., all costs associated with roadways and grounds.])
- Qualitative general wear expectations in which the useful life stated by the asset manufacturer may be affordably extended to produce a longer service life
- Qualitative requirements that reflect cultural values and organizational strengths, weaknesses, opportunities and threats
- Qualitative aesthetic values that refer to design considerations such as materials quality, color choices or maintenance frequency
- Qualitative expectations relative to social values such as occupant satisfaction, community reputation and competitive standing



Consolidate in a balanced scorecard

Performance measurements summarized as KPIs allow the facility manager to set measurable goals, based on the strategic facility plan. The balanced scorecard (BSC) allows the facility manager to position goal requirements as KPIs that are strategically relative to the strategic facility plan.

The entire organization—facility, management, executives and finance—can be viewed through the balanced scorecard perspective to develop metrics, collect data, and analyze options that transform strategic planning into measureable results determined from viable alternatives. In this way the balanced scorecard supports facility management in a business as well as operational manner.

The facility manager uses the balanced scorecard method and related metrics to assess how well goals are being achieved and to optimize the performance of the entire infrastructure. This helps the facility manager operate the facility with greater confidence and smarter business sense.

A balanced scorecard is based on the premise that intangible assets such as employees, information technology and image are important to an organization's success. To accomplish this, the balanced scorecard views the organization from four perspectives that facility managers turn into self-assessing questions:

- **Stakeholder** measures reflect the satisfaction of customers or occupants.
Ask: What do customers/occupants think of facility management performance?
- **Internal** measures reflect performance of key business processes.
Ask: Do we excel at what we do and apply best practices?
- **Learning and growth** measures reflect on the organization's learning curve or capabilities and position.
Ask: Is our staff adequately trained and capable?
- **Financial** measures reflect fiscal performance.
Ask: Are we taking care of assets?

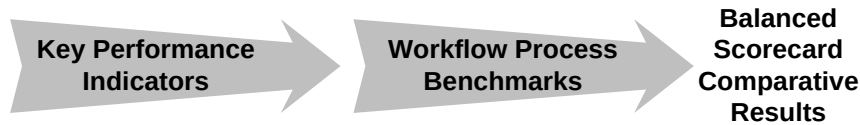
With these four perspectives, the BSC helps compare short- and long-term objectives, outcomes desired and the performance aspects that drive those outcomes as objective/subjective measures.

In an ideal planning cycle, the BSC is derived from the entire organization's strategic plan, the more specific strategic facility plan and the work plan process written into SLAs. In this way, the entire approach to managing the facility is aligned with and directed to the facility mission. The results of the



work log tell the facility manager how well O&M activities are achieving KPIs to meet BSC goals. The relationships are illustrated in Exhibit 6-6.

Exhibit 6-6: Performance Relationships



Compare to work log results

The facility manager incorporates several aspects to relate work log results (or workflow process benchmarks) to the BSC. These aspects may be summarized as follows:

- The approach begins with knowledge of the strategic facility plan.
- That plan identifies measurable goals for facility management.
- Metrics assess the O&M of assets relative to, for example, total cost of ownership and most favorable life-cycle cost.
- These metrics provide measurable goals that are defined as key performance indicators and included in the BSC.

The tracked results of the work plan log may be compiled within the CMMS to provide current, accurate, complete information on all accomplished O&M and related activities. By analyzing work log results (work plan data or work orders) and summarizing or comparing that information as KPIs in the BSC, the facility manager can:

- Compare planned to actual performance.
- Evaluate the metrics.
- Identify underperforming elements.

Compare planned to actual performance

A current and complete inventory/register should contain measurable cost and performance information taken from work logs and operations, maintenance and repair inputs. With this information, the facility manager updates TCO information to:

- Determine history and trends based on actual experience, manufacturer specifications and expected useful life.
- Compare planned performance to actual performance to reveal gaps or challenges.
- Review alternative systems and components as a possible baseline for future replacements or renovation.



By having the means to predict the total cost of ownership for an asset and measure that prediction against actual performance, the facility manager can make decisions to improve productivity and minimize costs.

Evaluate the metrics

Facility managers produce and use several potential metrics to produce a world-class facilities organization. Some of the more common ones include the following:

- **Facility condition index (FCI).** This assessment objectively benchmarks current and projected building condition, often using inputs and outputs from a CMMS (computerized maintenance management system). An FCI also helps the facility manager supply information to owners/decision makers about renewal costs compared to building replacement value.
- **Planned maintenance to corrective maintenance.** This KPI indicates how adequately current preventive maintenance (PM) is sustaining systems and equipment relative to the cost of unplanned corrective activities and repairs.
- **Preventive maintenance completion and percentage of re-work.** These rates indicate how efficiently work is being accomplished according to schedule and right the first time.
- **Work order aging and response time.** These measures indicate how quickly a problem is addressed and resolved. They may also help the facility manager recognize the capabilities of current staff or contractors relative to work load.
- **Customer/occupant satisfaction.** This KPI approach (which is further discussed as the final topic in this chapter) is useful for providing clear and factual metrics relative to how well O&M is serving occupant needs.
- **Custodial level of service.** Any occupant service or maintenance program such as custodial, groundskeeping, information technology or even security that has specified service level requirements may be compared against actual performance to determine gaps and weaknesses.
- **System reliability, unscheduled downtime and productivity.** These KPIs help a facility manager recognize how complete and adequate PM is and its effect on operations.



- **Top ten work orders by type.** This KPI identifies the most commonly occurring problems or activities in order to find ways to improve on performance.
- **Mean time between failures (MTBF).** This KPI estimates average time between anticipated failures as a system operates and may incorporate the time required for repair and downtime (unplanned maintenance) if a failure occurs. This helps a facility manager build O&M schedules that maintain the highest productivity at the lowest PM cost.
- **Percentage root cause analysis.** Root cause analysis (RCA) is not only a method for determining why things fail or problems occur but can also be used as a summary to indicate if analyses are useful to stop problems from recurring. Generally, the greater application of RCA produces better results.

The KPIs a facility manager establishes should reflect the strategic facility plan and the recognized mission of the entire organization. They can relate to O&M for building systems and equipment as well as quality of occupant services. In this way, the intentions of the owner/decision maker stipulated in the SFP are supported by the O&M work plan.

Identify underperforming elements

By measuring and comparing actual performance to planned performance, the facility manager is also alerted to assets that are not meeting expectations. These may be called variances, and they measure or otherwise indicate the difference between expected outcomes and actual results. Underperformance may be occurring for reasons such as inadequate or incorrect maintenance schedule; work time estimates; or work difficulty that is different from expectations, causing higher labor, materials, or other maintenance costs.

The facility manager recognizes assets that are not performing as expected and develops ways to set improvement goals, develop corrective action, monitor the approach and continue to measure results.

Recommend future asset purchases

Technologies, processes, compliance regulations and standards are always changing. For example, in many countries, building codes are updated and published in a new edition every year.

The facility manager stays current on all aspects of facility management. They become particularly relevant when considering adding new equipment, systems or materials or replacing those that are nearing the end of their useful life.



By collecting work log data and summarizing it into trends, the facility manager can begin to predict when an asset may need to be replaced or its function substituted with a renovation. By staying current and researching new alternatives for an asset, the facility manager can begin to develop an estimate for the total overall cost of ownership for that asset supported by life-cycle cost analysis and, over time, by balanced scorecard measurements of key performance indicators.

The facility manager analyzes material, equipment and systems to determine the most cost-efficient choices. Relative to performance measurements, these considerations can include the following:

- Best durability level at lowest overall cost. Flexibility may be considered along with durability so that, for example, a series of 50-ton chillers may be a better alternative than a 300-ton chiller.
- Cost efficiency that considers total cost of ownership when comparing and selecting lighting, boilers, roofs, windows, control systems, flooring, fixtures and so forth.
- When renovations or capital improvements are being planned, the facility manager supports a team approach represented by the range of individuals who own, manage, occupy and support the facility. Together they can reach decisions with the widest acceptance at the least cost that considers important criteria such as:
 - Life safety, barrier-free access and building standard requirements.
 - Repair versus replacement; upgrades versus renovation.
 - Type of space, services, furnishings, fixtures and equipment the facility should provide.

✦ Topic 3: Monitor Occupant Satisfaction

Occupant satisfaction described

Occupant satisfaction is important because it indicates customer approval, and the approval and satisfaction of customers is critical to the success of any business, public and private. Serving customers/occupants well is a vital step toward success and, ultimately, a facility's ability to fulfill its mission.

The comfort, confidence and productivity of occupants is at the core of a facility's purpose. The delivery of excellent operations and maintenance service should be seamless and invisible.



Quality occupant service is motivated and determined by several considerations, including the following:

- Occupants/customers determine what defines quality, not the facility manager or owner.
- There should be continuous improvement, not simply sustaining the current status.
- The facility manager and owner/executive management need to be willing to measure occupant satisfaction.
- The facility manager, maintenance staff and contractors should be willing to be measured through occupant surveys and other feedback.
- The facility manager should use a total quality management approach to determine, measure, monitor and improve occupant satisfaction standards.
- Staff and contractors should be held accountable to the expectations of the facility manager and the wishes of occupants.
- Staff and contractors are aware of, understand, and are knowledgeable about how to communicate with occupants, how to respond to them, and how to include feedback in work plans or surveys.

IFMA researchers have often summarized the main complaints facility managers hear from occupants. Over the years the types of complaints remain similar and vary only in their ranking among the top ten. A general summary of typical occupant complaints is shown in Exhibit 6-8.

Exhibit 6-8: Typical Top Complaints to Facility Managers

- | | |
|---|--------------------------------|
| • It is too cold; it is too hot. | • Poor indoor air quality. |
| • Custodial service is poor. | • No privacy in my workspace. |
| • Not enough conference rooms; not enough space for me. | • Inadequate parking. |
| | • Computer problems. |
| | • Too noisy; high noise level. |

Service quotient defined

A measurement of occupant satisfaction may also be called a service quotient or, more broadly, a customer service index. Relative to O&M, a service quotient is a measurement of how satisfied occupants are with their service at a given point in time. In this way, a service quotient describes the quality of occupant satisfaction as the goal; it can be expressed as a quantity or proportion increase or decrease when measured over time.



The foundation of that measurement starts by assessing the facility's current level of occupant satisfaction. Just as an inventory/register of all assets is needed to competently manage operations and maintenance, an accounting of occupant opinion is needed as a starting point to know how that O&M is perceived.

If a facility manager has never requested or summarized occupant feedback on the status of O&M services, then the first satisfaction survey provides the basis to compare against future measurements. This starting point provides the standard so that any change in performance can be recognized.

A service quotient is distinct and original to the facility and its occupants. The initial survey sets the base. Ongoing surveys reveal improvements and weaknesses. Because the goal is continuous improvement, the facility manager uses inputs to help refine the work plan and approach to continually improve occupant satisfaction.

Gathering satisfaction data

Occupant satisfaction data may be gathered informally through conversations or solicited suggestions as well as more formally through surveys and work plan analyses. Thorough, complete and periodic requests for feedback are generally an excellent approach that gives the facility manager a better understanding of where and how to respond through operations and maintenance.

Occupant surveys can be understood generally through the approach, sources, benefits and measurements.

Approach

Occupant satisfaction should be reviewed regularly. Standards-setting bodies suggest that occupants should be surveyed at least annually to achieve a level of 6 or a higher rating on a 1 to 10 facility management requirement scale.

The survey approach can be robust or simple depending on what the facility manager and executive management decide. Typically, written or interview surveys include ratings based on a range of questions that include:

- Did we do what you wanted done?
- Did we do it in a timely manner acceptable to you?
- How well did we do without disrupting your work?
- How well did we clean up and conduct ourselves?
- Did we check to see if you were satisfied?



The survey should be designed so that the facility manager can understand what occupants want and need. Communications with occupants should demonstrate that they are valued, heard and responded to. Survey inputs are two-directional: occupants are asked for their feedback and that feedback is returned to occupants as actions, communications and responses that indicate improvements are being made.

The facility manager uses the results of occupant surveys to foster competencies in staff and contractors that support the highest satisfaction levels possible. It is important that maintenance personnel be informed and engaged in a service commitment.

Major areas that the facility manager may stress relative to O&M services include the following:

- Establish policies and procedures that support occupant needs.
- Train personnel in communications skills as well as trades.
- Develop cross-trade qualifications to improve service delivery as well as effectiveness.
- Develop competency in systems that include electrical, electronic, HVAC equipment and piping as well as minor carpentry.
- Support operations and maintenance activities by efficiently supplying the necessary tools, parts, materials and equipment; through minimally disruptive work scheduling; and with a budget sufficient to permit quality levels of service response.

Sources

In addition to feedback from occupants, the facility manager may use other sources to develop a more comprehensive picture of occupant requirements and the facility's ability to respond to those requirements. These sources may include:

- O&M operational data summarized from work logs and summary reports.
- Input from contractors performing work within the facility.
- Input from other service providers (such as local telephone services, utilities providers, hired experts and others).
- Input from O&M staff and other department employees regarding their challenges, concerns, feedback from occupants and so forth.
- Visitors and guests who can offer their first impressions of the facility and its operation.



Benefits

Occupant surveys offer several benefits:

- Performed regularly, they demonstrate trends and reveal strengths and weaknesses.
- They provide direct input on the state of operations and maintenance perceived by users themselves for the most direct understanding of occupant needs and wishes.
- They indicate to occupants the desire and promise of responsive and satisfactory facility management as long as occupant requests and concerns receive attention and action.
- When survey outcomes produce O&M response, occupants gain confidence in service providers as well as services rendered, fostering a higher sense of professionalism and interest. Good service is a competitive advantage created by excellent operations and maintenance.
- Signs of occupant dissatisfaction can be identified and corrected early before they become a concern to senior management.
- Underperforming systems, spaces and service providers can be identified so that corrective actions can be taken.

Measurement

While some people might consider occupant satisfaction as merely a collection of opinions, the facility manager uses this feedback to measure occupant responses against work activities and outcomes. The facility manager can track key performance indicators to compare trends against the service quotient.

For routine work (such as predictive and preventive maintenance, service requests, and regular services such as custodial, waste management, cafeteria and so forth), the facility manager can set standards based on occupant input and service level expectations and then monitor the work through site visits, work logs and staff/contractor feedback.

Key performance indicators for O&M can be developed to assess for gaps or weaknesses by measuring:

- Planned activities (tasks, projects) compared to unplanned. This can help indicate the proficiency of maintenance planning and perhaps even the reliability of specific current assets.
- Cost cycles to identify when the cost of maintaining an asset becomes more than the life-cycle cost of replacing it.
- Proportion of completed work orders (number scheduled versus number finished). This can help indicate work planning, accuracy of scheduling and work expectations and readiness of contractors or staff.



- Total work completed at a given time compared to estimated time for that work (by trade, by contractor, by area). This can help indicate work readiness, worker training, scheduling efficiency and other work performance factors.
- Planned preventive maintenance hours compared to total expended hours. This can help indicate maintenance planning and work product, accuracy of service level expectations, and perhaps reliability or service costs on specific assets.
- Estimated costs compared to actual costs to determine proportion over and under budget related to type of work or area of facility or occupant type. This can help indicate areas of inefficiency, inaccurate estimates, repeat visits and so forth.
- Contractor service and preventive maintenance proficiency by comparing service or work orders by trade, occupant area or period-over-period (or historic) averages relative to:
 - Work orders scheduled versus completed and number outstanding.
 - Cost per service order compared to industry standards or historic averages.
 - Occupant satisfaction service quotient for a particular service or maintenance routine.
- Occupant satisfaction in general to update feedback, develop trends by comparing to historic service quotients, recognize increase/decrease in occupant responses to surveys, and discover changes in perceptions of community and competitive standing.

The facility manager may decide to allow occupants to remain anonymous, which would encourage more candid responses. However, it can be helpful to know what specific projects or work activities an occupant may be responding to as a way to compare inputs against the work plan. This can reveal other levels of detail such as:

- Correlations to service providers working on specific activities and trends among individuals or teams.
- Responses or dissatisfaction correlated to specific work activities or challenging work plans.
- Correlations among work plans, worker competencies, occupant areas and/or service levels.



Satisfaction improvement

The facility manager's goal relative to occupant satisfaction is to continually improve and raise the service quotient score. To do this, the facility manager:

- Provides the mechanisms and engagement that encourage early feedback on problems.
- Learns what occupants expect through surveys and active solicitations of opinion from all sources.
- Understands what occupants depend on, are aware of and otherwise use of the infrastructure.
- Develops this understanding into key performance indicators.
- Identifies critical interfaces with occupants (such as through maintenance staff, at moments of service interruption and so forth.)
- Knows how to precisely define what is wrong in order to understand what to improve.
- Understands how to respond to problems in ways that support occupant expectations, enhance relationships and demonstrate appreciation of occupant needs.
- Identifies exchanges or procedures that are not working because of ignorance, poor communication, lack of response, lost feedback and so forth.
- Observes performances through performance measures and ongoing feedback.
- Identifies opportunities to improve efficiency and effectiveness.
- Implements and manages/oversees changes.
- Validates improvements through occupant feedback and process results.

This cycle is the basis for continued improvement for higher occupant satisfaction reflected in an ever-improving service quotient.



Progress Check Questions

Directions: Read each question and respond in the space provided. Answers and page references follow the questions.

1. How does total cost of ownership (TCO) relate to life-cycle cost (LCC)?
☐ a. They are identical terms.
☐ b. TCO is an annual update of the entire life-cycle cost of an asset.
☐ c. TCO includes all estimated life-cycle (cradle-to-grave) costs plus other factors.
☐ d. TCO is only an accounting tool; LCC is only a maintenance tool.
2. What is the purpose of a degradation curve relative to life-cycle cost analysis?
☐ a. It indicates unforeseen gaps such as carrying charges or hidden competitive costs.
☐ b. It accounts for replacement and decommissioning costs.
☐ c. It factors in inflation, technology improvements and obsolescence.
☐ d. It estimates the rate and probability at which an asset may become impaired in intended function.
3. A standard rule for predicting the useful life of all assets is that
☐ a. There is no standard rule.
☐ b. All assets deteriorate over time.
☐ c. Different assets deteriorate at different rates.
☐ d. All of the above
4. What is the typical relationship between a strategic facility plan (SFP) and the O&M approach?
☐ a. They are not related; SFP is for the entire facility strategy and O&M is just schedules.
☐ b. O&M activities should have outcomes that support the SFP.
☐ c. They are the same in that the SFP schedules O&M.
☐ d. O&M measurements determine with the SPF plan should be.

The strategic facility plan determines tactics and the workflow process executes the plan. Match the following assessment stages to their tool description.

- | | |
|---------------------------------------|---------------------------|
| 5. _____ Know what to measure | a. Work log results |
| 6. _____ Know how to consolidate | b. Balanced scorecard |
| 7. _____ Know how to track and record | c. Performance indicators |



8. How does the SMART approach relate to key performance indicators (KPIs)?
- ☐ a. KPIs may be developed using the SMART approach.
 - ☐ b. SMART is a KPI.
 - ☐ c. KPIs measure performance; SMART measures effectiveness.
 - ☐ d. SMART has nothing to do with KPIs.
9. A facility condition index, mean time between failures, work order aging and custodial level of service are all examples of what?
- ☐ a. Failed or out-of-specification conditions
 - ☐ b. Engineering predictive and preventive maintenance tasks
 - ☐ c. Typical key performance indicators (KPIs)
 - ☐ d. Life-cycle cost analysis tools
10. Which of these is the best way the facility manager develops plans that reflect occupant expectations?
- ☐ a. By surveying occupants
 - ☐ b. By hiring more contract services
 - ☐ c. By reducing disruptions during moves and relocations
 - ☐ d. By installing higher-quality fixtures, furnishings and equipment
11. What is the best description of a service quotient for facilities management?
- ☐ a. Total annual costs of a facility's O&M for occupant services
 - ☐ b. Standard percentage for customer satisfaction; usually 95 to 98 percent
 - ☐ c. Measurement of occupant satisfaction
 - ☐ d. Difference in energy costs after a new device or system is installed



Progress check answers

1. c (p. 167)
2. d (p. 171)
3. d (p. 171)
4. b (p. 176)
5. c (p. 177)
6. b (p. 177)
7. a (p. 177)
8. a (p. 178)
9. c (p. 183)
10. a (p. 187)
11. c (p. 186)



Bibliography

The following resources were used during the development of the *Operations and Maintenance* course.

“Asset Lifecycle Model for Total Cost of Ownership Management—Framework, Glossary and Definitions,” www.ifma.org/know-base/fm-knowledge-base/knowledge-base-details/asset-lifecycle-model-for-total-cost-of-ownership-management.

ASTM International. *ASTM Standards for Whole Building Functionality and Serviceability*, 3rd edition. West Conshohocken, Pennsylvania: ASTM International, 2009.

Atkin, Brian, and Adrian Brooks. *Total Facilities Management*, 3rd edition. Chichester, United Kingdom: Wiley-Blackwell, 2009.

“Balanced Scorecard,” www.ifma.org/about/strategic-plan/balanced-scorecard.

Cost Planning and Estimating for Facilities Maintenance. Kingston, Massachusetts: RSMeans, 1996.

Cotts, David G., Kathy O. Roper, and Richard P. Payant. *Facility Management Handbook*, 3rd edition. New York: AMACOM, 2010.

Craighead, Geoff. *High-Rise Security and Fire Life Safety*, 3rd edition. Boston: Butterworth-Heinemann/Elsevier, 2009.

Hodges, Christopher P. *Operations and Maintenance: An IFMA Competency-Based Course*. Houston, Texas: International Facility Management Association (IFMA), 2005.

Kaiser, Harvey H. *The Facilities Audit*. Alexandria, Virginia: The Association of Higher Education Facilities Officers (APPA), 1993.

Kubba, Sam. *Property Condition Assessments*. New York: McGraw-Hill, 2008.

Lewis, Bernard, and Richard Payant. *The Facility Manager’s Emergency Preparedness Handbook*. New York: AMACOM, 2003.

Liska, Roger, and Judith Liska. *Building Maintenance: Forms, Checklists and Procedures*. Englewood Cliffs, New Jersey: Prentice Hall, 1998, 2001.

Martin, David M. *The A–Z of Facilities and Property Management*. London: Thorogood Publishing, 2006.



Payant, Richard, and Bernard Lewis. *Facility Manager's Maintenance Handbook*, 2nd edition. New York: McGraw-Hill, 1999, 2007.

“Quality Management Principles.” Geneva, Switzerland: International Organization for Standardization, 2012.

Rondeau, Edmond P., Robert Kevin Brown, and Paul D. Lapidès. *Facility Management*, 2nd edition. Hoboken, New Jersey: John Wiley and Sons, 2006.

Spiegel, Ross, and Dru Meadows. *Green Building Materials: A Guide to Product Selection and Specification*, 2nd edition. Hoboken, New Jersey: John Wiley and Sons, 2006.

Westerkamp, Thomas. *Maintenance Manager's Standard Manual*, 3rd edition. Anaheim, California: BNi Publications, 2007.

Zemke, Ron, and John A. Woods, editors. *Best Practices in Customer Service*. Amherst, Massachusetts: HRD Press, 1999.



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